

DATE	SYM	WAS	BY
		D-305501	

Pag. 0 cont. en 1
 Total de pags. 69

DIAGRAMA INPUT/OUTPUT

REFERENCIAS:
 TRABAJAR ESTE PLANO CON LOS PLANOS:
 5615-26-0034 (T1526AB1.xls) * LISTA DE MATERIALES
 5615-23-0030 (T1523ABE.dwg) * DIAGRAMA DE BORNERAS
 5615-24-0033 (T1524AB1.dwg) * CANALIZACIONES
 5615-21-0029 (T1521ABD.dwg) * LAY OUT
 5615-25-0032 (T1525ABG.dwg) * LISTA DE CABLES
 PLANO TECHINT. N° 2456-F-DF-550002 * JOB: 55067

SE AGREGAN SEÑALES, PAGS. 06; 26; 29; 32; 43; 55; 62		DYGE	DYGE	DNP	03-08-01
EMISION PARA CONSTRUCCION		12-06-01			
REV.	DESCRIPCION	EJEC.	CONT.	APROV.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		N° PROVEEDOR: D-308691		
ZONA:		SIDERCROM			
SECTOR:		LINEA LITTELL			
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E			
PARTES:		SCROLL SHEETING LINE			
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO	ESCALAS	5615-23-0028			
DIBUJO					
CONTROL					
APROB					
IMPULSION:		MAG		CAD: T152400.DWG	
		<		FILE: T1523ABC	

BR-4E SCROLL SHEETING LINE #262
SIDERAR, ARGENTINA, SOUTH AMERICA
LITTELL ORDER #94228-00

DATE	SYM	WAS	BY
07/04/90	(A)	D-308691	JDL

TABLE OF CONTENTS

SHEET NO.	DESCRIPTION
	<u>SCHEMATIC DIAGRAMS</u>
01	TABLE OF CONTENTS/LINE SPECIFICATIONS
02	NETWORK DIAGRAM
03	SINGLE LINE DIAGRAM
04	SINGLE LINE DIAGRAM
05	3 PHASE CONTROL POWER CIRCUIT
06	3 PHASE CONTROL POWER CIRCUIT
07	3 PHASE CONTROL POWER CIRCUIT
08	3 PHASE CONTROL POWER CIRCUIT
09	PILER 1 & 2 MAIN CONVEYOR DRIVES
10	PILER 3 & 4 MAIN CONVEYOR DRIVES
11	PILER SIDE GUIDES & END STOPS DRIVES
12	PILER 1 & 2 CONVEYOR DRIVES
13	PILER 3 & 4 CONVEYOR DRIVES
14	CONTROL CIRCUIT - POWER DISTRIBUTION
15	CONTROL CIRCUIT - LINE EMERGENCY STOP & RESET
16	CONTROL CIRCUIT - SHEAR CLUTCH & BRAKE
17	THICKNESS GAUGE AND CENTERING GUIDE (OPTION)
18	PINHOLE DETECTOR (OPTION)
19	DSP DRIVES CHARGING UNIT
20	DSP 2400 - PAYOFF REEL
21	DSP 2400 - STRAIGHTENER
22	DSP 2400 - FEED ROLL
23	DSP 2400 - SHEAR
24	DSP 2400 - TAKEAWAY ROLL
25	UNICO I/O RACK/CPU MODULE/RTC MODULE
26	UNICO (4) SERIAL/MEMORY MODULES
27	UNICO DATA ACQUISITION MODULE/SERVO MODULE (SPD REF. MOD.)
28	UNICO I/O RACK SLOT N - 24VDC INPUTS & SLOT P - 24VDC OUTPUTS
29	UNICO I/O RACK SLOT T - 24VDC INPUTS & SLOT T - 24VDC OUTPUTS
30	PHOENIX BLOCK 1
31	PHOENIX BLOCK 2
32	GE PLC SYSTEM CONFIGURATION
33	OPERATOR'S CONSOLE LOCAL SLOT 3 & 4 - 24VDC INPUTS
34	OPERATOR'S CONSOLE LOCAL 0 SLOT 5 & 6 - 24VDC OUTPUTS
35	BUCKLE DETECT
36	INSPECTION STATION RACK 1 SLOT 1 & 2 - 24VDC INPUTS
37	INSPECTION STATION RACK 1 SLOT 3 & 4 - 24VDC INPUTS
38	INSPECTION STATION RACK 1 SLOT 5 & 6 - 24VDC INPUTS
39	INSPECTION STATION RACK 1 SLOT 7 & 8 - 24VDC OUTPUTS
40	INSPECTION STATION RACK 1 SLOT 9 & 10 - 24VDC OUTPUTS
41	MAIN CONTROL PANEL RACK 2 SLOT 1 & 2 - 24VDC INPUTS
42	MAIN CONTROL PANEL RACK 2 SLOT 3 & 4 - 120VAC/220VAC ISOL. INPUTS
43	MAIN CONTROL PANEL RACK 2 SLOT 5 & 6 - 24VDC INPUTS
44	MAIN CONTROL PANEL RACK 2 SLOT 7 & 8 - CONTACT/24VDC OUTPUT
45	MAIN CONTROL PANEL RACK 2 SLOT 9 - 24VDC INPUT
46	MAIN CONTROL PANEL RACK 2 SLOT 10 - 24VDC OUTPUT
47	MAIN CONTROL PANEL RACK 3 SLOT 1 - 24VDC INPUT
48	MAIN CONTROL PANEL RACK 3 SLOT 2 - 24VDC INPUT
49	MAIN CONTROL PANEL RACK 3 SLOT 3 - 24VDC INPUT
50	MAIN CONTROL PANEL RACK 3 SLOT 4 - 24VDC INPUT
51	MAIN CONTROL PANEL RACK 3 SLOT 5 & 6 - CONTACT/24VDC OUTPUT
52	MAIN CONTROL PANEL RACK 3 SLOT 7 & 8 - 24VDC OUTPUT/CONTACT
53	MAIN CONTROL PANEL RACK 3 SLOT 9 - 24VDC OUTPUT
54	STACKER RACK 4 SLOT 1 & 2 - 24VDC INPUTS
55	STACKER RACK 4 SLOT 3 & 4 - 24VDC INPUTS
56	STACKER RACK 4 SLOT 5/RACK 5 SLOT 1 - 24VDC OUTPUT/24VDC INPUT
57	STACKER RACK 5 SLOT 2 & 3 - 24VDC INPUTS
58	STACKER RACK 5 SLOT 4 & 5 - 24VDC INPUT/24VDC OUTPUT
59	CLASSIFIER PLC I/O RACK/ RTC MODULE/ CPU MODULE
60	CLASSIFIER SERIAL/ MEMORY MODULE/ SERVO MODULE
61	CLASSIFIER INPUT/ OUTPUT
62	OPERATOR CONSOLE COMPUTER, COLOR MONITOR, DIAGN. STAR MODULE
63	SYSTEM NOTES
64	SYSTEM NOTES
65	DRIVE PARAMETERS
66	MOTORIZED GAUGE TABLE (OPTION)
67	ADDITIONAL CUSTOMER CONSOLE

TABLE OF CONTENTS

SHEET NO.	DESCRIPTION
	<u>LINE LAYOUT DIAGRAMS</u>
01	LINE LAYOUT PLAN "B", "D" - SHEARING END
02	LINE LAYOUT PLAN "A", "D" - STACKER END
03	LIST OF SOLENOIDS AND SWITCHES
	<u>WIRING DIAGRAMS</u>
01	MAIN OPERATOR'S CONSOLE PUSHBUTTONS
02	DOWNENDER (OPTION), EDGE TRIMMER, AUX. & STRAIGHTENER PUSHBUTTONS
03	SHEAR & CATENARY PUSHBUTTONS
04	PILER 1 & 2 PUSHBUTTONS
05	PILER 3 & 4 PUSHBUTTONS
06	MAIN OPERATOR'S CONSOLE RACK #0
07	MAIN OPERATOR'S CONSOLE TERMINALS
08	COIL CAR & PAYOFF REEL TERMINALS
09	DOWNENDER (OPTION) & EDGE TRIMMER TERMINALS
10	STRAIGHTENER & THICKNESS GAUGE TERMINALS
11	INSPECTION & SHEAR RACK #1
12	INSPECTION STATION
13	SHEAR TERMINALS
14	SHEAR & FEED MOTORS
15	PILER 1 TERMINALS
16	PILER 1 & 2 MOTOR TERMINALS
17	PILER 1 & 2 RACK #4
18	PILER 2 TERMINALS
19	PILER 3 TERMINALS
20	PILER 3 & 4 RACK #5
21	PILER 4 TERMINALS
22	PILER 3 & 4 MOTOR TERMINALS
23	MOTORIZED GAUGE TABLE
24	PANEL 1 & 2 - LAYOUT
25	PANEL 3 & 4 - LAYOUT
26	PANEL 5 & 6 - LAYOUT
27	PANEL 7 - LAYOUT
28	PANEL 8 & 9 - LAYOUT
29	PANEL 1,2,3,4 - TERMINALS
30	PANEL 5 & 6 - TERMINALS
31	PANEL 7 - TERMINALS
32	PANEL 8 & 9 - TERMINALS
33	ADDITIONAL CUSTOMER CONSOLE

DEVICE LOCATION KEY

MD or OC	- OPERATOR'S CONSOLE	P1	- PILER 1
DWN	- DOWNENDER	P1F	- PILER 1 FRONT PB STATION
CC	- COIL CAR	P1B	- PILER 1 BACK PB STATION
CCTRACK	- COIL CAR TRACK	P2	- PILER 2
R	- REEL	P2F	- PILER 2 FRONT PB STATION
ET	- EDGE TRIMMER	P2B	- PILER 2 BACK PB STATION
PR	- PINCH ROLL STAND	P3	- PILER 3
PRA	- PINCH ROLL AUX. PB STATION	P3F	- PILER 3 FRONT PB STATION
ST	- STRAIGHTENER	P3B	- PILER 3 BACK PB STATION
INSP	- INSPECTION TERMINAL BOX	P4	- PILER 4
SH	- SHEAR	P4F	- PILER 4 FRONT PB STATION
CAT	- SHEAR CATENARY	P4B	- PILER 4 BACK PB STATION
		P4INSP	- PILER 4 INSPECTION PB STATION

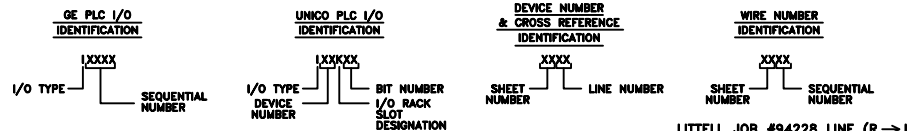
WIRE COLOR CODE:

BLK - 250VAC OR ABOVE
RED - 110/220VAC - HOT
WHT - 110/220VAC - COM (GROUNDED)
Y - AUXILIARY POWER
BL - +24VDC
BL/WHT - 24VDC COM (GROUNDED)
GRN/Y - GROUND

SWITCH SETTINGS



SHADED (DARKENED) AREA DENOTES DEPRESSED POSITION OF THE SWITCH



LINE SPECIFICATIONS

<u>LINE SPEED ENTRY SECTION</u>	MAX. RUNNING	600.0	FPW
		183.0	MPW
	MIN. RUNNING	75.0	FPW
		22.9	MPW
	JOG	30.0	FPW
		9.1	MPW
<u>SHEAR</u>	MAX. RUNNING (AT 30.85" FEED LENGTH)	225.0	SPW
	MIN. RUNNING	50.0	SPW
	SINGLE STROKE	50.0	SPW
	INCH	50.0	SPW
<u>MAIN CONVEYOR</u>	MAXIMUM	1050.0	FPW
		320.0	MPW
	MINIMUM	550.0	FPW
		167.0	MPW
<u>PILE CONVEYOR</u>	MAXIMUM	360.0	FPW
		110.0	MPW
	MINIMUM	160.0	FPW
		48.8	MPW
<u>FEED LENGTH</u>	MAXIMUM	46.0	IN
		1158.0	MM
	MINIMUM	18.0	IN
		457.0	MM
<u>FEED WIDTH</u>	MAXIMUM	40.0	IN
		1016.0	MM
	MINIMUM	24.0	IN
		610.0	MM
<u>COIL DATA</u>	O.D. MAXIMUM	72.0	IN
		1828.8	MM
	I.D.	16.5	IN
		419	MM
	WEIGHT	30,000.0	LBS
		13,600.0	KGS

OPTION	YES	NO	NOTES
POWER SUPPLY - THREE PHASE			380VAC, 50HZ
FLOOR PLAN			D
DOWNENDER		NO	
STORAGE RACK	YES		
STANDARD REEL		NO	
TRAVERSING REEL	YES		
AUXILIARY PINCH ROLL STAND		NO	
EDGE TRIMMER BASE	YES		
CENTER GUIDE DETECTOR	YES		
SCRAP CHOPPER	YES		
STRAIGHTENER ROLLS INSPECTION HATCH		NO	
THICKNESS GAUGE	YES		RADIOMETRIC
THICKNESS MARKER		NO	
CHART RECORDER		NO	
PINHOLE DETECTOR	YES		SEMS
PINHOLE MARKER		NO	
QUICK DIE CHANGE		NO	
STACKER GUIDES AUTO POSITIONING	YES		
MOTORIZED GAUGE TABLE	YES		
REMOTE DIAGNOSTIC SOFTWARE PACKAGE	YES		
DATA REPORTING PACKAGE		NO	

Form containing project information, including company name (LITTELL INTERNATIONAL INC.), project name (SIDERAR PLANTA SIDEROCROM), and various identification numbers (D-308691, 5615-23-0028).

DATE	SYM	WAS	BY
		D-305501	

BR-4E

STACKER

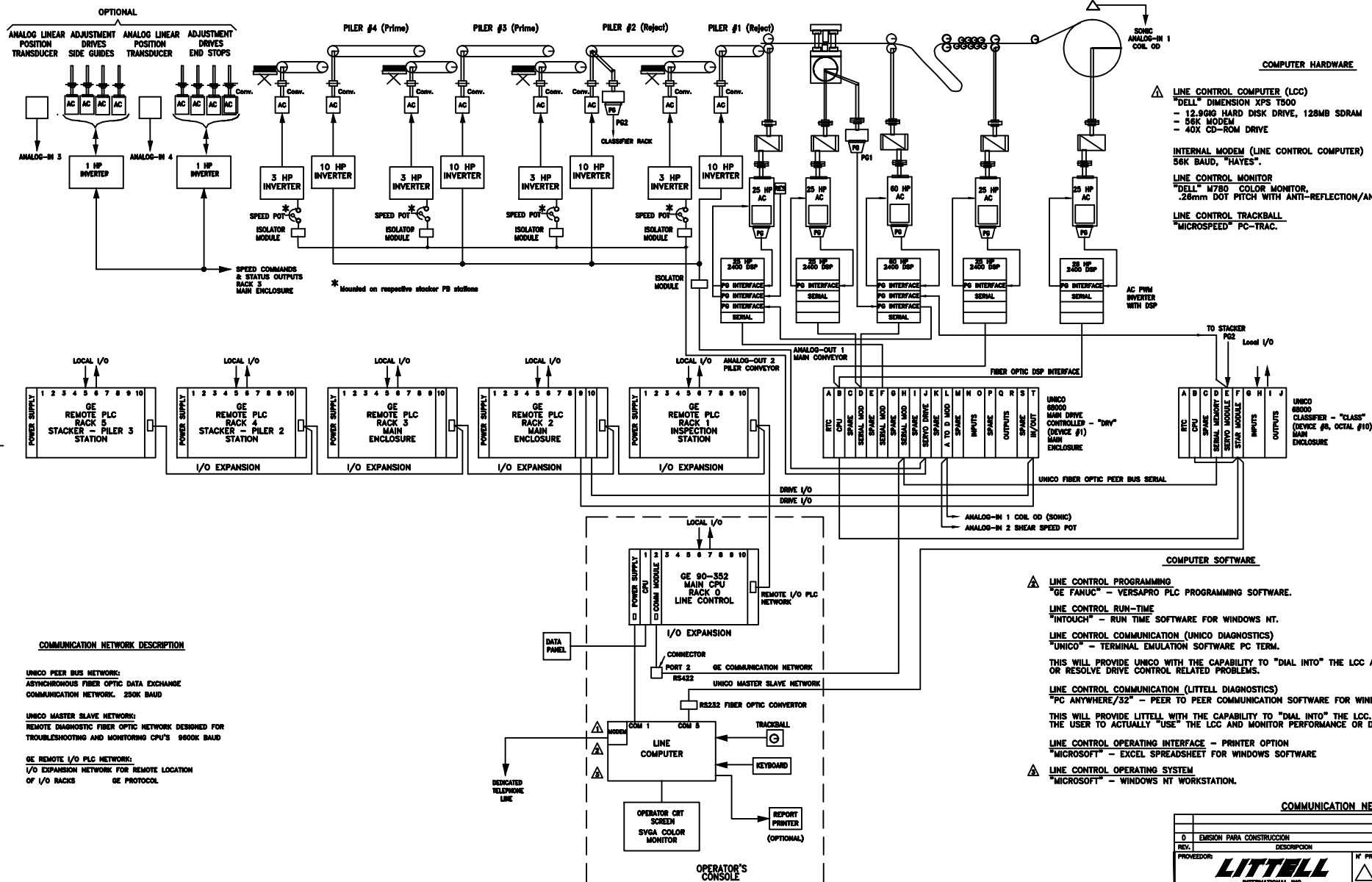
TAKEAWAY

SHEAR

FEED ROLL

STRAIGHTENER

PAYOFF



COMMUNICATION NETWORK DESCRIPTION

UNICO PEER BUS NETWORK:
ASYNCHRONOUS FIBER OPTIC DATA EXCHANGE
COMMUNICATION NETWORK. 250K BAUD

UNICO MASTER SLAVE NETWORK:
REMOTE DIAGNOSTIC FIBER OPTIC NETWORK DESIGNED FOR
TROUBLESHOOTING AND MONITORING CPU'S 9000K BAUD

GE REMOTE I/O PLC NETWORK:
I/O EXPANSION NETWORK FOR REMOTE LOCATION
OF I/O RACKS GE PROTOCOL

LINE CONTROL PROGRAMMING
"GE FANUC" - VERSAPRO PLC PROGRAMMING SOFTWARE.

LINE CONTROL RUN-TIME
"INTOUCH" - RUN TIME SOFTWARE FOR WINDOWS NT.

LINE CONTROL COMMUNICATION (UNICO DIAGNOSTICS)
"UNICO" - TERMINAL EMULATION SOFTWARE PC TERM.

THIS WILL PROVIDE UNICO WITH THE CAPABILITY TO "DIAL INTO" THE LCC AND DIAGNOSE
OR RESOLVE DRIVE CONTROL RELATED PROBLEMS.

LINE CONTROL COMMUNICATION (LITTELL DIAGNOSTICS)
"PC ANYWHERE/32" - PEER TO PEER COMMUNICATION SOFTWARE FOR WINDOWS 95/NT.

THIS WILL PROVIDE LITTELL WITH THE CAPABILITY TO "DIAL INTO" THE LCC. THIS SOFTWARE ALLOWS
THE USER TO ACTUALLY "USE" THE LCC AND MONITOR PERFORMANCE OR DIAGNOSE SYSTEM PROBLEMS.

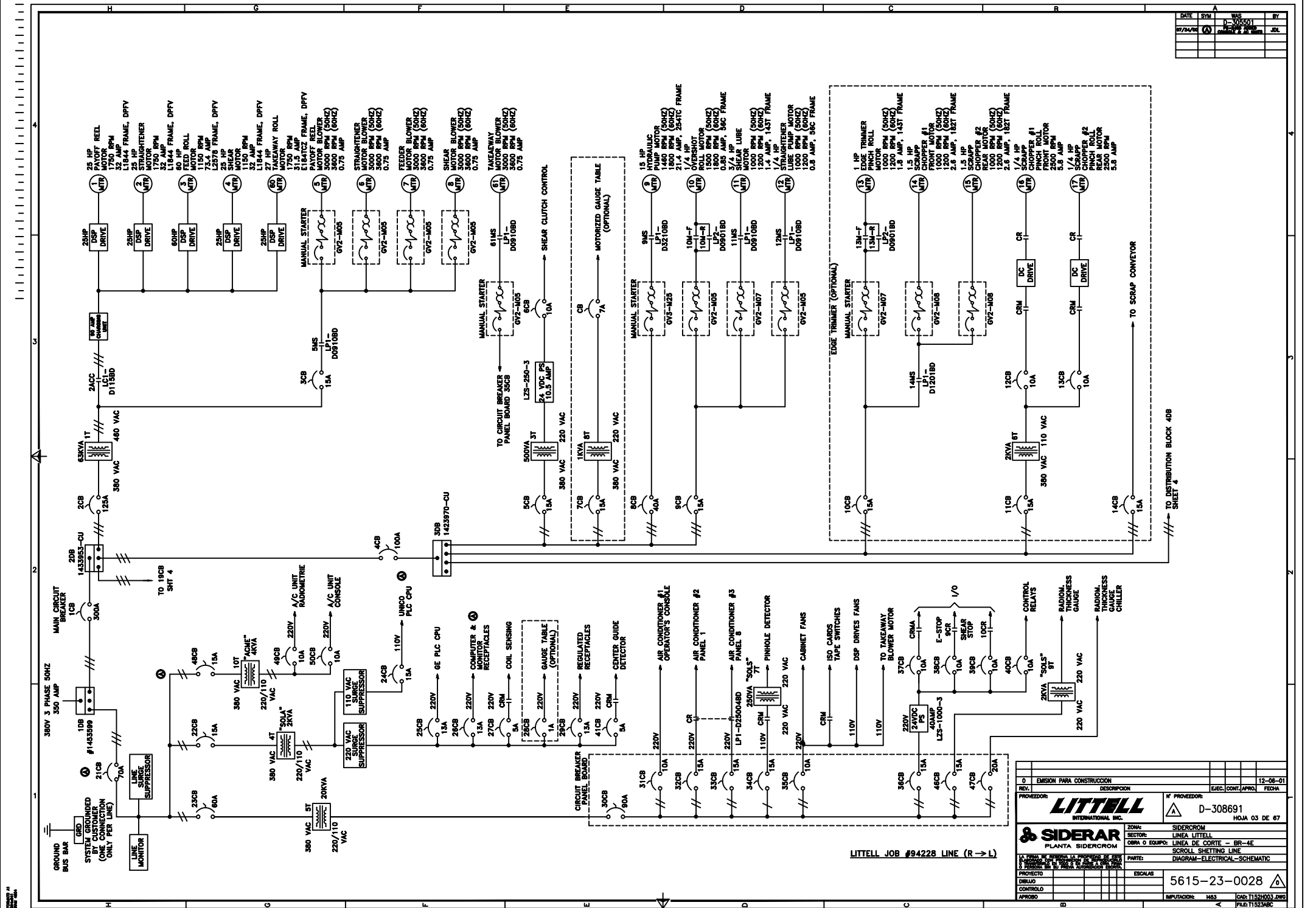
LINE CONTROL OPERATING INTERFACE - PRINTER OPTION
"MICROSOFT" - EXCEL SPREADSHEET FOR WINDOWS SOFTWARE

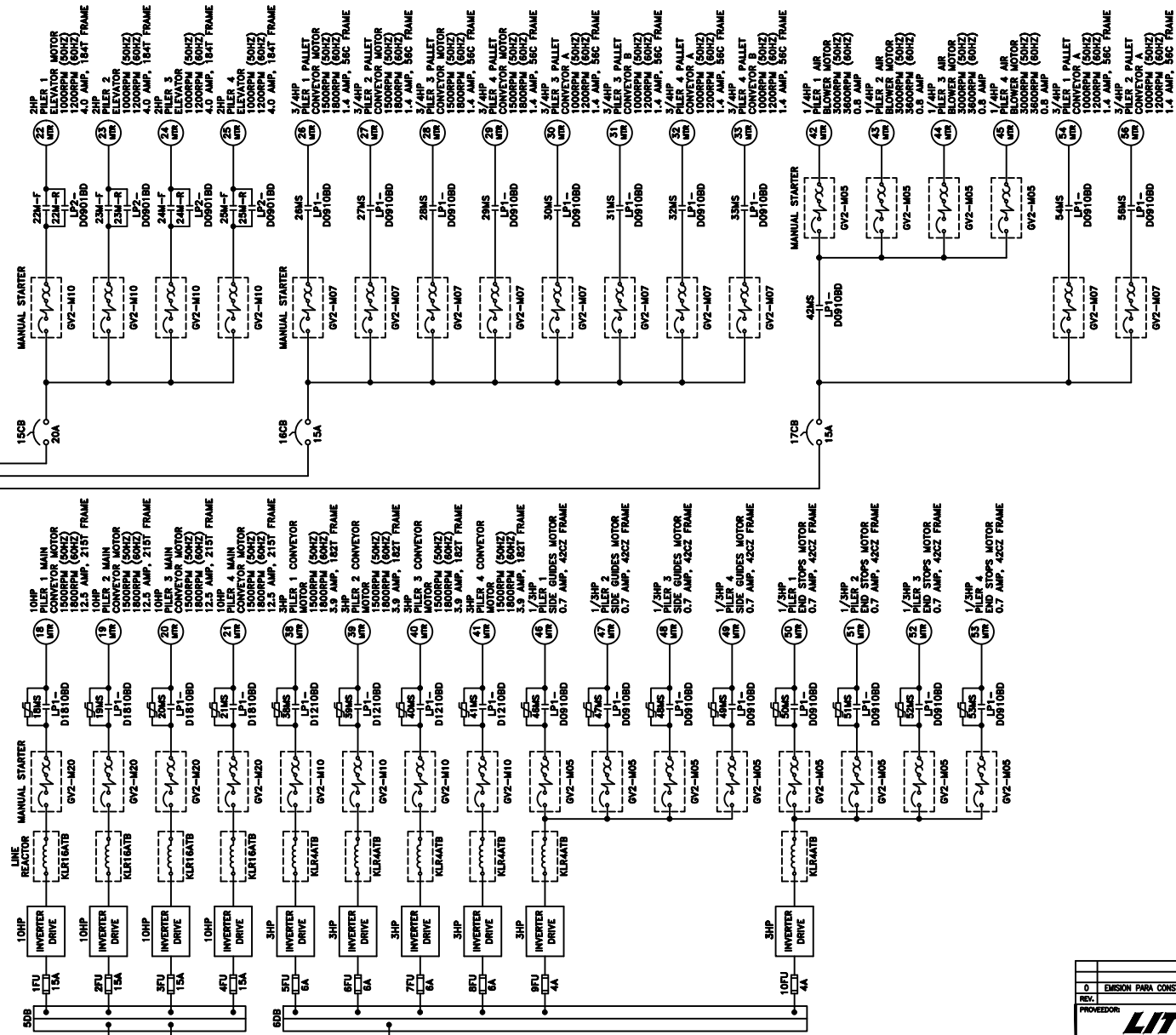
LINE CONTROL OPERATING SYSTEM
"MICROSOFT" - WINDOWS NT WORKSTATION.

COMMUNICATION NETWORK DIAGRAM

REV.	DESCRIPTION	ELEC. CONT. APPRO.	FECHA
0	EMISION PARA CONSTRUCCION		12-06-01
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVIDER:	D-308691 HOJA 02 DE 67
ZONA:	SIDERAR	SECTOR:	LINEA DE CORTE
OBRA O EQUIPO:	PLANTA SIDERAR	PARTE:	SCROLL SHEETING LINE
PROYECTO:		ESCALAS:	DIAGRAM-ELECTRICAL-SCHEMATIC
DISEÑO:			5615-23-0028
CONTROL:			
APROBADO:			

LITTELL JOB #94228 LINE (R → L)



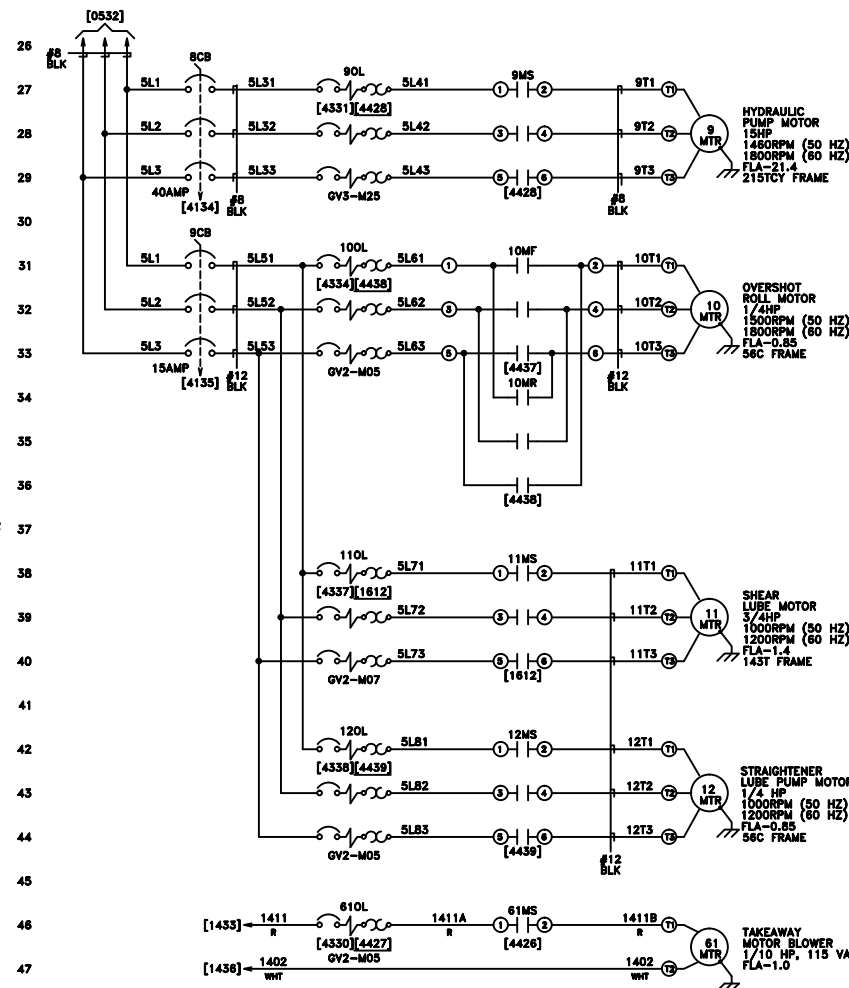
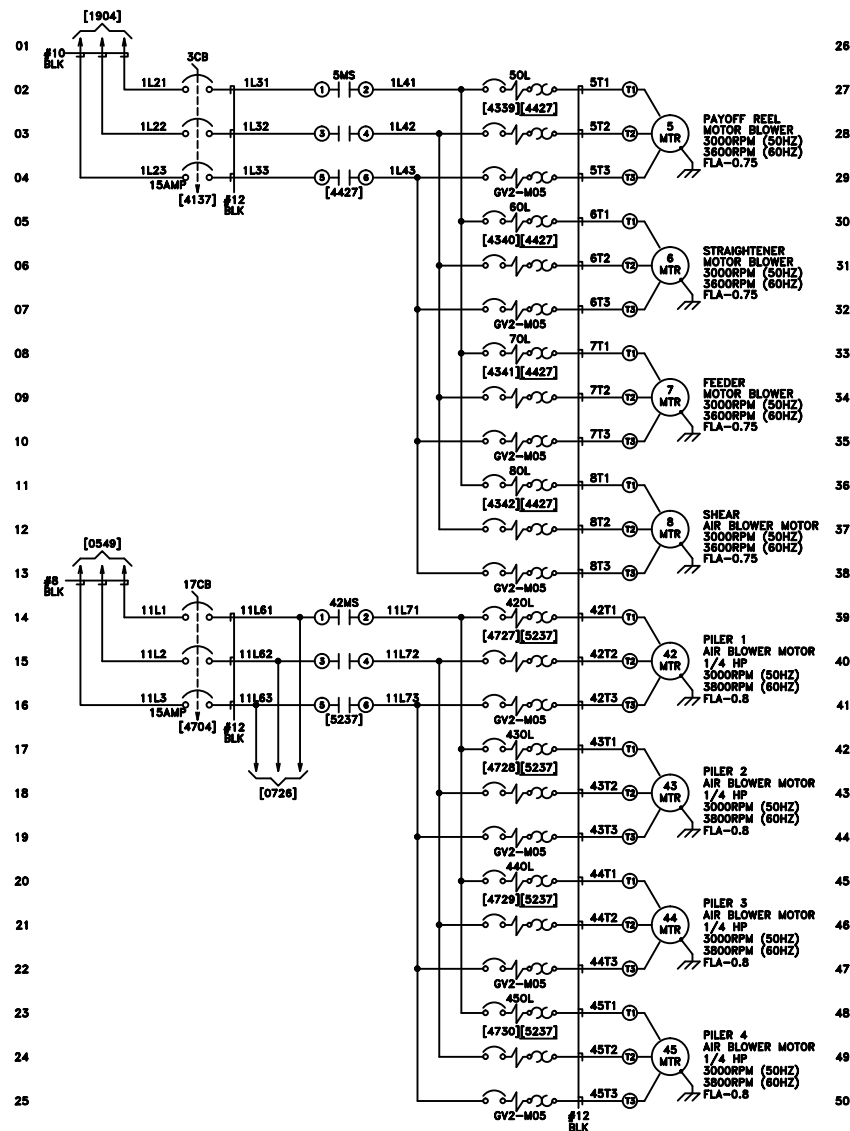


DATE	SYM	WAS	BY
		D-305501	

LITTELL JOB #94228 LINE (R → L)

O EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	N° PROVEEDOR:	D-308691
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE - BR-4E	
OBRA O EQUIPO:		SCROLL SHEETING LINE	
PARTES:		DIAGRAM - ELECTRICAL - SCHEMATIC	
PROYECTO	ESCALAS	5615-23-0028	
DISEÑO			
CONTROL			
APROBADO			
IMPULSION: HAS		CARG: T152H004.DWG	
		FILE: T1523ABC	

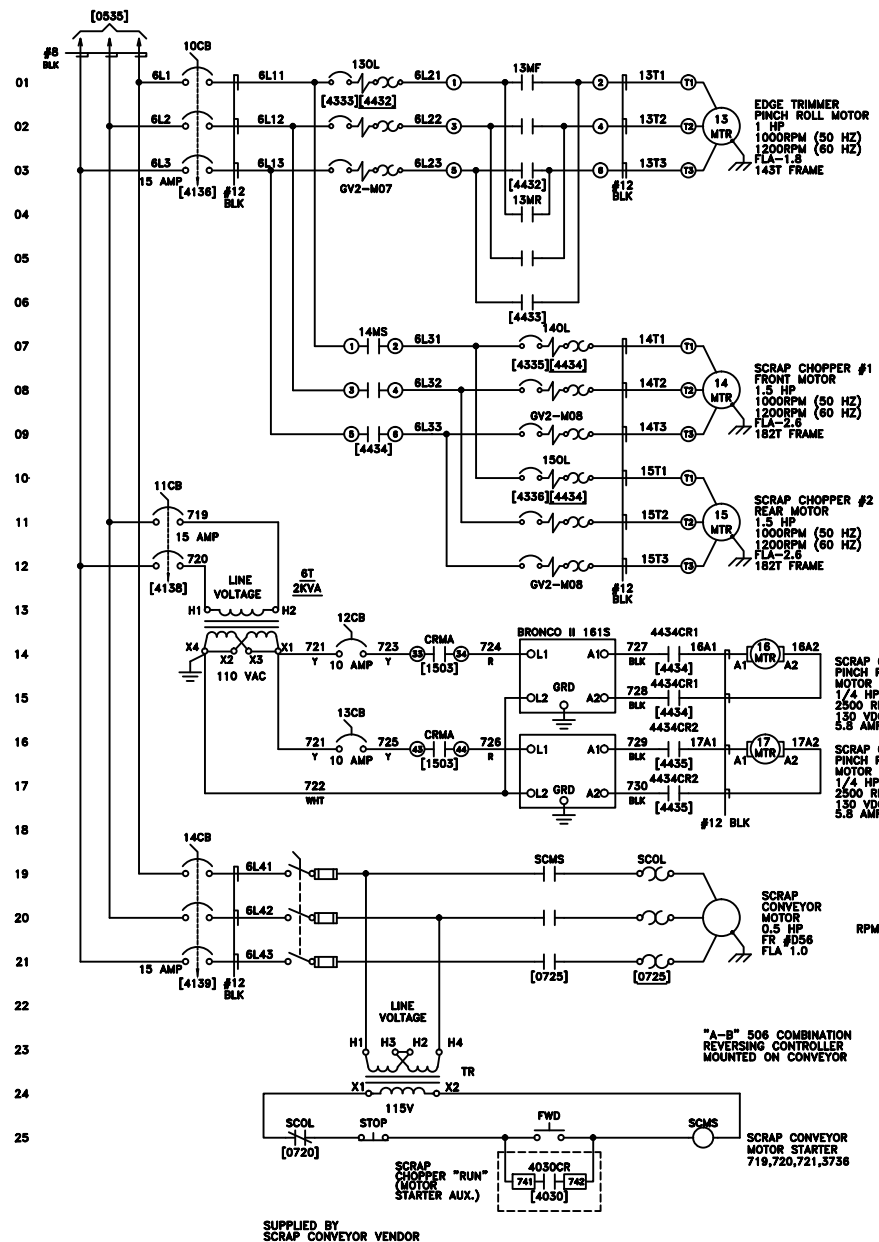
DATE	SYM	WAS	BY
		D-305501	



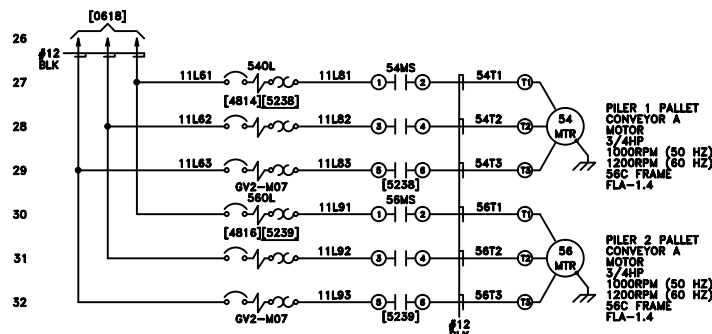
LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROCROM		HORA 06 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPULSION: HAS	CAD: T1520008.DWG	
CONTROL:	FILE: T152348C		

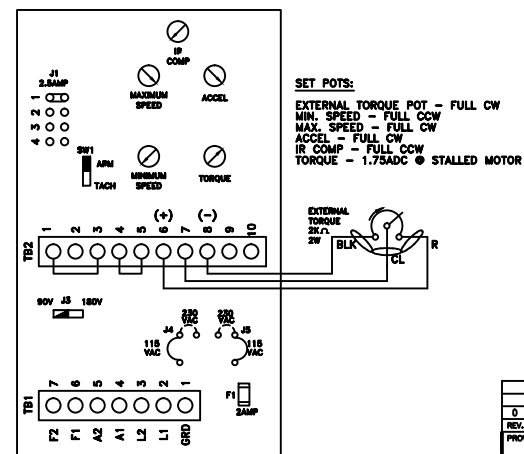
DATE	SYM	WAS	BY
		D-305501	



SUPPLIED BY
SCRAP CONVEYOR VENDOR



DETAILS OF "SECO-WARNER ELECTRIC" DC DRIVE "BRONCO II 1615"



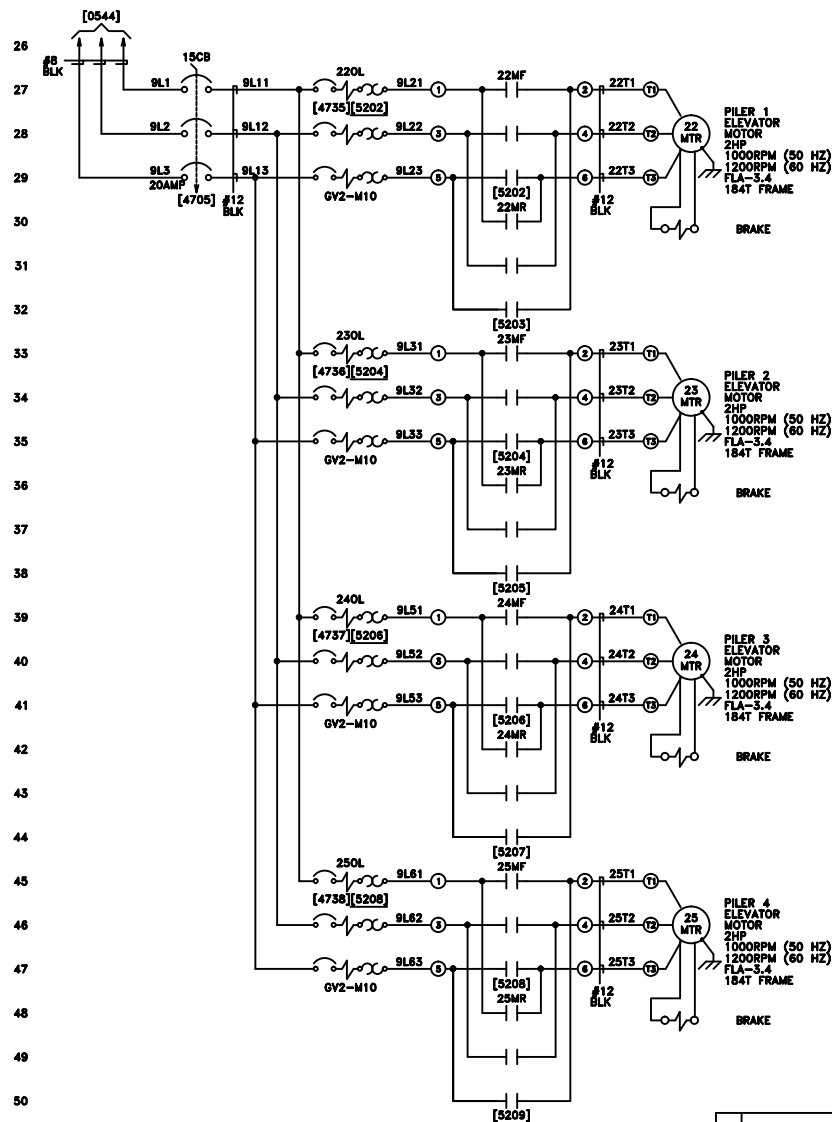
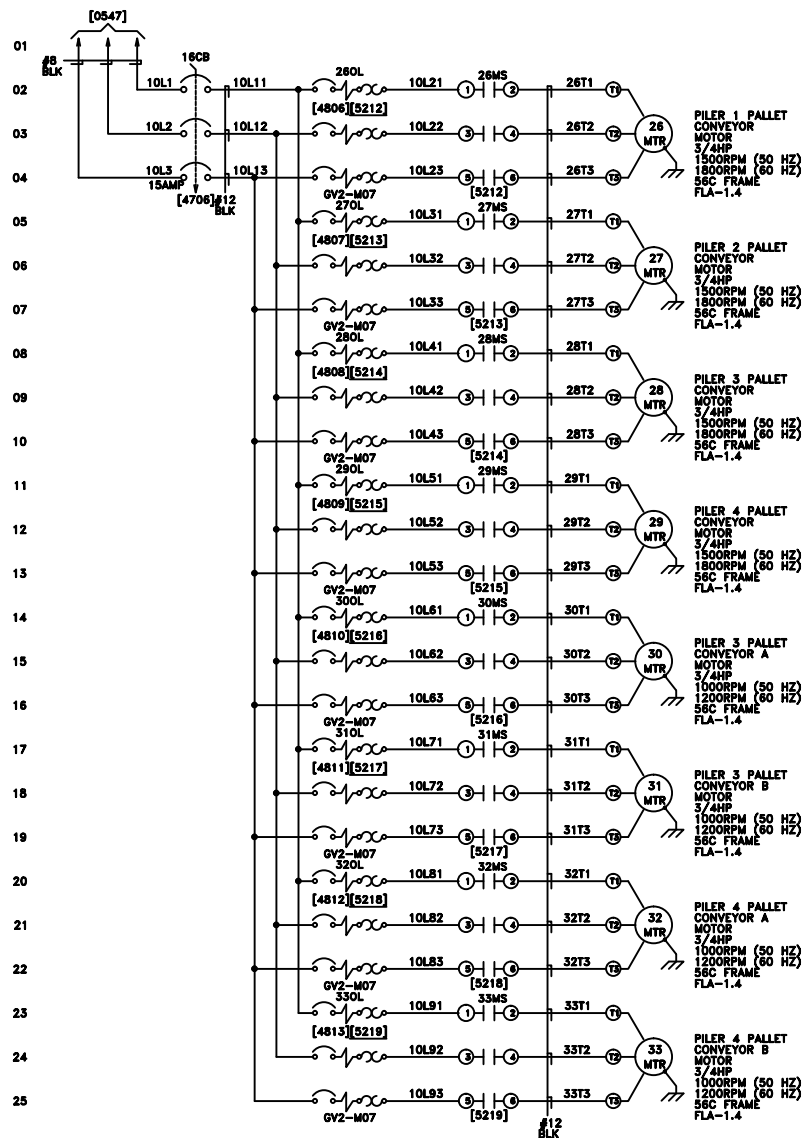
LITTELL P.N. 275397

LITTELL JOB #94228 LINE (R → L)

EDGE TRIMMER
AND SCRAP CONVEYOR
(OPTIONAL)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	ELEC. CONT. APROV.	FECHA
PROVEEDOR:	SIDERCROM		HOJA 07 DE 67
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE - BR-4E	
OBRA O EQUIPO:		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:		5615-23-0028
DISEÑO:	IMPRESION:		HAS
CONTROL:	CADA:		1152407.DWG
APROB:	FECHA:		FILE 1152407

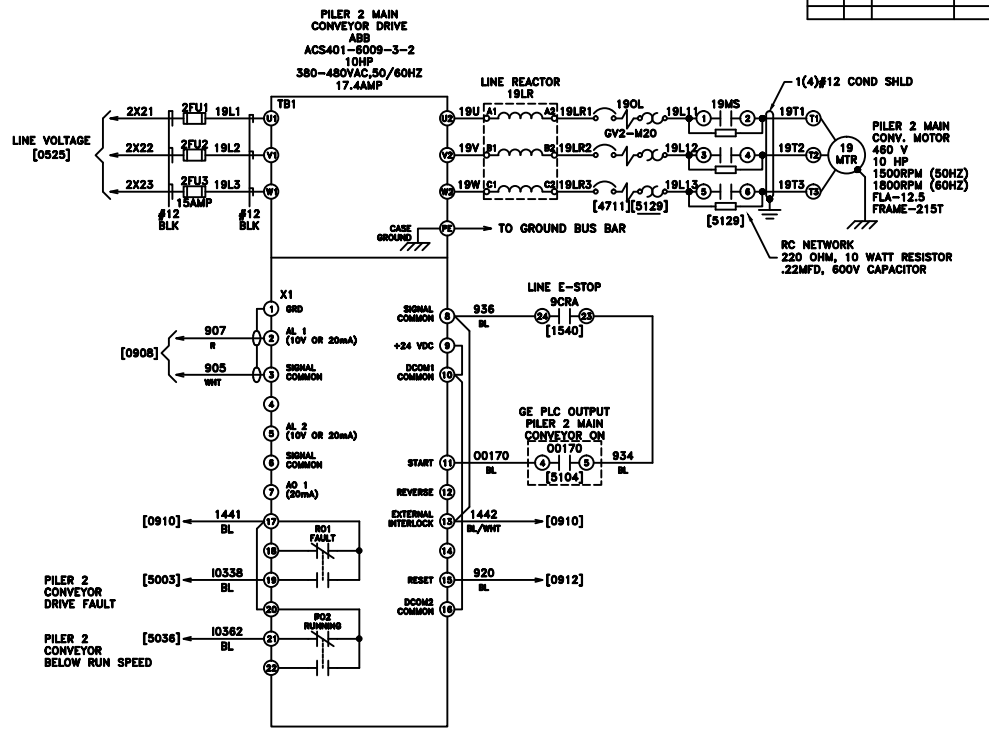
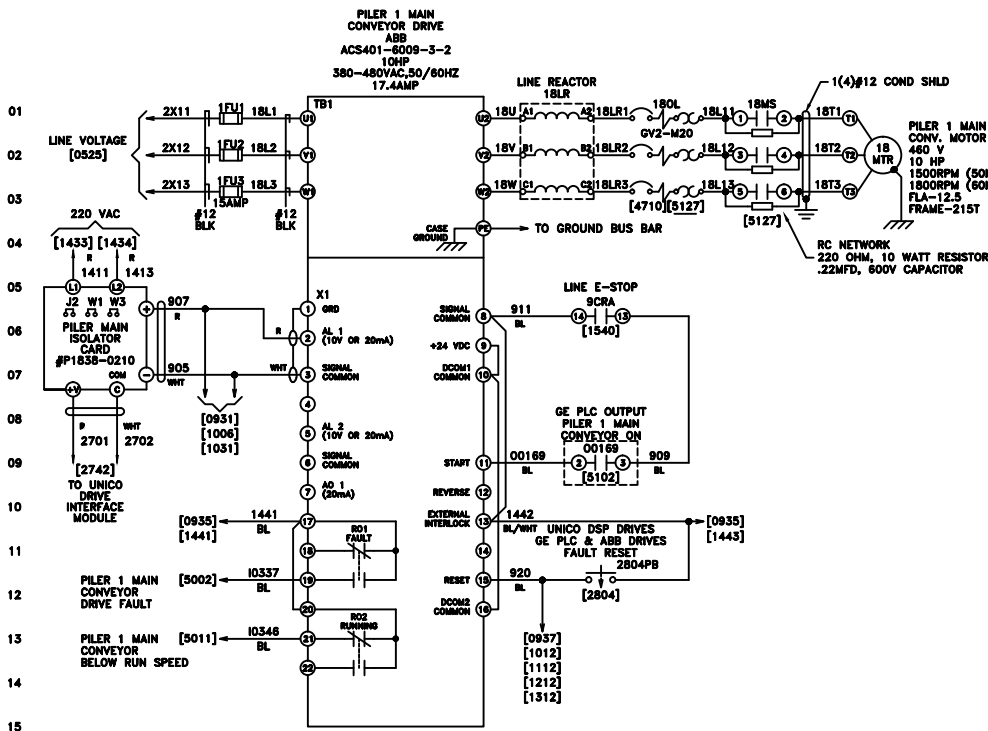
DATE	SYM	WAS	BY
		D-305501	



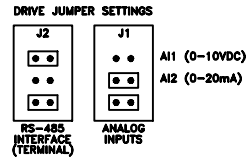
LITELL JOB #94228 LINE (R → L)

0 EMISION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	ELEC. CONT. APROV.	FECHA
PROVEEDOR:	LITELL INTERNATIONAL INC.		IF PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 08 DE 07	
SECTOR: LINEA LITELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PROYECTO: DIAGRAM-ELECTRICAL-SCHEMATIC			
DIBUJO	ESCALAS	5615-23-0028	
CONTROL	IMPULSION: HAS	CAD: T1524008.DWG	
APROB		FILE: T152348C	

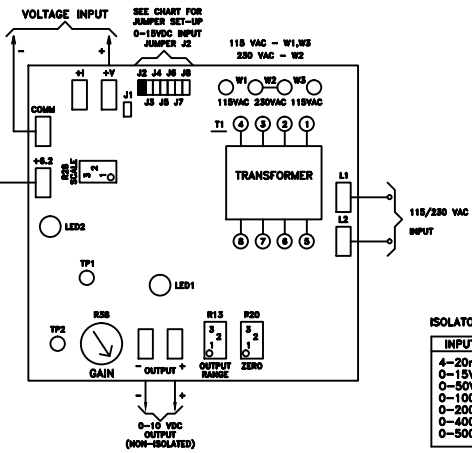
DATE	SYM	WAS	BY
		D-305501	



- NOTES:**
1. CONTACTS SHOWN IN UNPOWERED STATE
 2. JUMPERS ARE SET TO FACTORY CONFIGURATION UNLESS OTHERWISE SPECIFIED.



"EXTRON" #P1838-0210
 ISOLATOR CARD
 0-10 VDC - 0-10 VDC
 P.N. 274496
 (DETAILS)



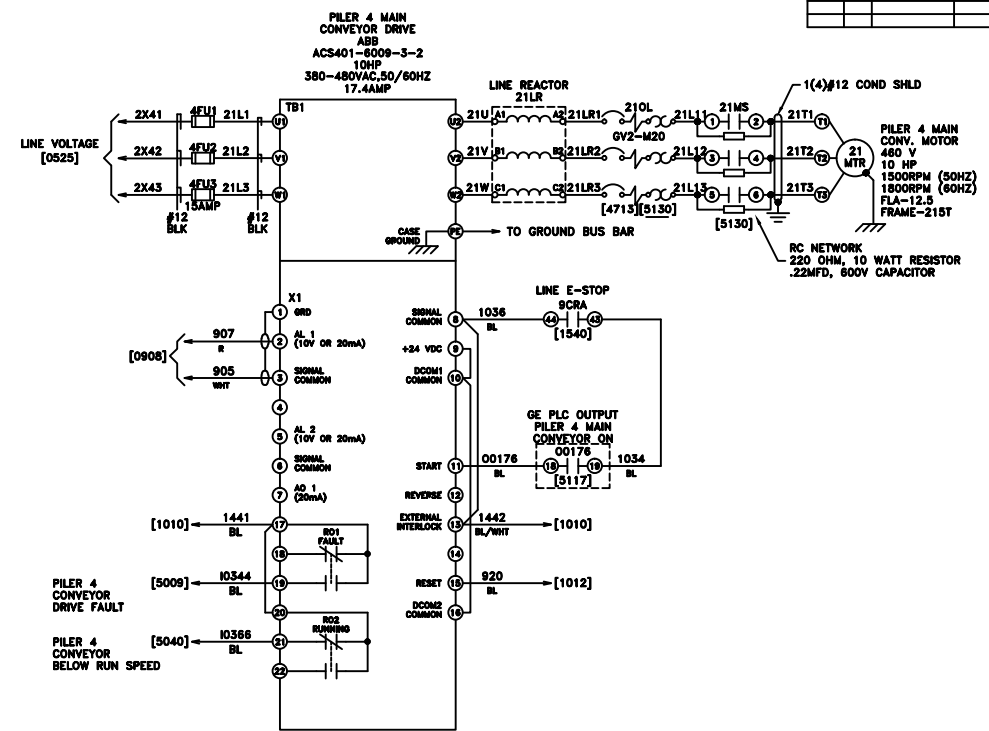
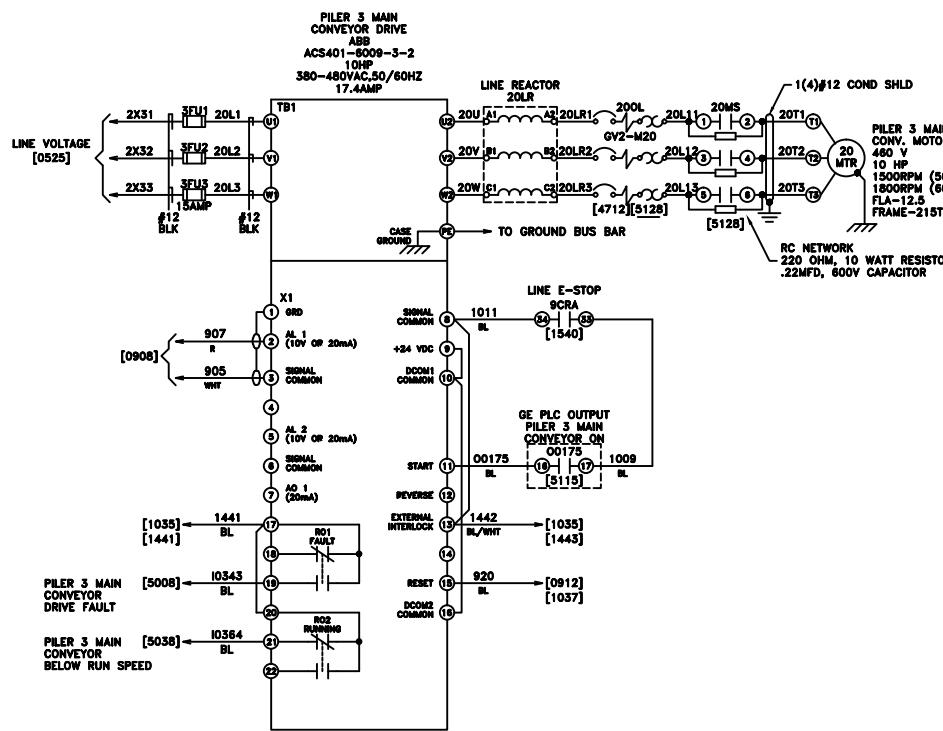
- 1) CHECK THAT APPROPRIATE JUMPERS ARE INSTALLED FOR LINE INPUT POWER (W1, W2, W3) FOR 115 VAC INSTALL JUMPERS W1 AND W3. FOR 230 VAC INSTALL JUMPER W2.
- 2) SET APPROPRIATE JUMPER (J1-J7) FOR INPUT VOLTAGE OR CURRENT. SET JUMPER J2 FOR 0-15 VDC VOLTAGE INPUT.
- 3) WITH INPUT SET AT MINIMUM LEVEL SET ZERO (R20) TO ZERO VOLTS AT OUTPUT TERMINAL.
- 4) WITH INPUT SET TO MAXIMUM LEVEL, ADJUST SCALING (R28) TO 4 VDC AT TP1 AND COMM.
- 5) ADJUST GAIN (R38) TO OBTAIN 4 VDC AT TP2 AND - OUTPUT TERMINAL.
- 6) ADJUST OUTPUT RANGE (R13) TO DESIRED MAXIMUM 10 VDC LEVEL AT OUTPUT TERMINAL.
- 7) RECHECK STEP 3.

PILER 1 & 2 MAIN CONVEYOR DRIVES

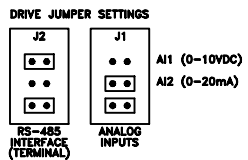
LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 09 DE 67	
SECTOR: LINEA DE CORTE		BR-4E	
OBRA O EQUIPO: LINEA DE CORTE		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAS	CAD: T1550009.DWG
CONTROL:	FILE:	T155348C	

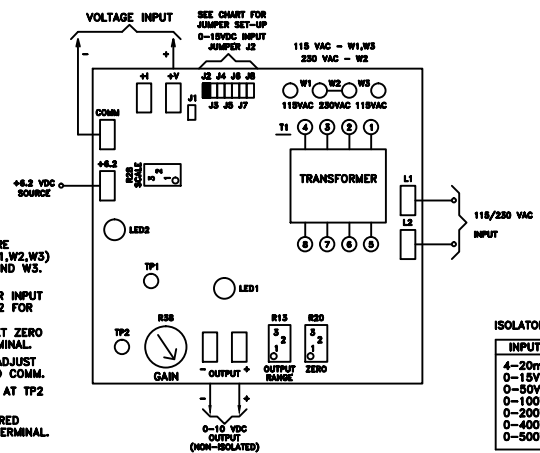
DATE	SYM	WAS	BY
		D-305501	



- NOTES:
1. CONTACTS SHOWN IN UNPOWERED STATE
 2. JUMPERS ARE SET TO FACTORY CONFIGURATION UNLESS OTHERWISE SPECIFIED.



"EXTRON" #P1838-0210
ISOLATOR CARD
0-10 VDC - 0-10 VDC
P.N. 274498
(DETAILS)



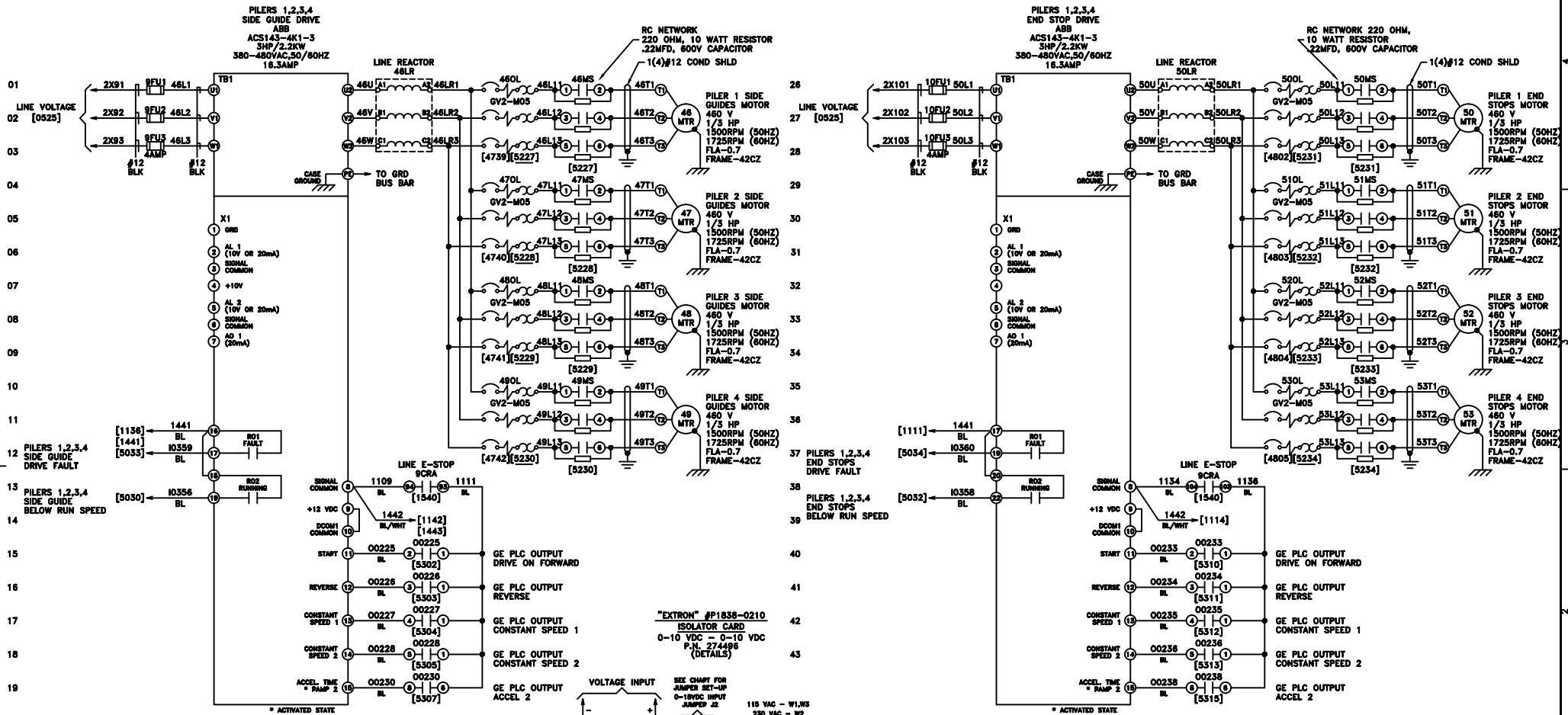
INPUT	JUMPER TO USE
4-20mA	J1
0-15VDC	J2
0-50VDC	J3
0-100VDC	J4
0-200VDC	J5
0-400VDC	J6
0-500VDC	J7

PILER 3 & 4 MAIN CONVEYOR DRIVES

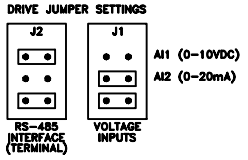
0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691 HOJA 10 DE 87
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE - BR-4E	
OBRA O EQUIPO:		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	1155010.DWG	
APROBADO:	FECHA:	1155348C	

LITTELL JOB #94228 LINE (R -> L)

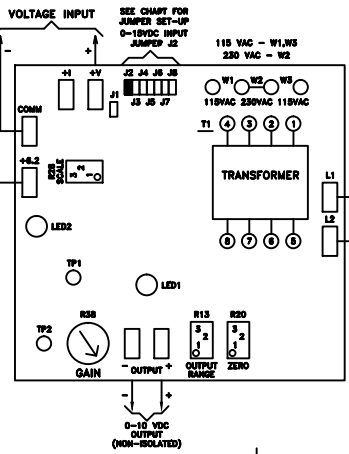
DATE	SYM	WAS	BY
		D-305501	



NOTES:
 1. CONTACTS SHOWN IN UNPOWERED STATE
 2. JUMPERS ARE SET TO FACTORY CONFIGURATION UNLESS OTHERWISE SPECIFIED.



- 1) CHECK THAT APPROPRIATE JUMPERS ARE INSTALLED FOR LINE INPUT POWER (W1, W2, W3) FOR 115 VAC. INSTALL JUMPERS W1 AND W3. FOR 220 VAC INSTALL JUMPER W2.
- 2) SET APPROPRIATE JUMPER (J1-J7) FOR INPUT VOLTAGE OR CURRENT. SET JUMPER J2 FOR 0-15 VDC VOLTAGE INPUT.
- 3) WITH INPUT SET AT MINIMUM LEVEL SET ZERO (R20) TO ZERO VOLTS AT OUTPUT TERMINAL.
- 4) WITH INPUT SET TO MAXIMUM LEVEL, ADJUST SCALING (R28) TO 4 VDC AT TP1 AND COMM.
- 5) ADJUST GAIN (R38) TO OBTAIN 4 VDC AT TP2 AND - OUTPUT TERMINAL.
- 6) ADJUST OUTPUT RANGE (R13) TO DESIRED MAXIMUM 10 VDC LEVEL AT OUTPUT TERMINAL.
- 7) RECHECK STEP 3.



ISOLATOR CARD JUMPER SETUP

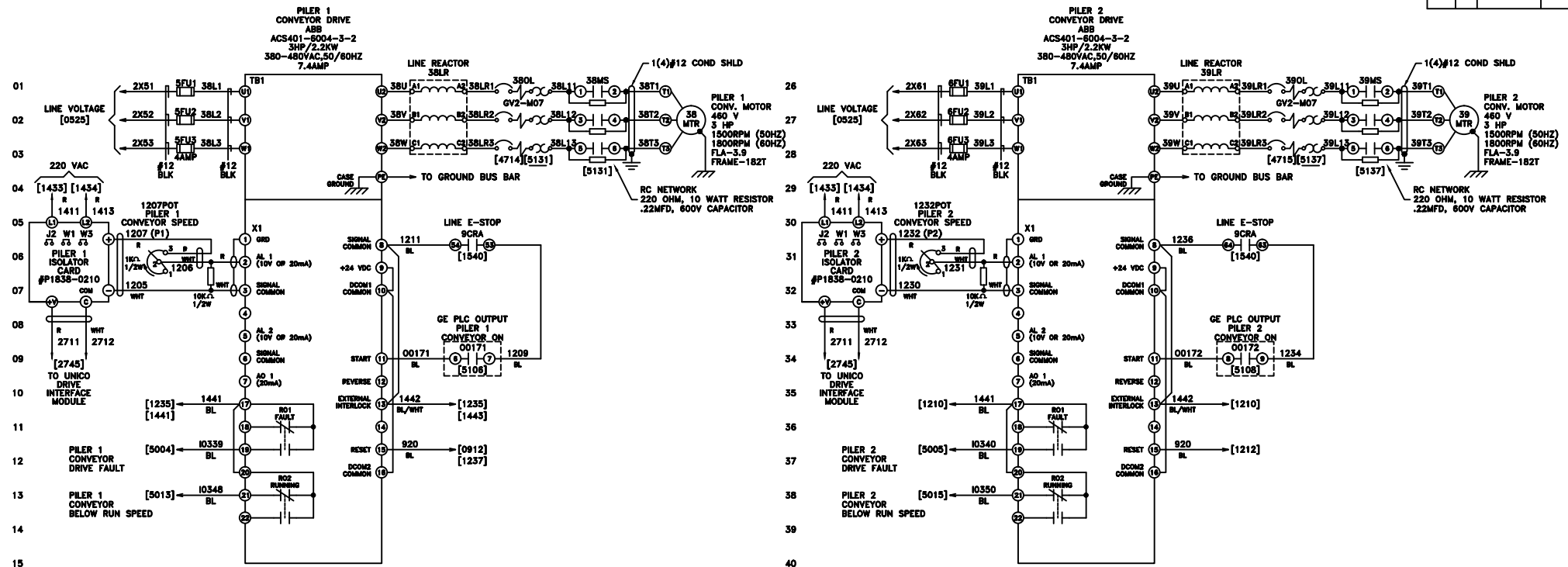
INPUT	JUMPER TO USE
4-20mA	J1
0-15VDC	J2
0-50VDC	J3
0-100VDC	J4
0-200VDC	J5
0-400VDC	J6
0-800VDC	J7

**PILER 1,2,3,4
SIDE GUIDE & END STOP DRIVES**

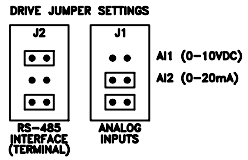
LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-01	
REV.	DESCRIPTION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		HOJA 11 DE 67
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE	
OBRA O EQUIPO:		BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		5615-23-0028	
DISEÑO:		IMPULSION: HAS	
CONTROL:		CARG: T155011.000	
APROBADO:		FILE: T155346C	

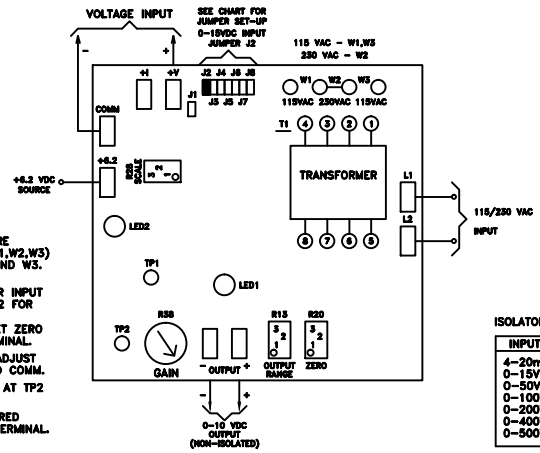
DATE	SYM	WAS	BY
		D-305501	



- NOTES:
- CONTACTS SHOWN IN UNPOWERED STATE
 - JUMPERS ARE SET TO FACTORY CONFIGURATION UNLESS OTHERWISE SPECIFIED.



"EXTRON" #P1858-0210
ISOLATOR CARD
0-10 VDC - 0-10 VDC
P.N. 274498
(DETAILS)



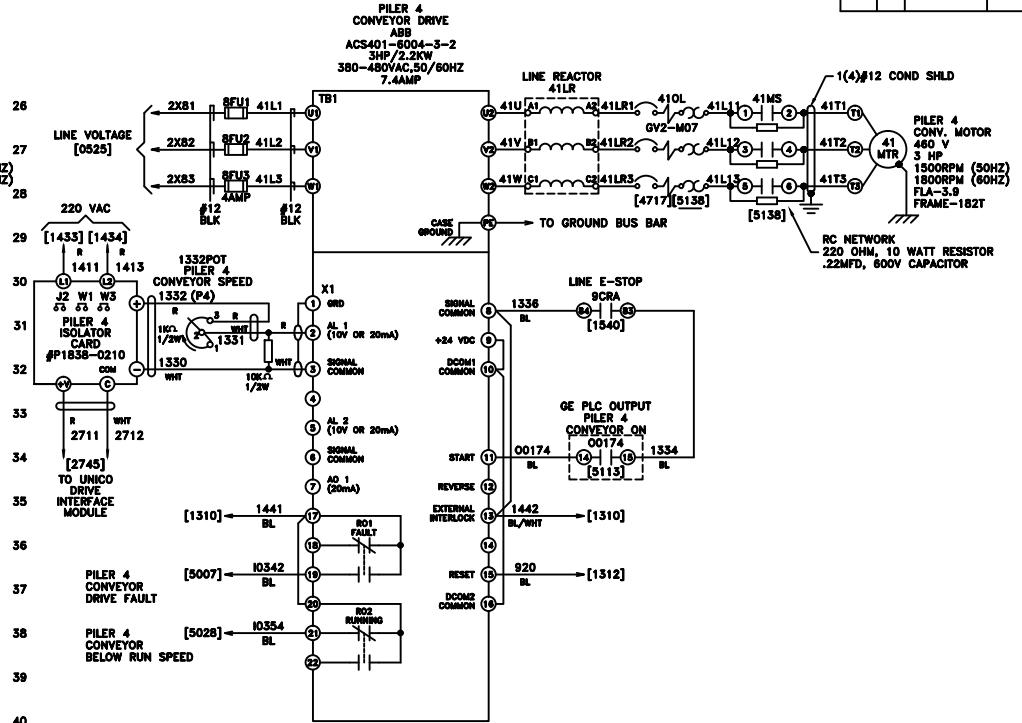
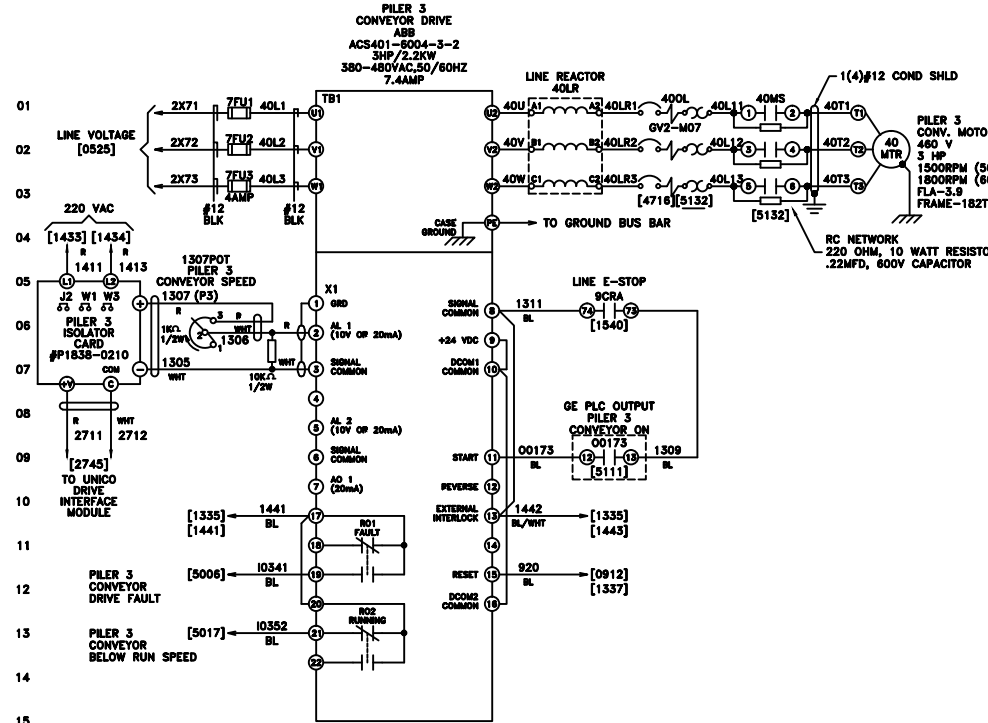
- CHECK THAT APPROPRIATE JUMPERS ARE INSTALLED FOR LINE INPUT POWER (W1,W2,W3) FOR 115 VAC INSTALL JUMPERS W1 AND W3. FOR 220 VAC INSTALL JUMPER W2.
- SET APPROPRIATE JUMPER (J1-J7) FOR INPUT VOLTAGE OR CURRENT. SET JUMPER J2 FOR 0-15 VDC VOLTAGE INPUT.
- WITH INPUT SET AT MINIMUM LEVEL SET ZERO (R20) TO ZERO VOLTS AT OUTPUT TERMINAL.
- WITH INPUT SET TO MAXIMUM LEVEL, ADJUST SCALING (R28) TO 4 VDC AT TP1 AND COMM.
- ADJUST GAIN (R38) TO OBTAIN 4 VDC AT TP2 AND - OUTPUT TERMINAL.
- ADJUST OUTPUT RANGE (R13) TO DESIRED MAXIMUM 10 VDC LEVEL AT OUTPUT TERMINAL.
- RECHECK STEP 5.

PILER 1 & 2
CONVEYOR DRIVES

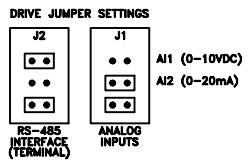
LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 12 DE 67	
SECTOR: LINEA DE CORTE		BR-4E	
OBRA O EQUIPO: LINEA DE CORTE		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

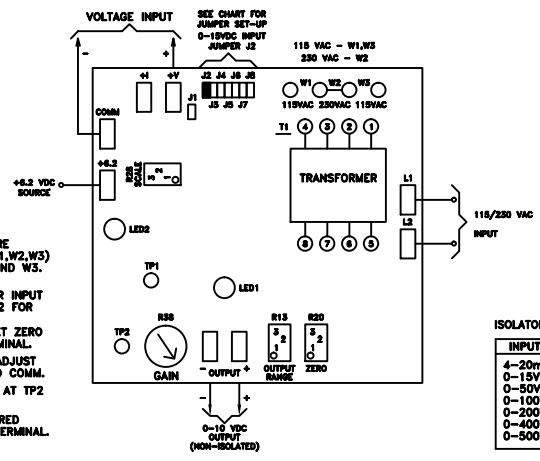
DATE	SYM	WAS	BY
		D-305501	



- NOTES:
1. CONTACTS SHOWN IN UNPOWERED STATE
 2. JUMPERS ARE SET TO FACTORY CONFIGURATION UNLESS OTHERWISE SPECIFIED.



"EXTROM" #P1838-0210
ISOLATOR CARD
0-10 VDC - 0-10 VDC
P.N. 274496
(DETAILS)



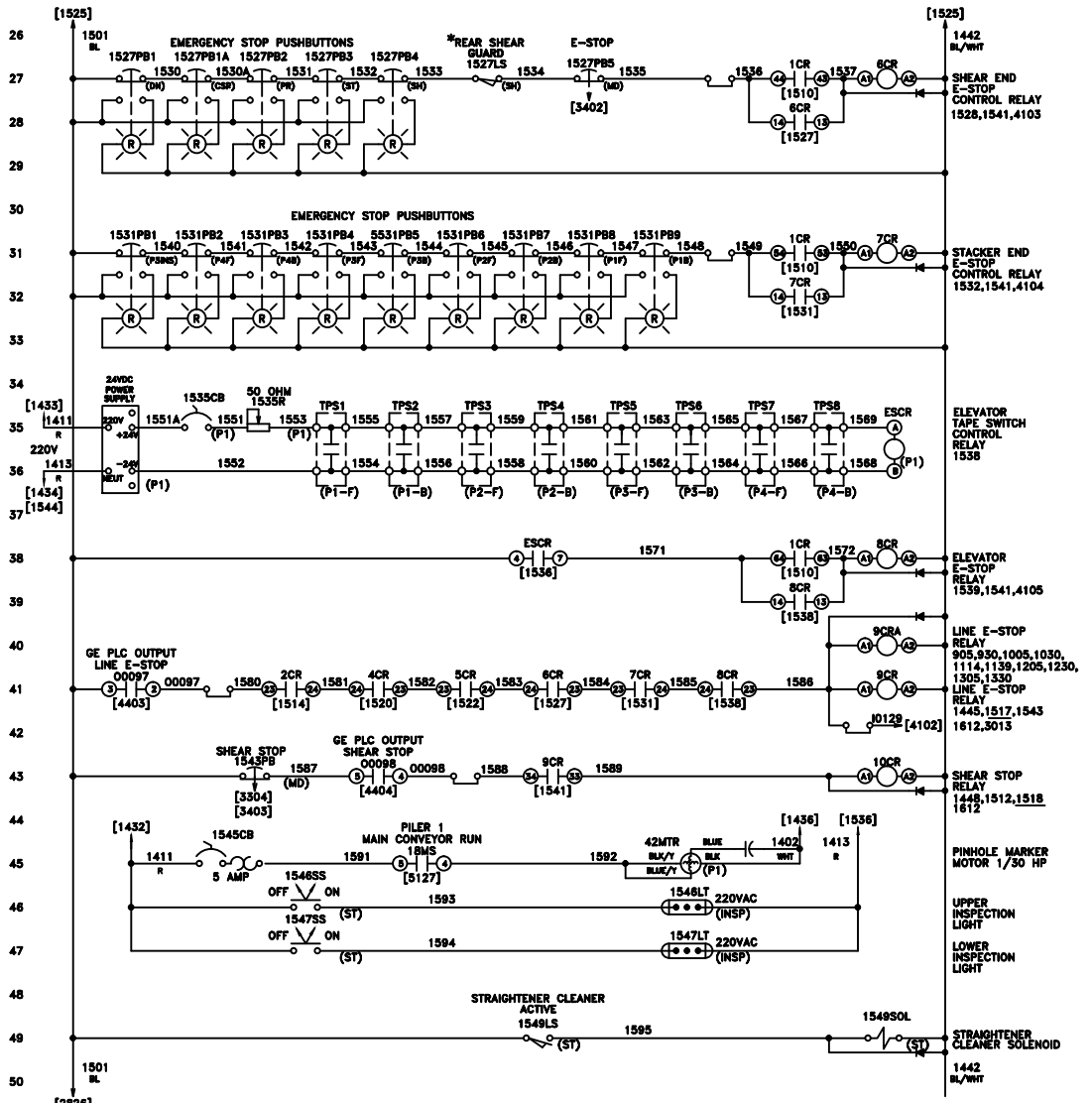
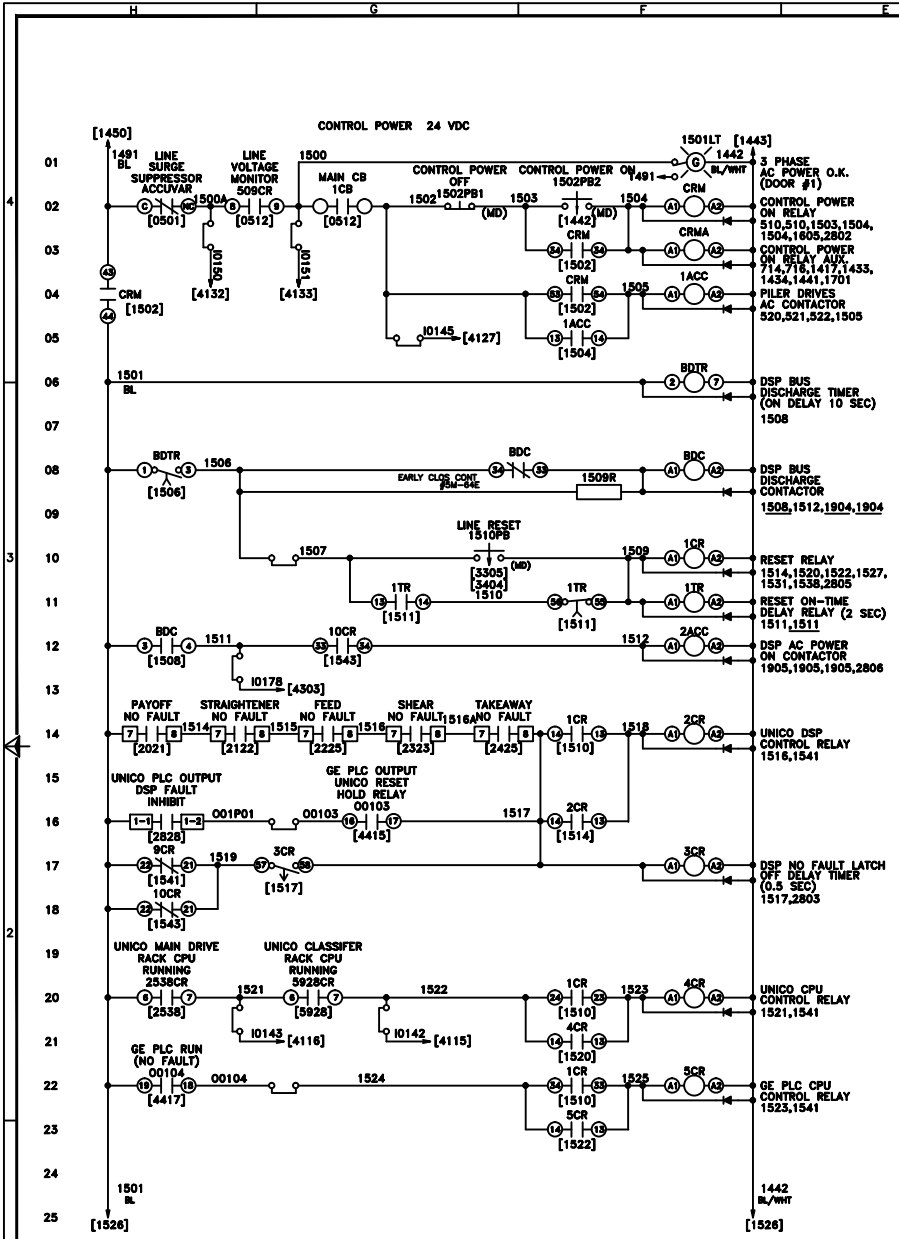
INPUT	JUMPER TO USE
4-20mA	J1
0-10VDC	J2
0-50VDC	J3
0-100VDC	J4
0-200VDC	J5
0-400VDC	J6
0-500VDC	J7

LITTELL JOB #94228 LINE (R → L)

PILER 3 & 4 CONVEYOR DRIVES

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691
ZONA:		HOJA 13 DE 67	
SECTOR:		SIDERCROM	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DISEÑO:		ESCALAS:	
CONTROLADO:		5615-23-0028	
APROBADO:		IMPRESION: HAS	
		CADA: T155013.DWG	
		FILE: T155348C	

DATE	SYM	WAS	BY
		D-305501	

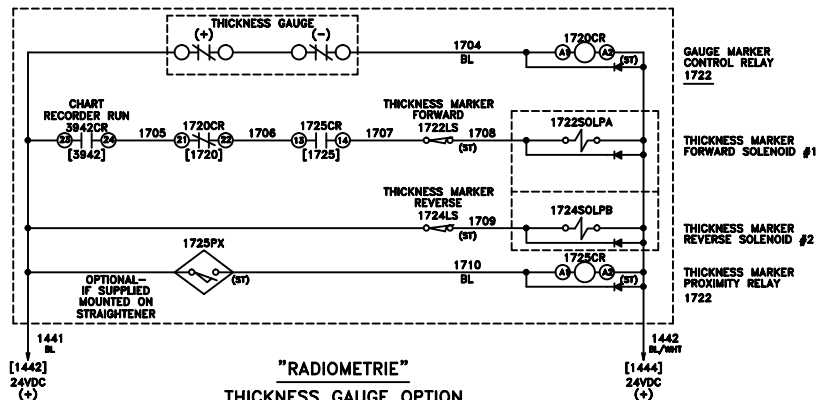
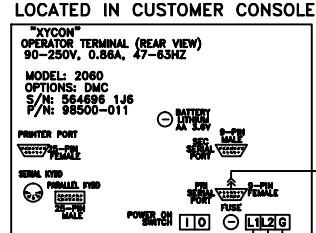
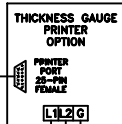
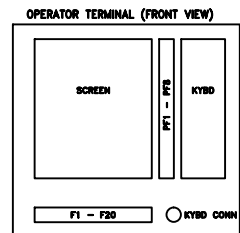
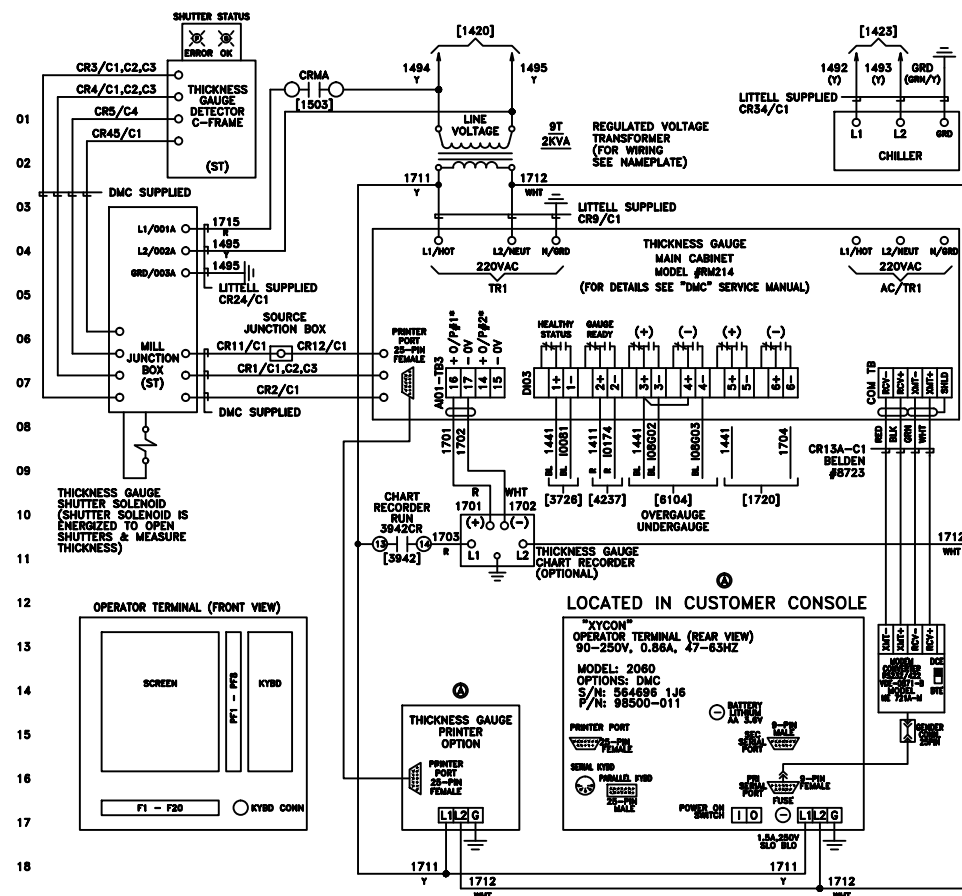


* NOTE: GUARD SWITCH(S) SHOWN WHEN GUARD(S) IS CLOSED.

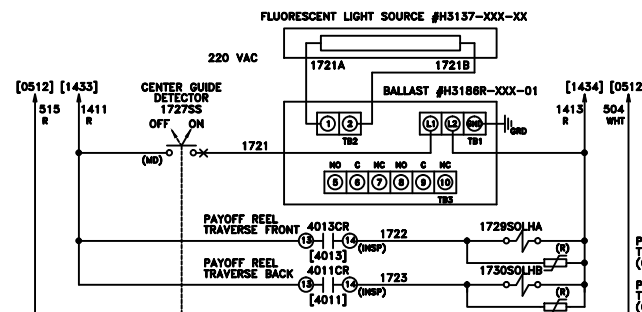
LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IF PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 15 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
SCROLL SHEETING LINE			
DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DIBUJO:			
CONTROL:			
APPROB:			

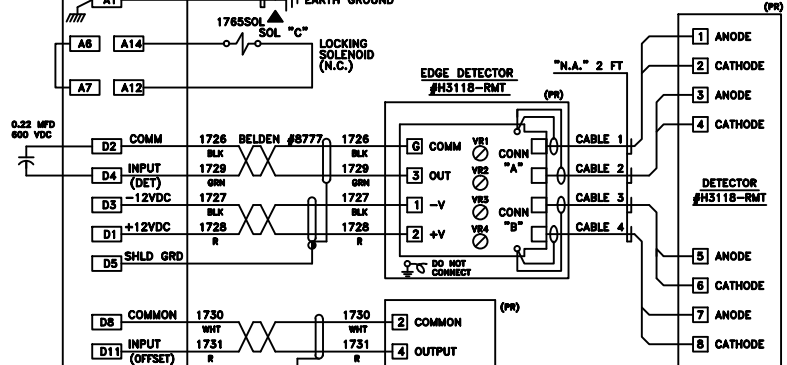
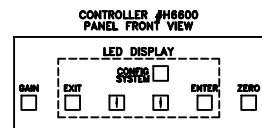
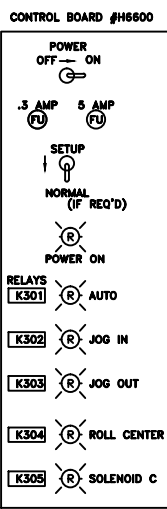
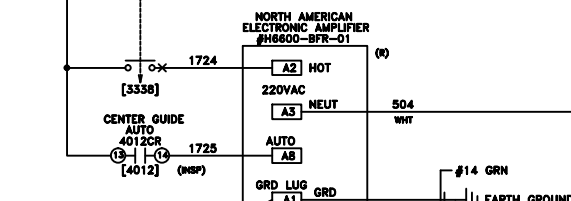
DATE	SYM	WAS	BY
07/24/91	(A)	D-308501	JDL



"RADIOMETRIC"
THICKNESS GAUGE OPTION
FOR DETAILS SEE SERVICE MANUALS



NORTH AMERICAN MFG. CO.
NOTES:
▲ INDICATES COMPONENTS MOUNTED ON HYDRAULIC UNIT & WIRED AT FACTORY
● RUN IN SEPARATE CONDUIT OR SHIELDED CABLE



NOTE: GROUND TO BUILDING GROUND BOUNDING SYSTEM

STRIP OFFSET CONTROL STATION #H6229

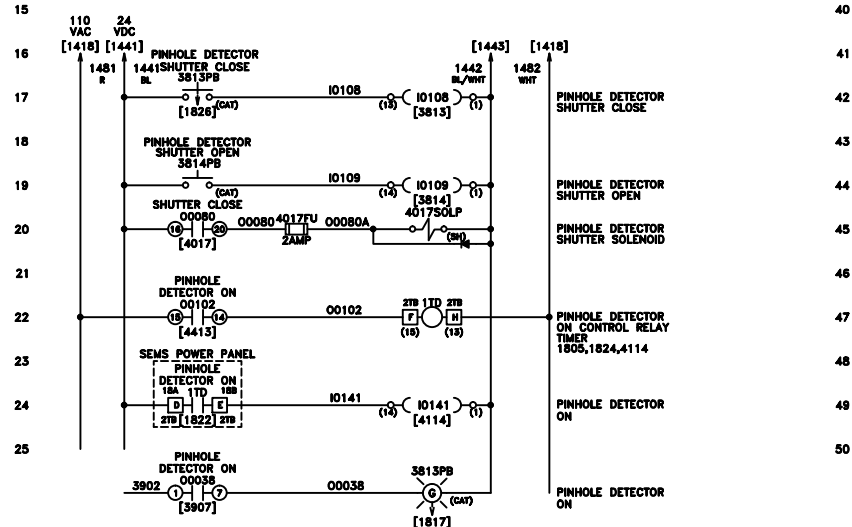
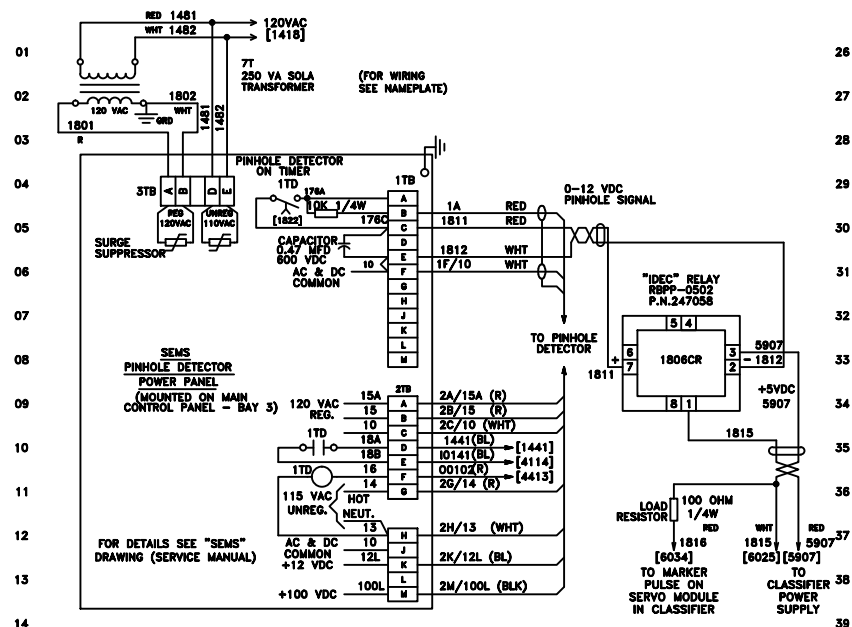
"NORTH AMERICAN"
CENTER GUIDE DETECTOR OPTION
FOR DETAILS SEE SERVICE MANUAL
(OPTIONAL)

LITTELL JOB #94228 LINE (R → L)

CONFIGURATION SETUPS FOR CENTER GUIDE DETECTOR			
		ADDITIONAL FEATURES	
CALIBRATE DETECTOR	DET1	H3118	INTEGRAL TERM 0
	DET2	NONE	DERIVATIVE TERM 2
GUIDE DIRECTION		NORMAL	DITHER ADJUST 17%
CONFIG ROLL CTR/FDBK		NONE	AUTO SPEED ADJUST 100%
OFFSET INPUT SELECT		H6229	MANUAL SPEED ADJUST 100%
			REMOTE METER
			AMPLIFIER OUTPUT
			LOCK KEYBOARD
			GAIN 100
			PILOT PRESSURE 220 PSI
			SYSTEM PRESSURE 800 PSI

0 EMISSION PARA CONSTRUCCION		12-06-91	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVIDER: D-308691
ZONA: PLANTA SIDERCROM		SIDERCROM	
SECTOR: LINEA DE CORTE - BR-4E		LITTELL	
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		SCROLL SHEETING LINE	
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO	ESCALAS	5615-23-0028	
DISEÑO			
CONTROL			
APROBADO			

DATE	SYM	WAS	BY
		D-305501	

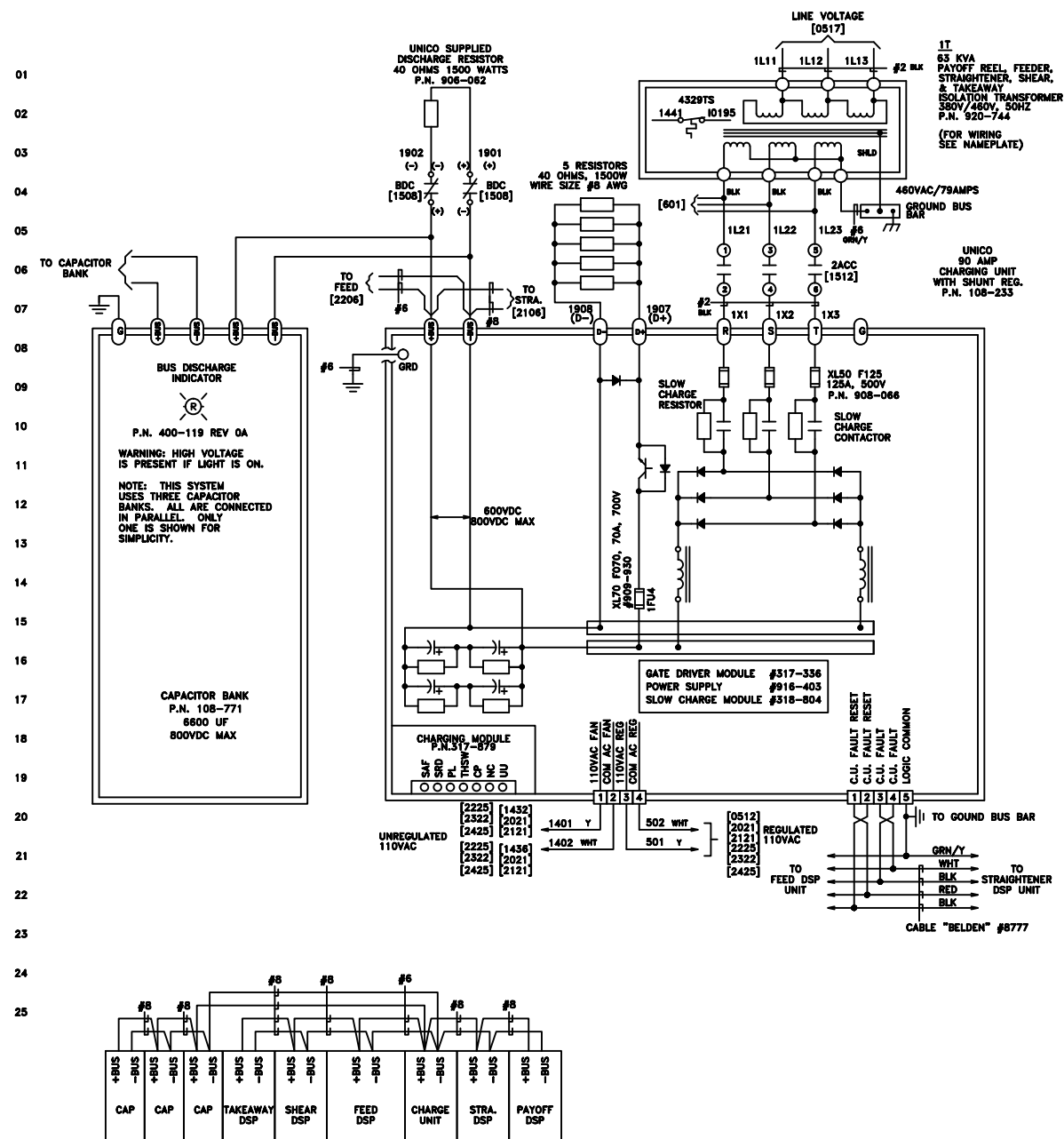


"SEMS" PINHOLE DETECTOR

LITTELL JOB #94228 LINE (R → L)

0 EMISION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 18 DE 67	
SECTOR: LINEA DE CORTE		OBRAS O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	11/23/81	

A			
DATE	SYM	WAS	BY
		D-305501	



"UNICO" DRIVE SYSTEM
SC-2397
90AMP CHARGING UNIT

0		EMISION PARA CONSTRUCCION		12-06-01	
REV.		DESCRIPCION		EJEC. CONT. APPRO. FECHA	
PROVEEDOR:				N° PROVEEDOR:  D-308691 HOJA 19 DE 67	
 SIDERAC PLANTA SIDERCHOM		ZONA: SIDERCHOM		SIDERCHOM LINEA LITTELL	
		OBRA O EQUIPO: LINEA DE CORTE -- BR-4E		SCROLL SHEETING LINE DIAGRAM--ELECTRICAL--SCHEMATIC	
LA EMPRESA SE RESERVA LA PROPIEDAD DE LOS DISEÑOS Y PROYECTOS QUE SE ELABORAN PARA EL CLIENTE. EL CLIENTE NO DEBE REPRODUCIR, COPIAR, VENDER, CEDER, O TRANSFERIR LOS DISEÑOS Y PROYECTOS A TERCERAS PERSONAS SIN EL CONSENTIMIENTO ESCRITO DE LA EMPRESA.		PARTES: ESCALAS		5615-23-0028 	
PROYECTO DISEÑO CONTROL APROBADO				INSTRUCCIONES: H43 CAD: T152H019.DWG FILE: T153.SAC	

LITTELL JOB #94228 LINE (R→L)

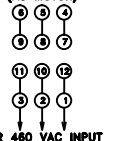
DATE	SYM	WAS	BY
		D-305501	

DSP PROGRAM LOCATION IS ON DSP CONTROL MODULE.
THESE IC'S ARE LOCATED UNDER THE SYNC SERIAL
MODULE. SYNC SERIAL MODULE MUST BE REMOVED
TO SET AT THE IC'S. ALL POWER MUST BE SHUT
OFF BEFORE CHECKING PROGRAM NUMBERS.

804-266.343 LOW U28 P.N. 913-788	804-266.343 MD U27 P.N. 913-788	804-266.343 MD U27 P.N. 913-788	804-266.343 MD U27 P.N. 913-788
--	---------------------------------------	---------------------------------------	---------------------------------------

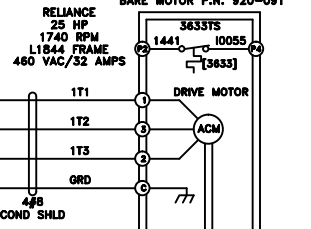
PAYOFF REEL
SERIES 2400
DSP DRIVE 25 HP, 460 VAC, 34 AMP
P.N. 109-181

MOTOR LEAD CONNECTIONS
(AC MOTOR)



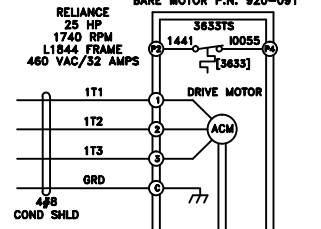
LINE R → L

PAYOFF REEL
AC MOTOR ASSEMBLY P.N. 707-533
BARE MOTOR P.N. 920-091

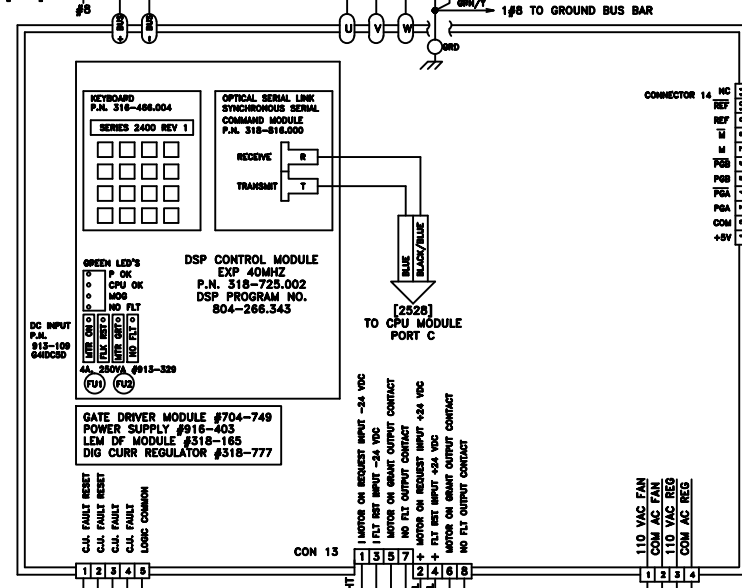


LINE L → R

PAYOFF REEL
AC MOTOR ASSEMBLY P.N. 707-533
BARE MOTOR P.N. 920-091



FROM
STRAIGHTENER
[2106]



CONNECTOR 14

CABLE P.N. 202-273

COUPLING
P.N. 309-298

2048 PPR
P.N. 905-731

CABLE P.N. 202-273

COUPLING
P.N. 309-298

2048 PPR
P.N. 905-731

OPTICAL SERIAL LINK
SYNCHRONOUS SERIAL COMM. MODULE - DETAILS
P.N. 318-816

PORT A
FIBER OPTICS
TRA/REC
U19

PORT B

PAYOFF REEL DSP DRIVE

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-08-81	
REX.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691
ZONA: SIDERAR		HORA 20 DE 87	
SECTOR: LINEA DE CORTE - BR-4E		OBRAS O EQUIPO: PLANTA SIDERAR	
PARTES: SCROLL SHEETING LINE		DIAGRAM - ELECTRICAL - SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	1152486	

DSP PROGRAM LOCATION IS ON DSP CONTROL MODULE.
THESE IC'S ARE LOCATED UNDER THE SYNC SERIAL MODULE. SYNC SERIAL MODULE MUST BE REMOVED TO SET AT THE IC'S. ALL POWER MUST BE SHUT OFF BEFORE CHECKING PROGRAM NUMBERS.

804-266.343
LOW U22
P.N. 915-788

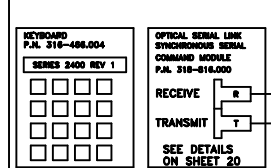
804-266.343
MED U27
P.N. 915-788

804-266.343
HIGH U28
P.N. 915-788

STRAIGHTENER
SERIES 2400
DSP DRIVE 25 HP, 480 VAC, 34 AMP
P.N. 109-181

FROM CHARGING UNIT [1906]

TO PAYOFF REEL DSP UNIT [2006]



DSP CONTROL MODULE
P.N. 318-725.002
DSP PROGRAM NO. 804-266.343

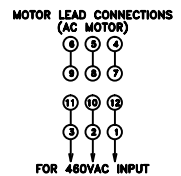
GREEN LED'S
P OK
CPU OK
MOD
NO FLT
SERIAL

GATE DRIVER MODULE #704-748
POWER SUPPLY #918-403
LEM OF MODULE #318-165
DIG CURR REGULATOR #318-777

C.A. FAULT RESET
C.A. FAULT
C.A. FAULT
LOGIC COMMON

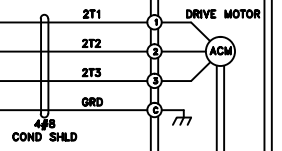
FROM CHARGING UNIT

TO PAYOFF REEL DSP UNIT



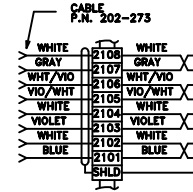
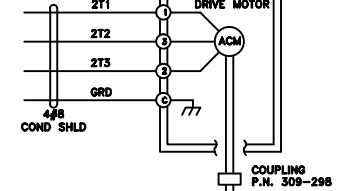
STRAIGHTENER
AC MOTOR ASSEMBLY P.N. 707-533
BARE MOTOR P.N. 920-091

RELIANCE
25 HP
1740 RPM
L1844 FRAME
460 VAC/32 AMPS

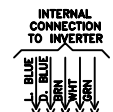


STRAIGHTENER
AC MOTOR ASSEMBLY P.N. 707-533
BARE MOTOR P.N. 920-091

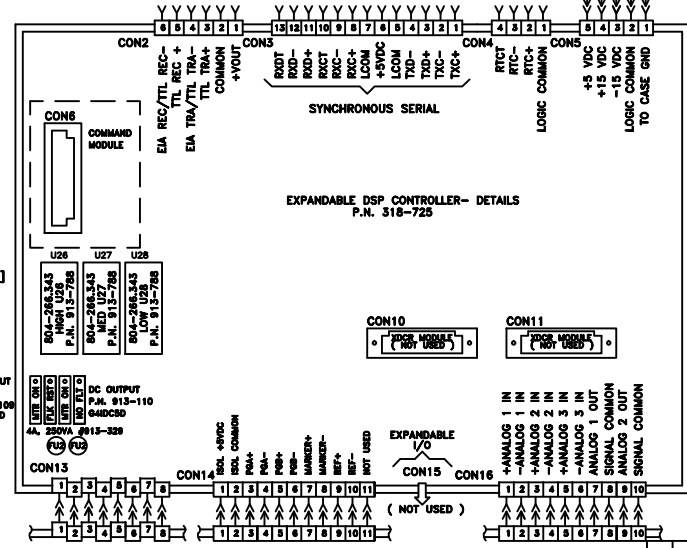
RELIANCE
25 HP
1740 RPM
L1844 FRAME
460 VAC/32 AMPS



2048 PPR
P.N. 905-731



EXPANDABLE DSP CONTROLLER- DETAILS
P.N. 318-725



STRAIGHTENER DSP DRIVE

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR:	D-308691		HOJA 21 DE 67
ZONA: SIDERAR		SECTOR: LINEA DE CORTE - BR-4E	
OBRA O EQUIPO: PLANTA SIDERAR		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	11/25/80	

DATE	SYM	WAS	BY
		D-305501	

DSP PROGRAM LOCATION IS ON DSP CONTROL MODULE. THESE IC'S ARE LOCATED UNDER THE SYNC SERIAL MODULE. SYNC SERIAL MODULE MUST BE REMOVED TO GET AT THE IC'S. ALL POWER MUST BE SHUT OFF BEFORE CHANGING PROGRAM NUMBERS.

MOTOR LEAD CONNECTIONS (AC MOTOR)

LINE R → L
AC MOTOR ASSEMBLY P.N. 706-044
REF: 918-784

LINE L → R
AC MOTOR ASSEMBLY P.N. 706-044
REF: 918-784

FEED
SERIES 2400
DSP DRIVE 60 HP, 480 VAC, 77 AMP
P.N. 109-198

TO SHEAR [2306]
FROM CHARGING UNIT [1906]

#6 BLK
#6 BLK

TO SERIAL #3 PORT A

KEYBOARD P.N. 516-488-004
SERIES 2400 REV 2

OPTICAL SERIAL LINK
SYNCHRONOUS SERIAL
COMMAND MODULE P.N. 518-816-005

RECEIVE
TRANSMIT

GREEN LED'S
P OK
CPU OK
MOD
NO FLT

DSP CONTROL MODULE
P.N. 518-725-002
DSP PROGRAM NO. 804-268-343

DC INPUT
P.N. 513-109
64DCDD

4A 250VAC #913-320

GATE DRIVER MODULE #704-749
POWER SUPPLY #916-403
LEM DF MODULE #518-165
DIG CURR REGULATOR #518-777

TO SHEAR DSP UNIT

FROM CHARGING UNIT

TO TAKEAWAY ROLL DRIVE [2421]

TO TAKEAWAY ROLL OPEN SOLENOID

TO TAKEAWAY ROLL CLOSE SOLENOID

TO TAKEAWAY ROLL PERMISSIVE

TO TAKEAWAY ROLL CLOSE

TO TAKEAWAY ROLL OPEN

TO TAKEAWAY ROLL PERMISSIVE

TO TAKEAWAY ROLL CLOSE

TO TAKEAWAY ROLL OPEN

TO TAKEAWAY ROLL PERMISSIVE

TO TAKEAWAY ROLL CLOSE

TO TAKEAWAY ROLL OPEN

TO TAKEAWAY ROLL PERMISSIVE

TO TAKEAWAY ROLL CLOSE

TO TAKEAWAY ROLL OPEN

TO TAKEAWAY ROLL PERMISSIVE

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TO TAKEAWAY ROLL OPEN

TO TAKEAWAY ROLL PERMISSIVE

TO TAKEAWAY ROLL CLOSE

TO TAKEAWAY ROLL OPEN

DSP PROGRAM LOCATION IS ON DSP CONTROL MODULE. THESE IC'S ARE LOCATED UNDER THE SYNC SERIAL MODULE. SYNC SERIAL MODULE MUST BE REMOVED TO GET AT THE IC'S. ALL POWER MUST BE SHUT OFF BEFORE CHECKING PROGRAM NUMBERS.



LITTELL JOB #94228 LINE (R → L)

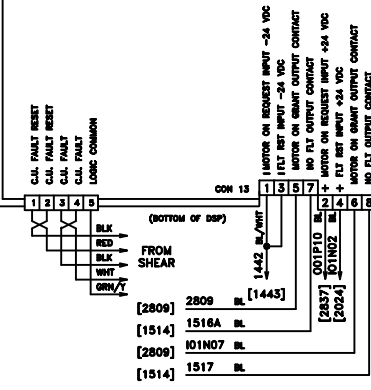
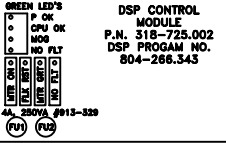
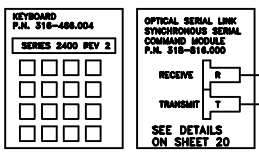
DATE	SYM	WAS	BY
		D-305501	

DSP PROGRAM LOCATION IS ON DSP CONTROL MODULE.
THESE IC'S ARE LOCATED UNDER THE SYNC SERIAL
MODULE. SYNC SERIAL MODULE MUST BE REMOVED
TO GET AT THE IC'S. ALL POWER MUST BE SHUT
OFF BEFORE CHECKING PROGRAM NUMBERS.

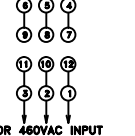
IC#	P.N.	IC#	P.N.
1	817-708	4	817-708
2	817-708	5	817-708
3	817-708	6	817-708

TAKEAWAY
SERIES 2400
DSP DRIVE
25 HP, 480 VAC, 34 AMP
P.N. 109-181

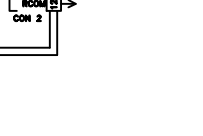
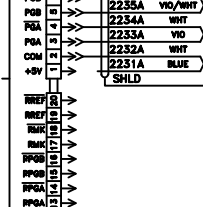
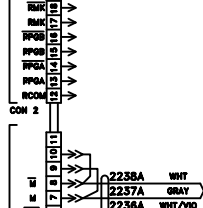
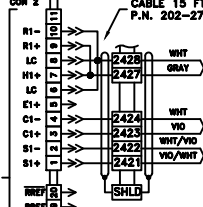
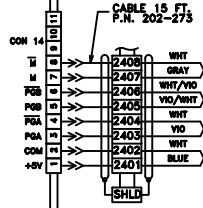
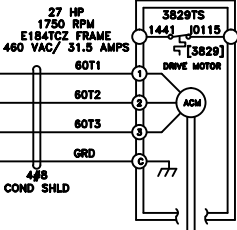
FROM
SHEAR
[2106]



MOTOR LEAD CONNECTIONS
(AC MOTOR)



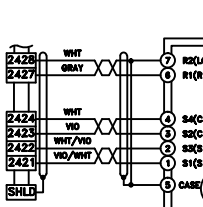
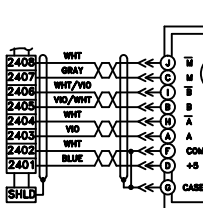
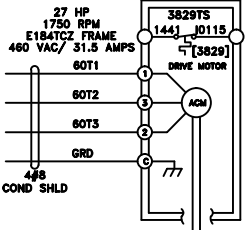
TAKEAWAY
AC MOTOR ASSEMBLY P.N. 707-801
REF: 920-485



PULSE GENERATOR
2048 PPR
P.N. 905-751

RESOLVER
14BIT/ 7KHz

TAKEAWAY
AC MOTOR ASSEMBLY P.N. 707-801
REF: 920-485

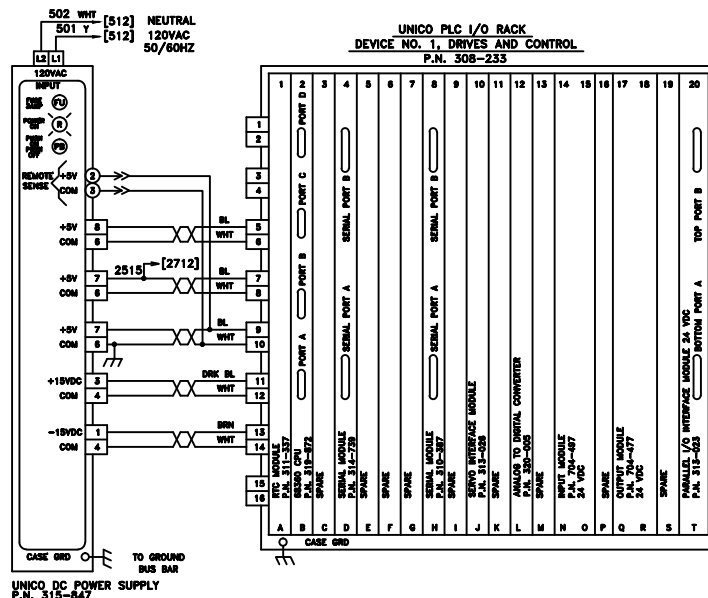


TAKEAWAY DSP DRIVE

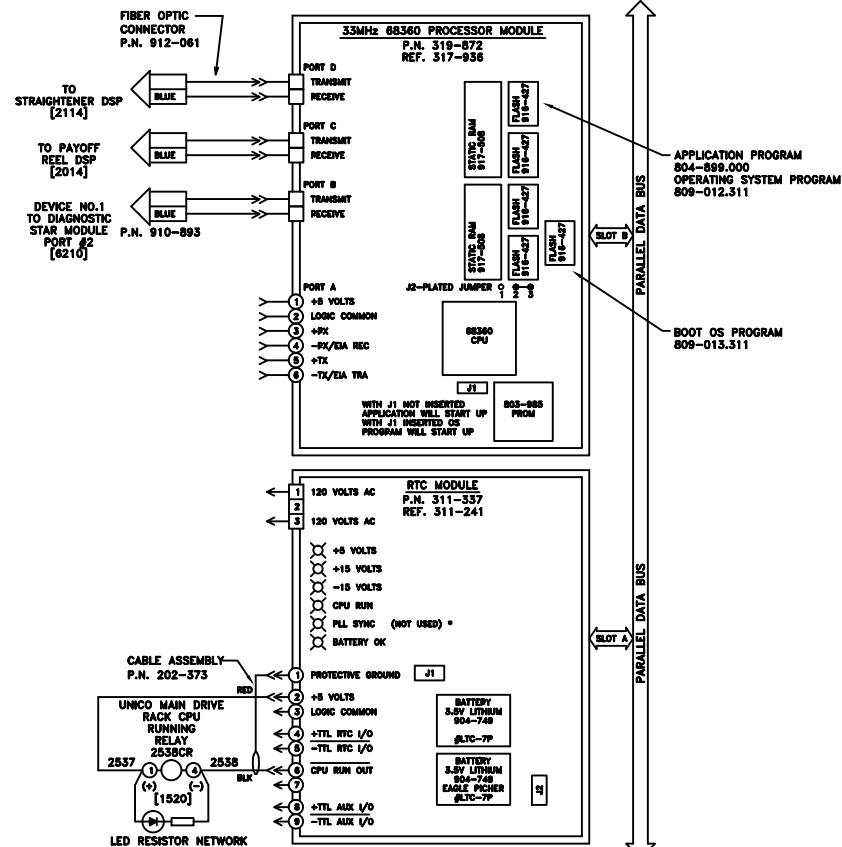
0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HOJA 24 DE 67	
ZONA: SIDERAR		SECTOR: LINEA DE CORTE	
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		SCROLL SHEETING LINE	
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO	ESCALAS	5615-23-0028	
DISEÑO			
CONTROL			
APROBADO			

LITTELL JOB #94228 LINE (R -> L)

DATE	SYM	WAS	BY
		D-305501	



* NOTE:
ALL I/O RACK SLOTS
ARE IDENTICAL. MODULES
CAN BE INSTALLED IN ANY
SLOT. I/O MODULES ARE
IDENTIFIED BY LOGICAL SLOT
NUMBER.
MODULES ARE SPACED INTENTIONALLY
TO IMPROVE HEAT DISSIPATION.



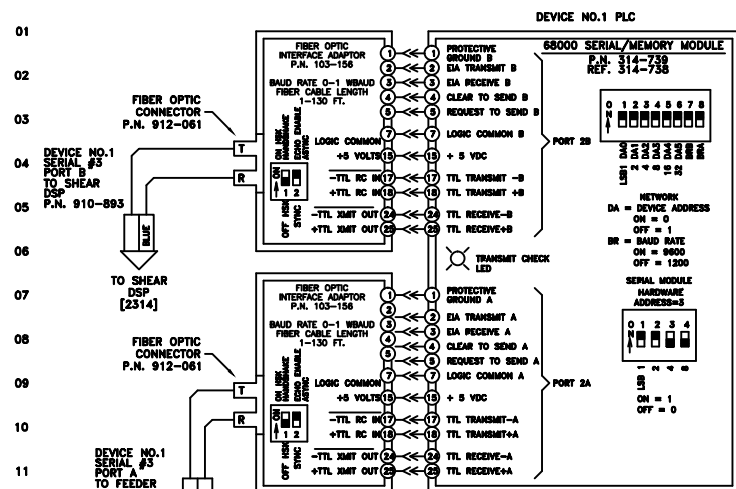
* NOTE:
THE PLL SYNC LED IS NOT USED WHEN 120 VAC
IS NOT APPLIED TO TERMINALS L1 AND L2 AT THE TOP
OF THE MODULE. THE LED MAY OR MAY NOT BE ON.

"UNICO" PLC I/O RACK DEVICE NO.1
DRIVES AND CONTROL
#SC-2397

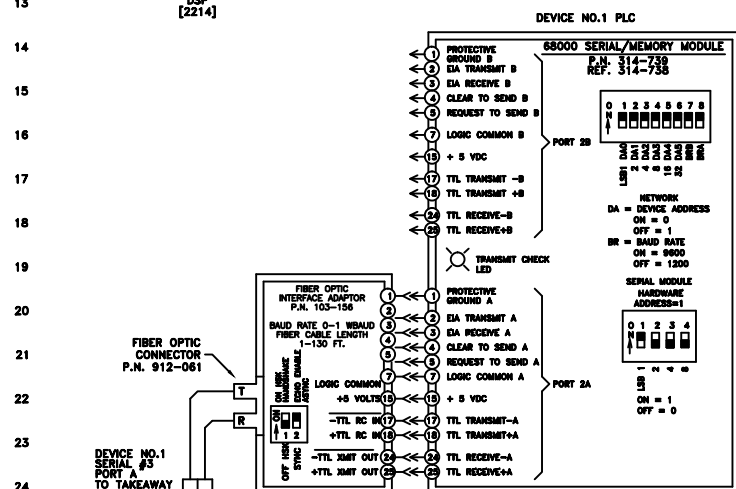
0 EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	ELEC. CONT. APRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 25 DE 67	
SECTOR: LINEA DE CORTE		BR-4E	
OBRA O EQUIPO: LINEA DE CORTE		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

LITTELL JOB #94228 LINE (R → L)

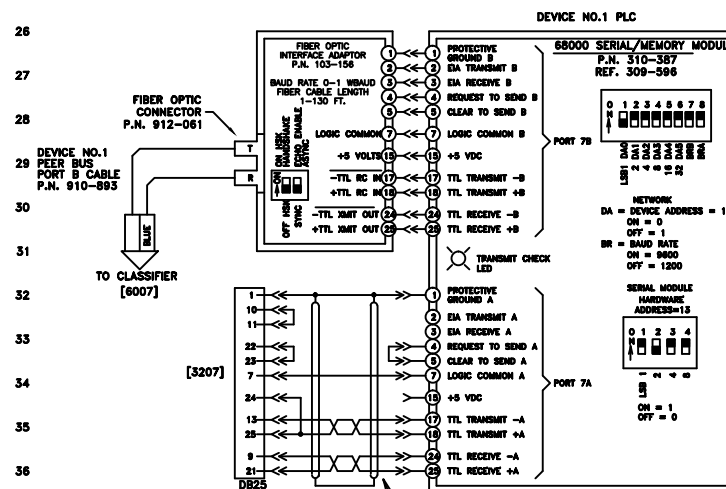
DATE	SYM	WAS	BY
		D-305501	



SHEAR AND FEEDER SERIAL MODULE TO DSP



TAKEAWAY SERIAL MODULE TO DSP



GE COMM COPROC MODULE PORT 2 (OPERATORS CONSOLE)

SERIAL CABLE P.N. 281396

FIBER OPTIC INTERFACE ADAPTOR P.N. 103-156 SWITCH SETTINGS

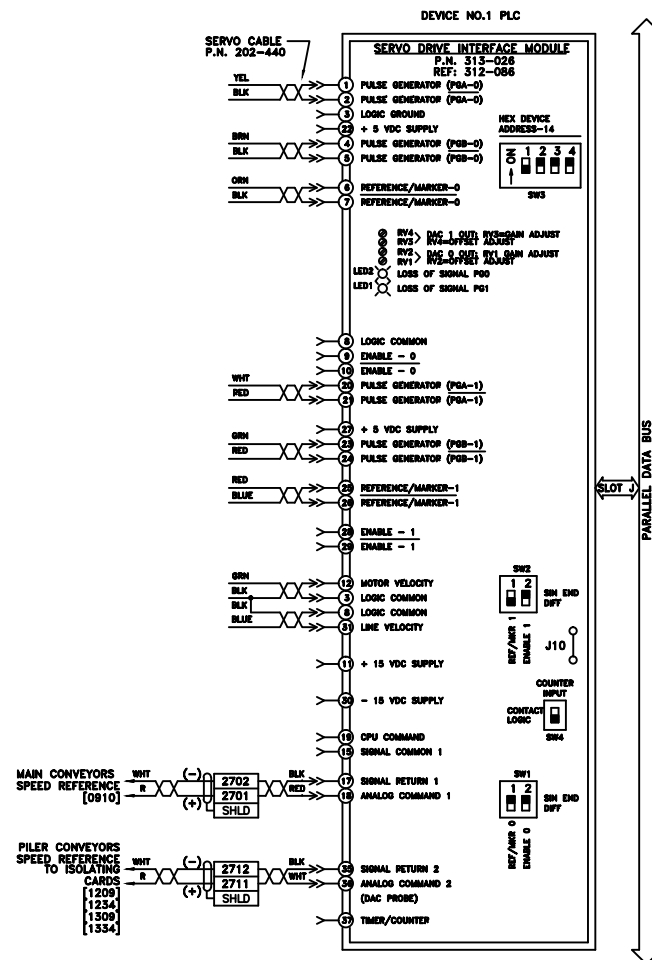
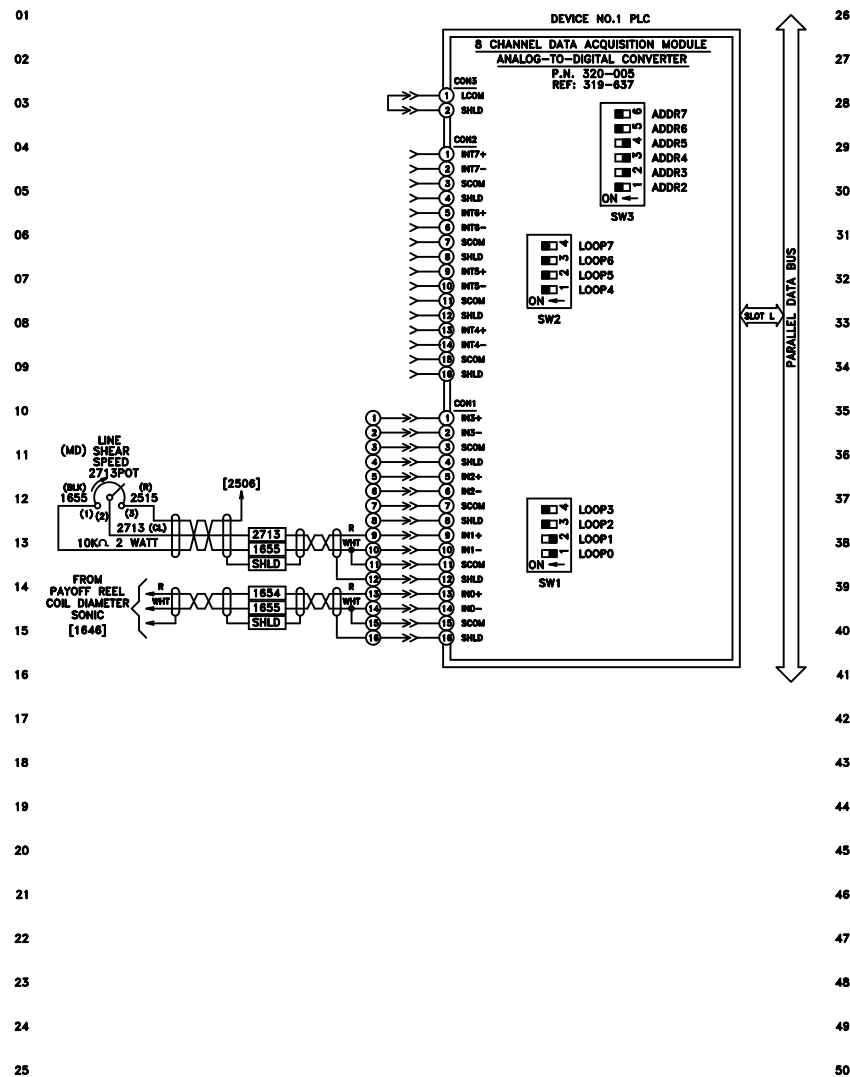
HANDSHAKE	SWITCH	ON	OFF	OFF
ECHO ENABLE	2	ON	OFF	ON
		ASYNC	PEER BUS	DSP

UNICO PLC I/O RACK - DEVICE NO.1
68000 SERIAL/MEMORY MODULE
(DOOR 3)

LITTELL JOB #94228 LINE (R → L)

0	EMISION PARA CONSTRUCCION			12-06-81
REV.	DESCRIPCION	ELEC. CONT. APROB.	FECHA	
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691	HOUA 26 DE 67
ZONA:	SIDERAR	SECTOR:	LINEA DE CORTE	
OBRA O EQUIPO:	LINEA DE CORTE - BR-4E	PARTE:	SCROLL SHEETING LINE	
PROYECTO:		ESCALAS:	5615-23-0028	
DIBUJO:		IMPULSION:	HAS	1554028.DWG
CONTROL:		PLA:	1152348C	
APROBADO:				

DATE	SYM	WAS	BY
		D-305501	

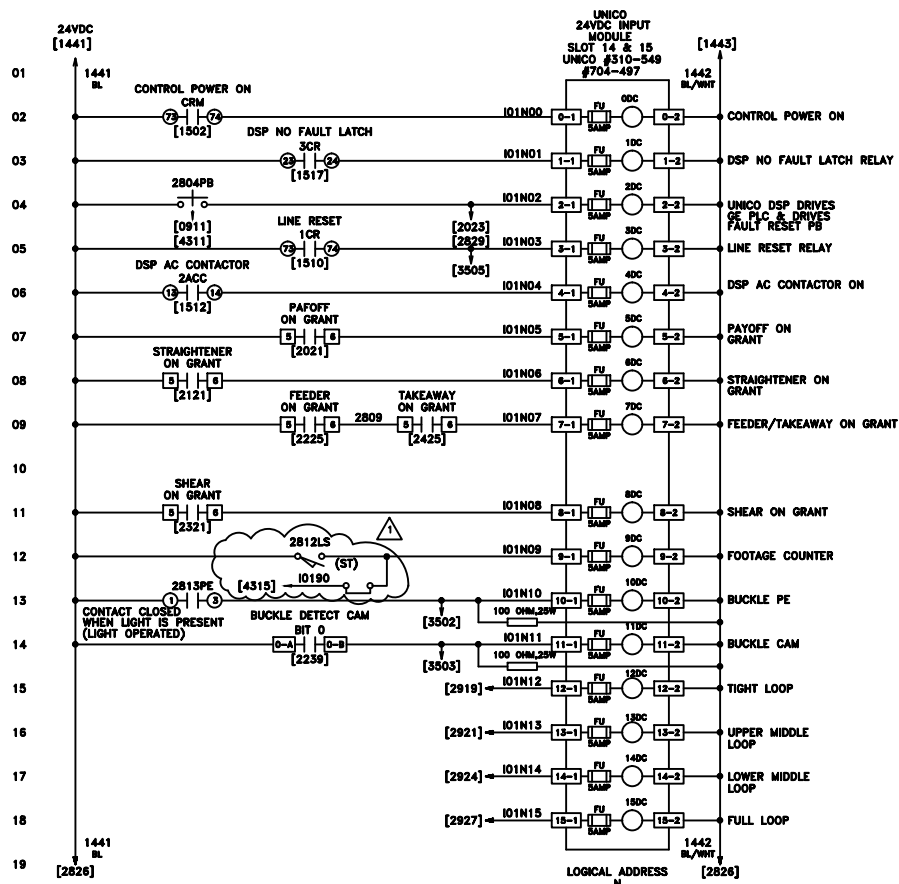


UNICO PLC I/O RACK - DEVICE NO.1
DATA ACQUISITION MODULE
SERVO DRIVE INTERFACE MODULE
(DOOR 3)

0	EMISION PARA CONSTRUCCION			12-06-81
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA	
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691	HOJA 27 DE 67
ZONA:	SIDEROCROM	SECTOR:	LINEA LITTELL	
OBRA O EQUIPO:	PLANTA SIDEROCROM	SECCION:	LINEA DE CORTE - BR-4E	
LA LINEA DE REFERENCIA LA PROYECCION DE TUBO		PARTE:	SCROLL SHEETING LINE	
ESCALAS		PROYECTO		
DISEÑO		CONTROLADO		
APROBADO		IMPRESION:	MAS	FILE T1524027.DWG
				FILE T1524027.DWG

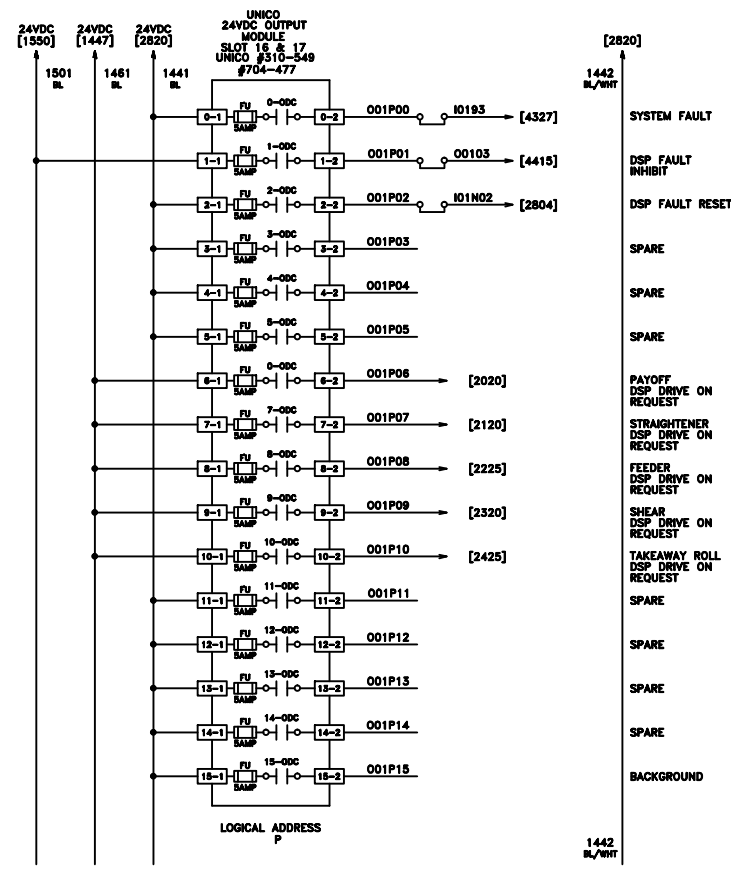
LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



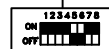
UNICO I/O RACK SLOT N-0

BIT 10 #914-324	BIT 11 #914-324	BIT 7 #914-324	BIT 3 #914-324
BIT 14 #914-324	BIT 10 #914-324	BIT 6 #914-324	BIT 2 #914-324
BIT 13 #914-324	BIT 9 #914-324	BIT 5 #914-324	BIT 1 #914-324
BIT 12 #914-324	BIT 8 #914-324	BIT 4 #914-324	BIT 0 #914-324



UNICO I/O RACK SLOT P & Q

BIT 10 #914-324	BIT 11 #914-324	BIT 7 #914-324	BIT 3 #914-324
BIT 14 #914-324	BIT 10 #914-324	BIT 6 #914-324	BIT 2 #914-324
BIT 13 #914-324	BIT 9 #914-324	BIT 5 #914-324	BIT 1 #914-324
BIT 12 #914-324	BIT 8 #914-324	BIT 4 #914-324	BIT 0 #914-324

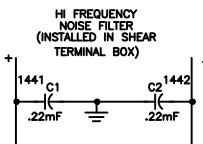
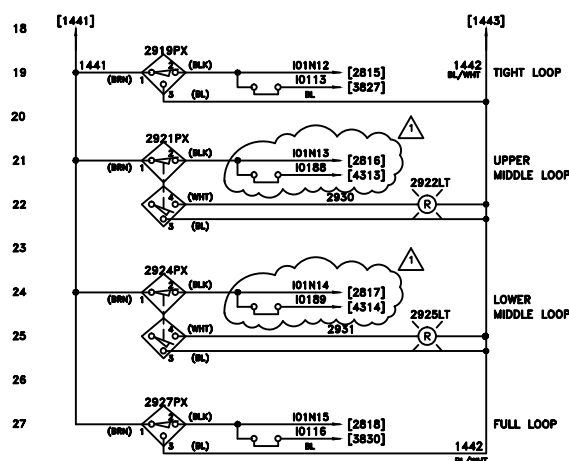
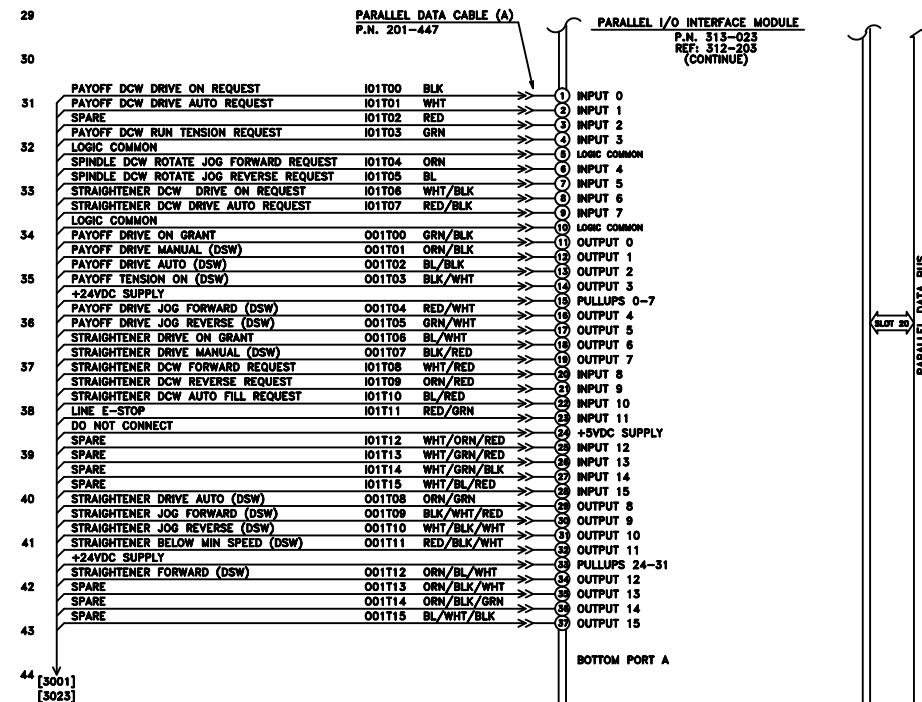


UNICO PLC I/O RACK - DEVICE NO.1
I/O MODULES
(DOOR 3)

SE AGREGA PUENTE IO1N09 E IO190		DYCE	DYCE	DFP	03-08-01
EMISION PARA CONSTRUCCION					
REDA	DESCRIPCION	EJEC.	CONT.	APROV.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		N° PROVEEDOR: D-308691		
ZONA:		SIDERCROM			
SECTOR:		LINEA DE CORTE			
OBRA O EQUIPO:		BR-4E			
PARTES:		SCROLL SHEETING LINE			
PROYECTO:		ESCALAS:			
DISEÑO:		5615-23-0028			
CONTROL:		IMPULSION: M43			
APROBADO:		CAD: T1524028.DWG			
		FILE: T1524028			

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



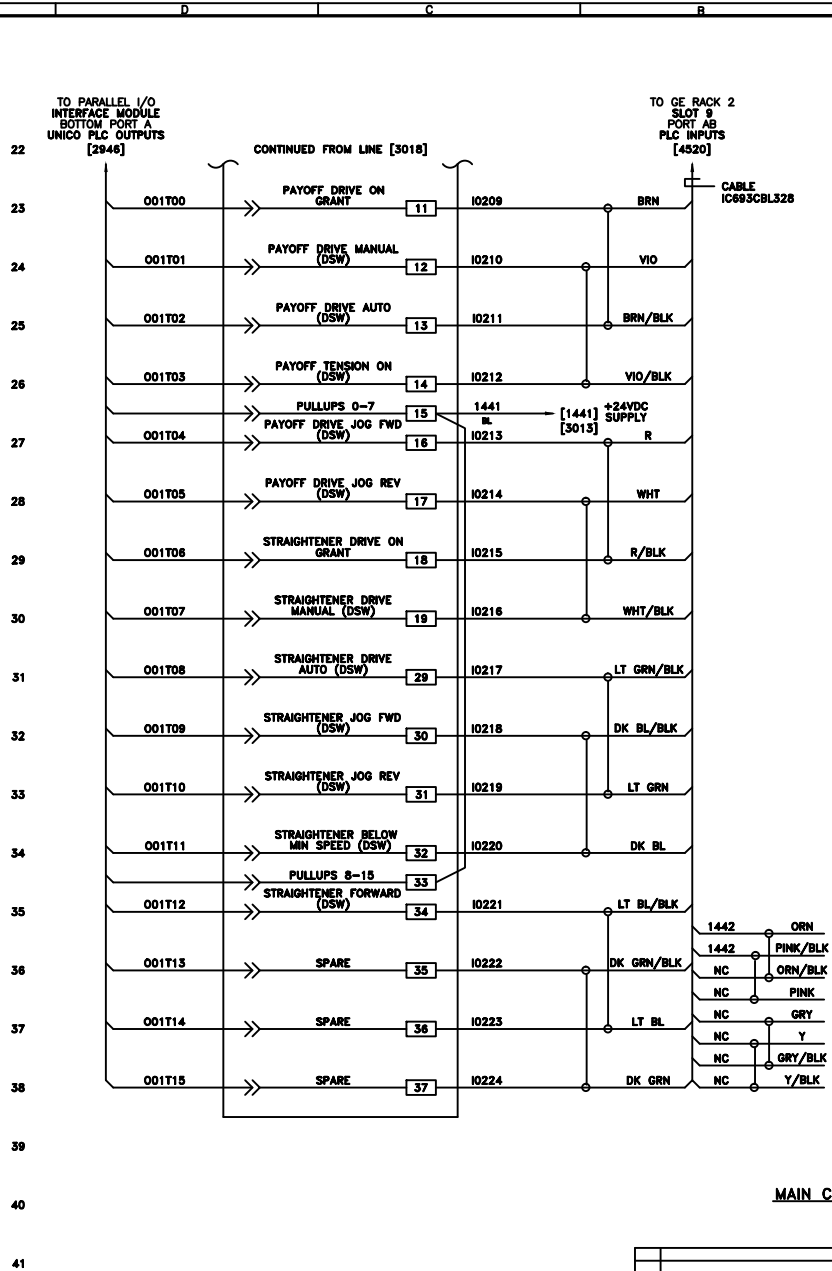
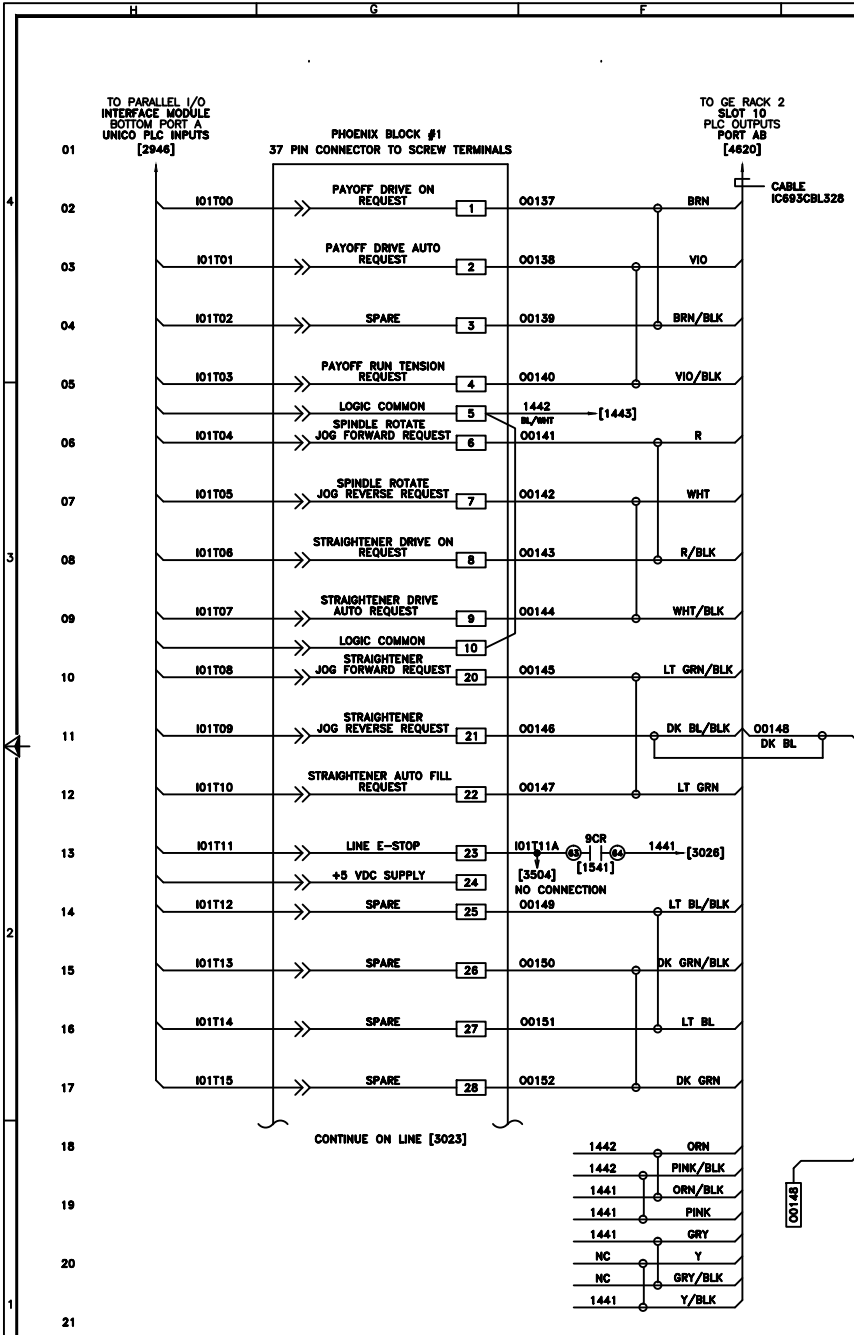
N.C. CONTACT OPENS WHEN COIL IS DETECTED

UNICO PLC I/O RACK - DEVICE NO.1
PARALLEL I/O INTERFACE MODULE
(DOOR 3)

SE AGREGAN PUENTES I01B8/I01N13, I01B9/I01N14		DWG: DWG	DWG: 03-08-01
EMISION PARA CONSTRUCCION		12-06-01	
REDA	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR: LITTELL INTERNATIONAL INC.	IT PROVEEDOR: D-308691	HOJA 29 DE 67	
ZONA: SIDERAR		SECTOR: LINEA DE CORTE	
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO	ESCALAS	5615-23-0028	
DISEÑO	IMPRESION: M33	CAD: T155H029.DWG	
CONTROLADO	PLA: T155348C		
APROBADO			

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	

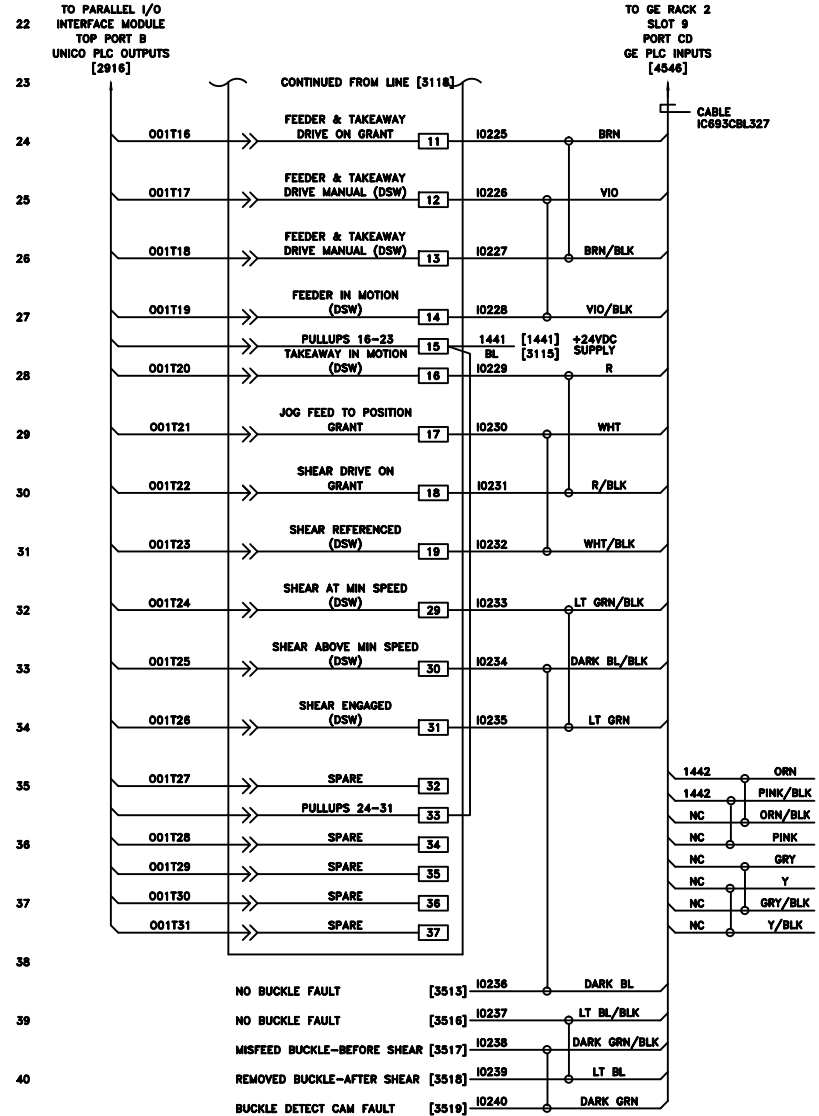
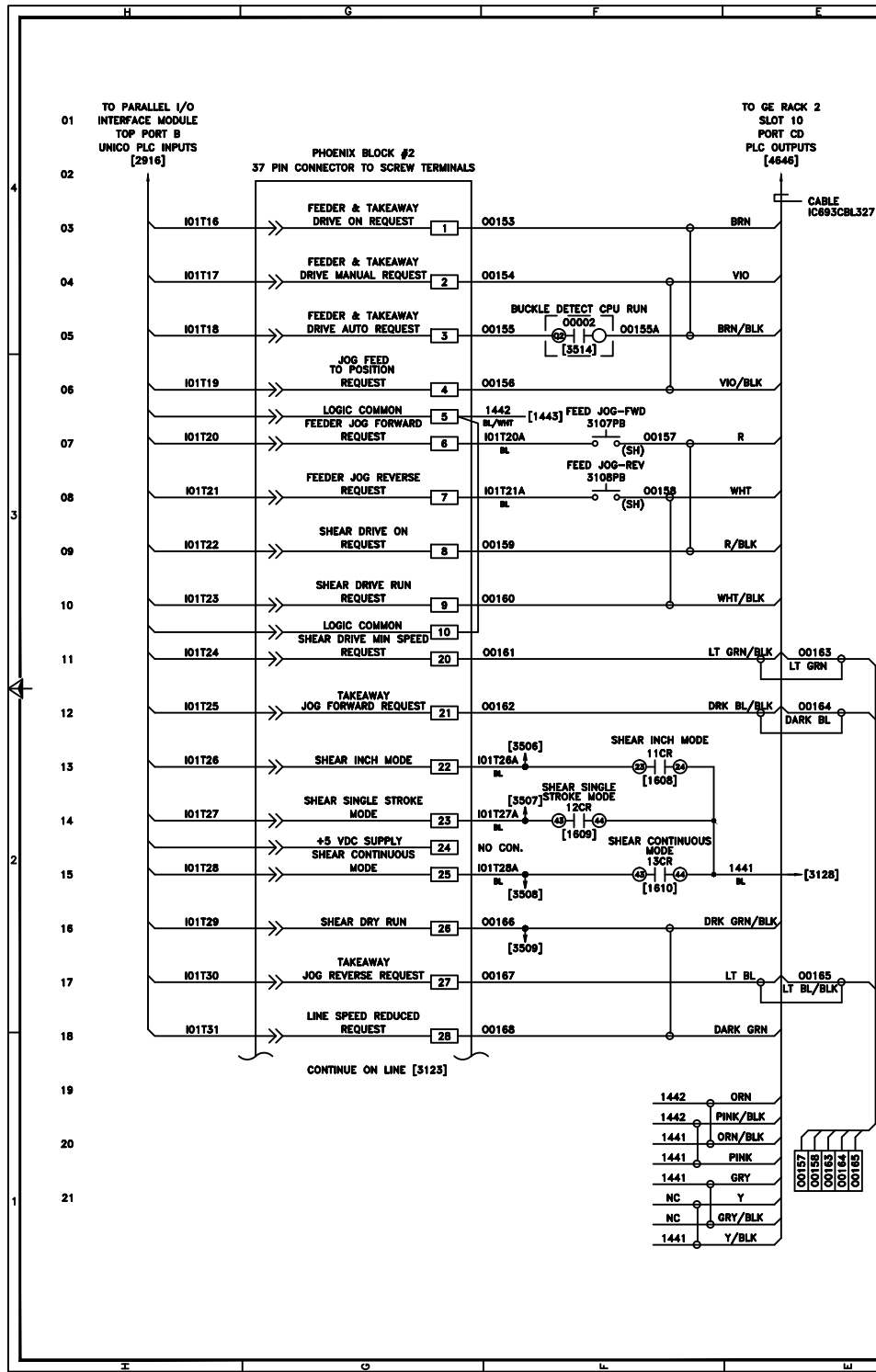


MAIN CONTROL ENCLOSURE

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HQA 30 DE 87	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
SCROLL SHEETING LINE			
PARTES:			
ESCALAS:		5615-23-0028	
PROYECTO:	IMPULSION:	MAS	CAE: T152340C.DWG
DISEÑO:	FILE:	T152340C	
CONTROL:			
APROB:			

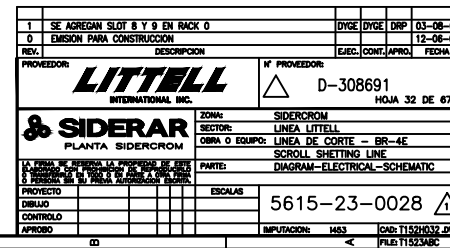
LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	

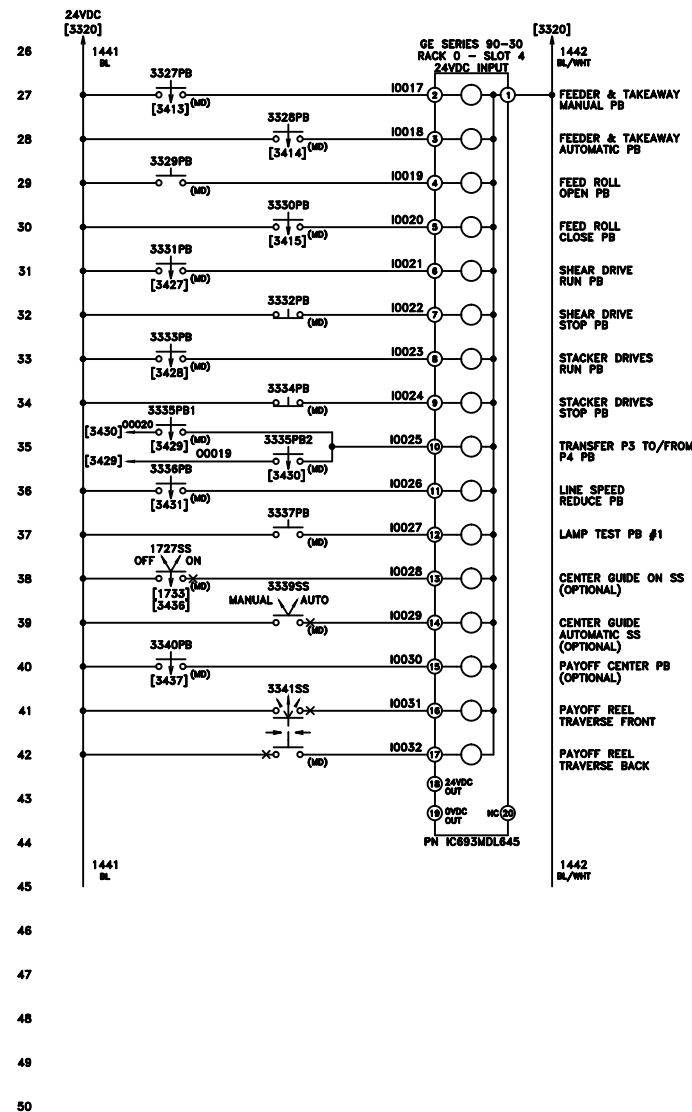
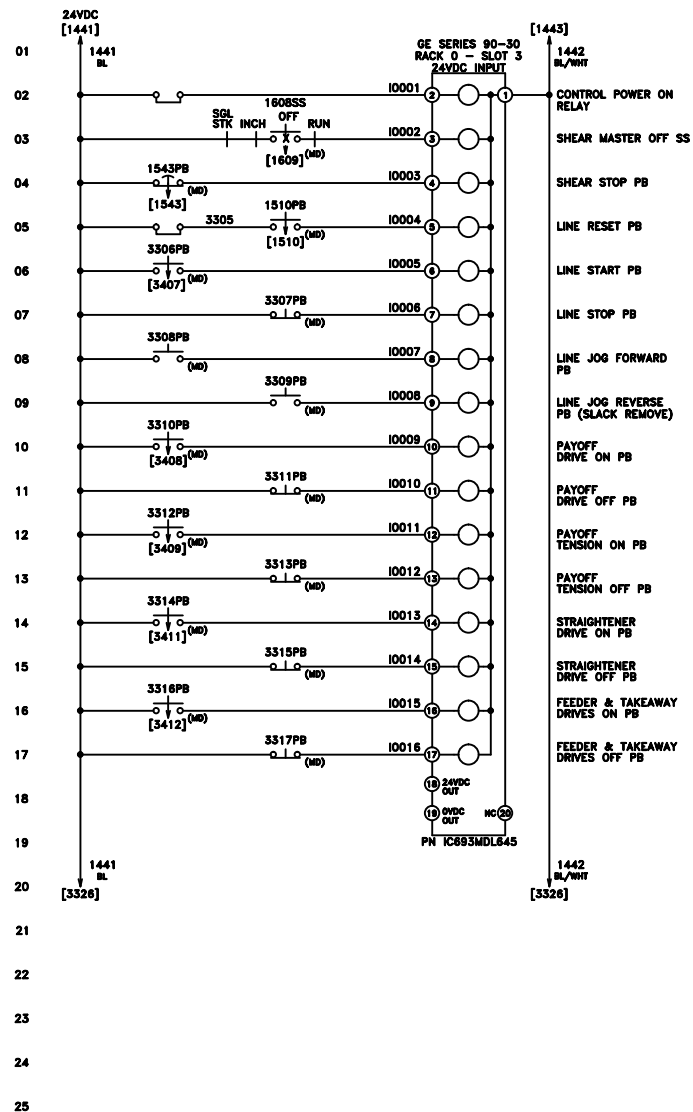


LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-01	
REV.	DESCRIPCION	EJEC.	CONT. APROB. FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691 HOJA 31 DE 67
ZONA: SIDEROCROM		SECTOR: LINEA DE CORTE	
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		SCROLL SHEETING LINE	
PARTES:		DIAGRAMA ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	



DATE	SYM	WAS	BY
		D-305501	

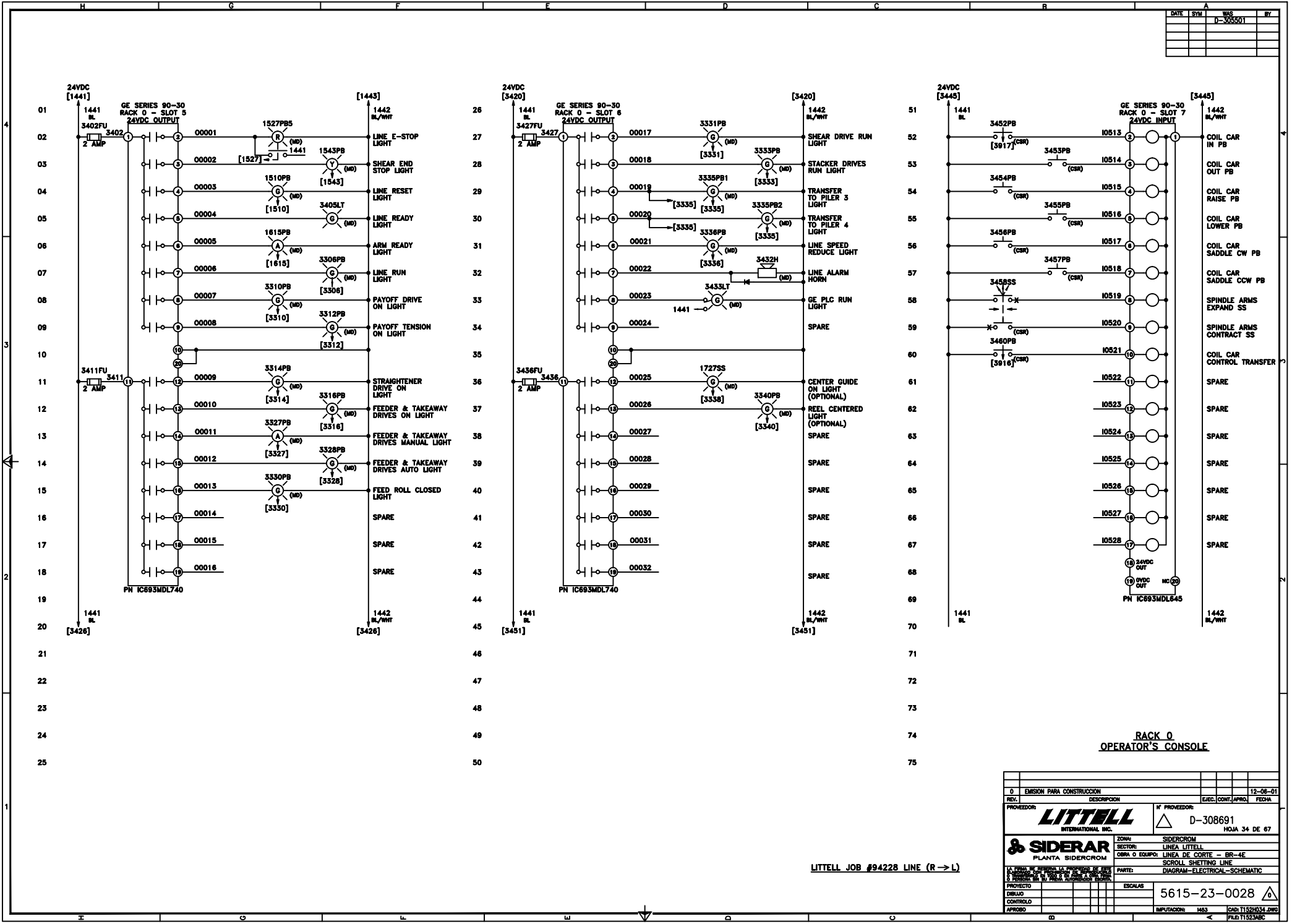


RACK 0
OPERATOR'S CONSOLE

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691 HOJA 33 DE 67
ZONA: SIDEROM		SECTOR: LINEA DE CORTE	
OBRA O EQUIPO: PLANTA SIDEROM		BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

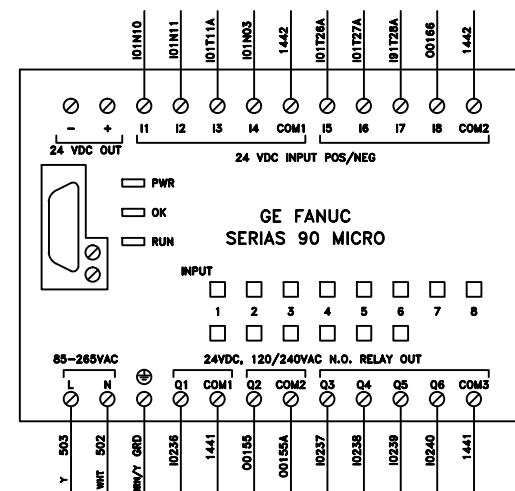
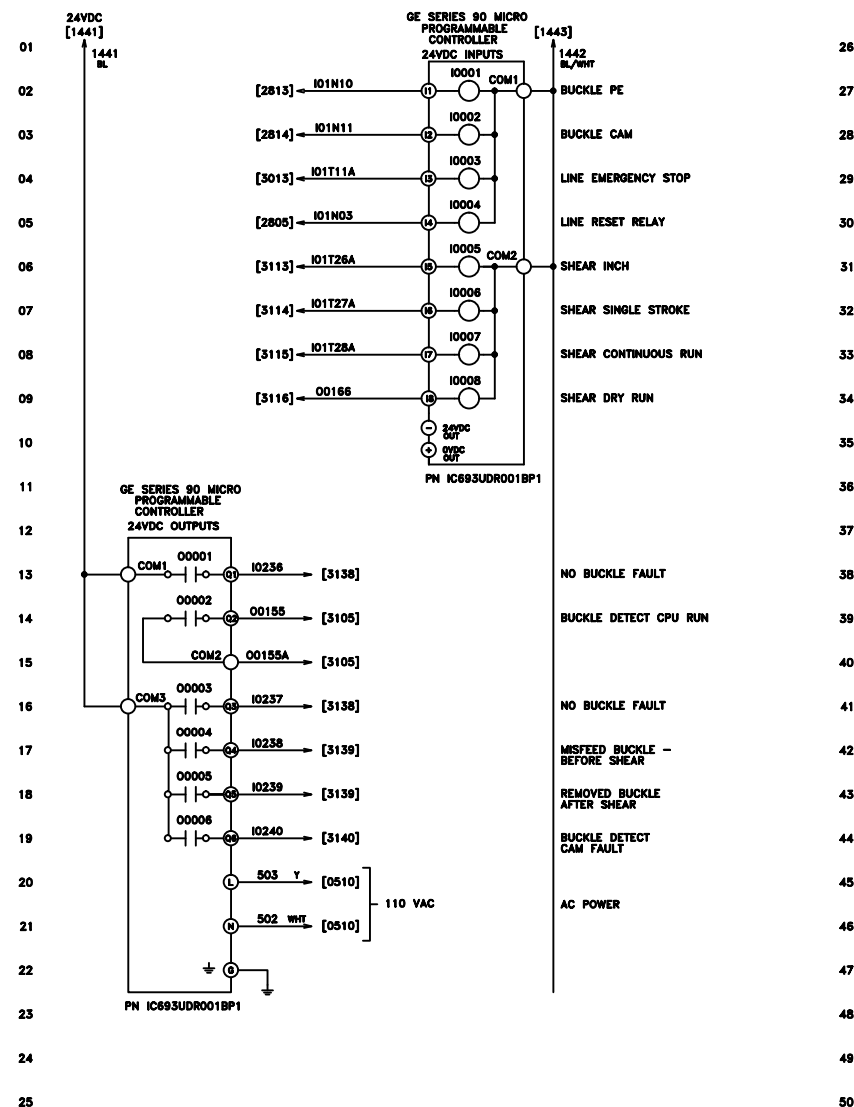
DATE	SYM	WAS	BY
		D-305501	



RACK 0
OPERATOR'S CONSOLE

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROV.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HOJA 34 DE 67	
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DIBUJO	ESCALAS	5615-23-0028	
CONTROLADO	IMPRESION:	HAB	
APROBADO	FILED:	T152348C	

LITTELL JOB #94228 LINE (R → L)

[illegible]

MODEL: IC 693 UDR 001 BP1

INPUTS: 24 VDC 8 mA MAX, 8 PT POS/NEG

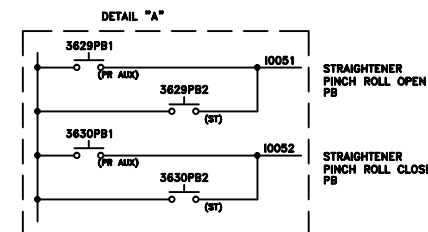
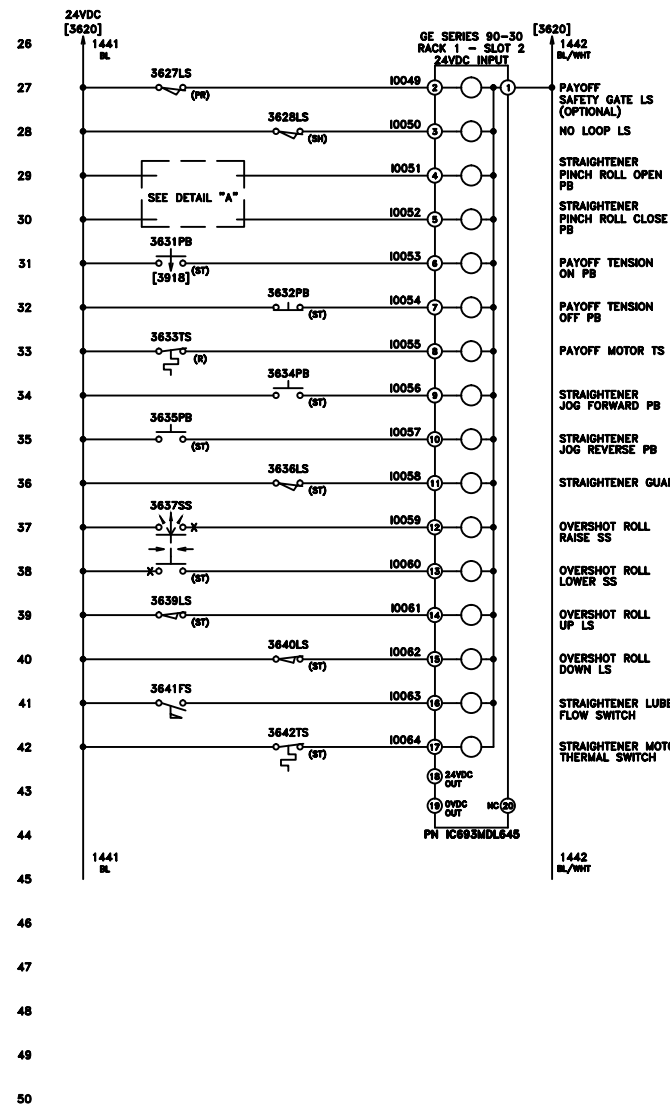
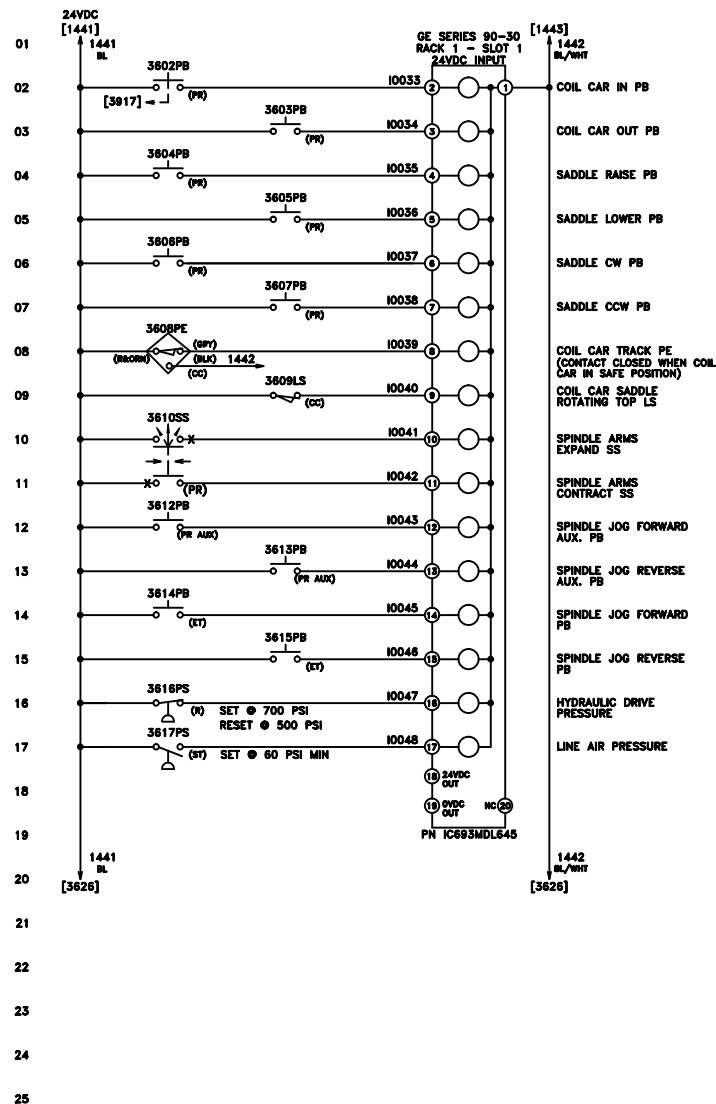
OUTPUTS: 24 VDC 120/240 VAC, 6PT
N.O. RELAY 2A PILOT DUTY

SOURCE: 110-240 VAC 20 VA MAX

BUCKLE DETECTOR PROGRAMMABLE CONTROLLER

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



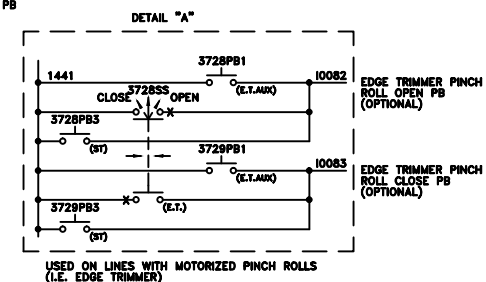
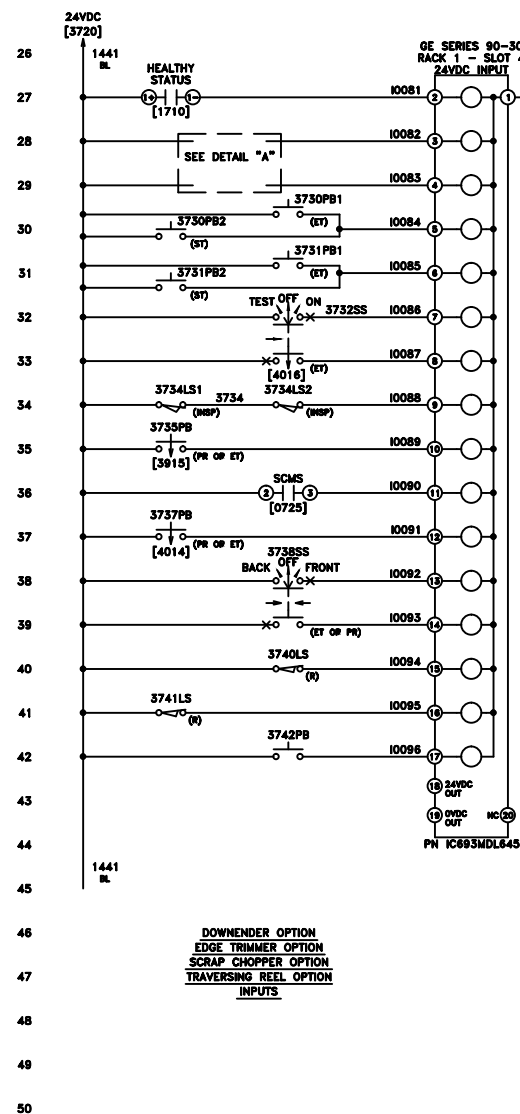
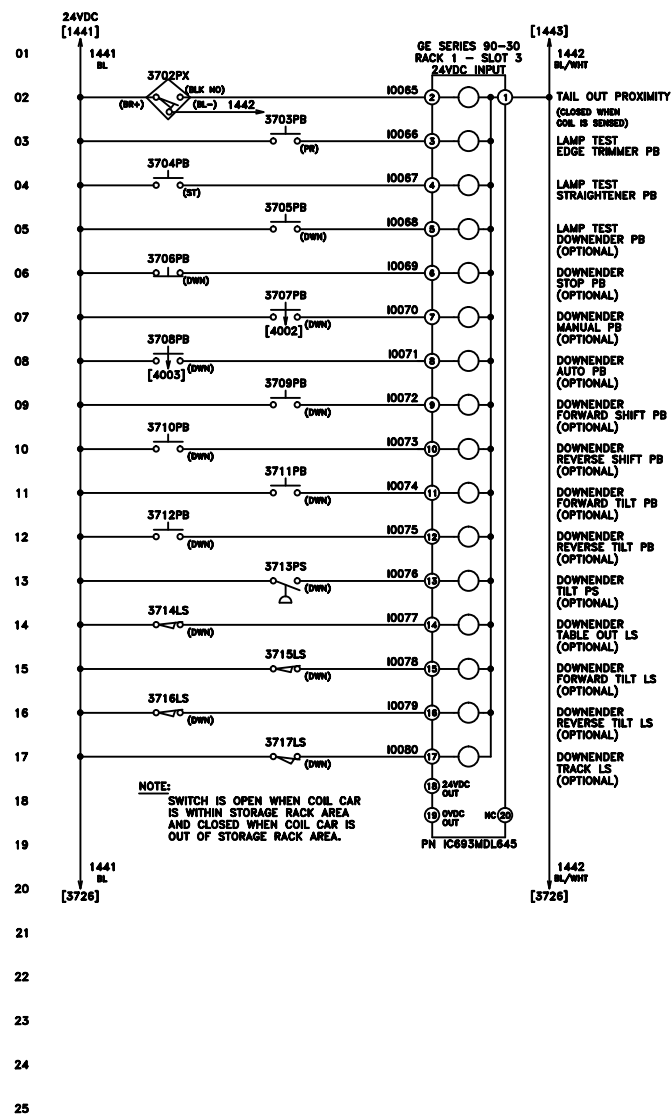
NOTE: - JUMPER 1441 TO 10048 IF PAYOFF REEL SAFETY GATES ARE NOT INSTALLED
- JUMPER 1441 TO 3636 IF THICKNESS MARKER IS INSTALLED
NOTE: INSTALLED IN STRAIGHTENER BASE

LITTELL JOB #94228 LINE (R → L)

RACK 1 INSPECTION STATION

0 EMISSION PARA CONSTRUCCION		12-06-01	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IF PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 36 DE 67	
SECTOR: LINEA DE CORTE		OBRAS O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	11/03/06	
APROBADO:	FECHA:	11/03/06	

DATE	SYM	WAS	BY
		D-305501	

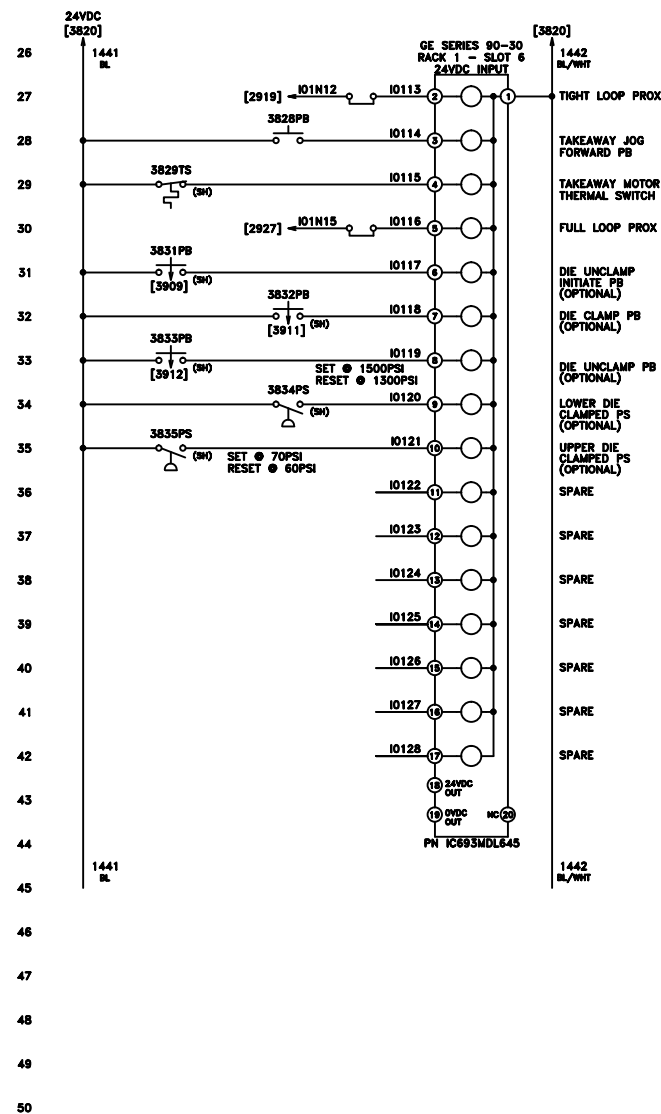
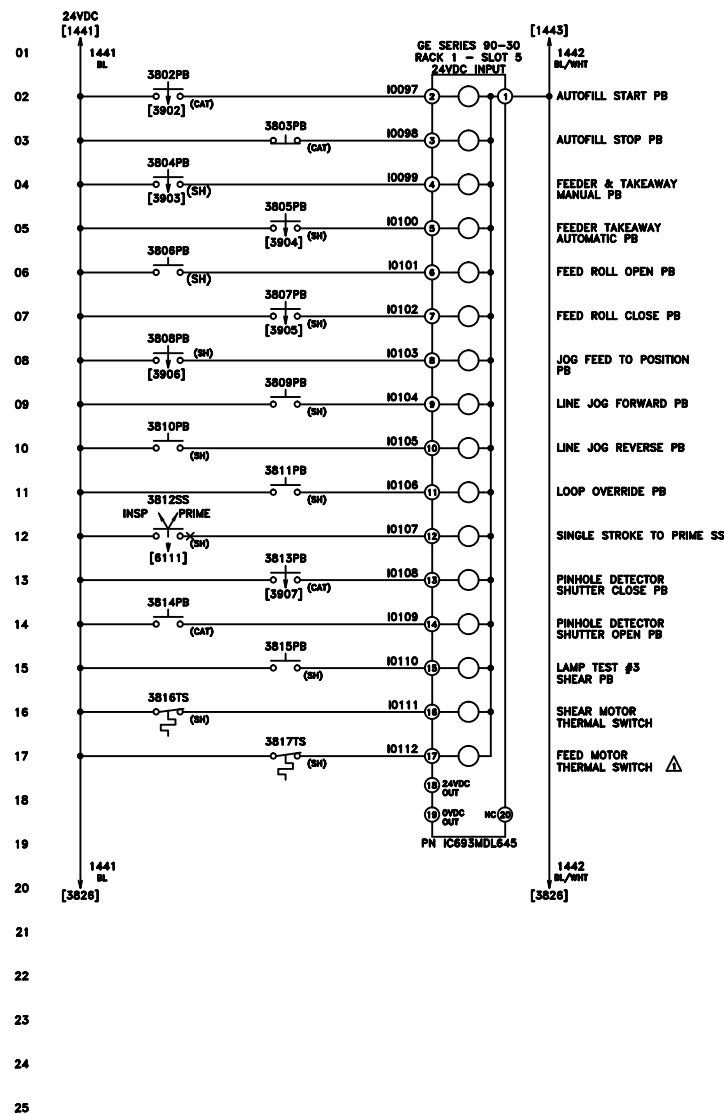


RACK 1 INSPECTION STATION

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APRO.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HQUA 37 DE 67	
ZONA: SIDEROCROM		SECTOR: LINEA LITTELL	
OBRA O EQUIPO: PLANTA SIDEROCROM		BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



RACK 1 INSPECTION STATION

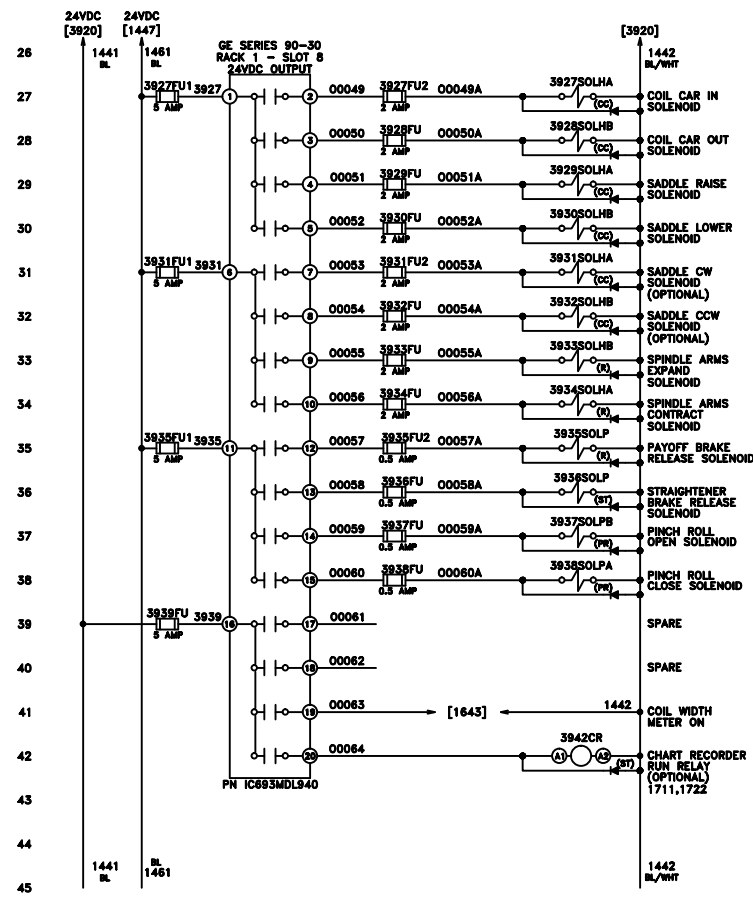
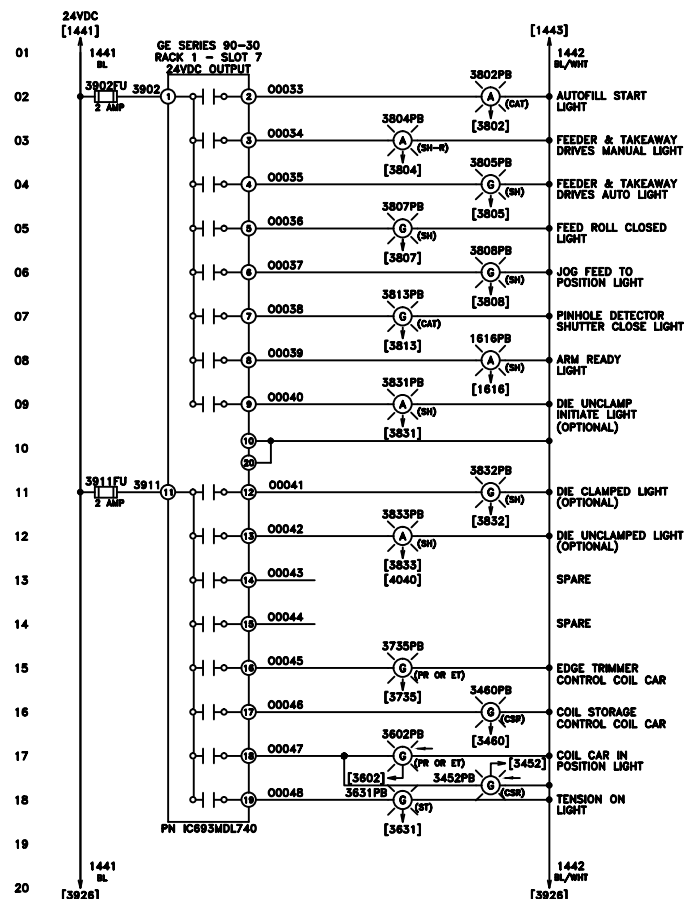
NOTE:

△ JUMPER 1441 TO IO112 IF FEED MOTOR THERMAL SWITCH IS NOT SUPPLIED

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 38 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DIBUJO:	IMPRESION:	HAB	
CONTROL:	FECHA:	11/5/83	
APROBADO:	FECHA:	11/5/83	

DATE	SYM	WAS	BY
		D-305501	

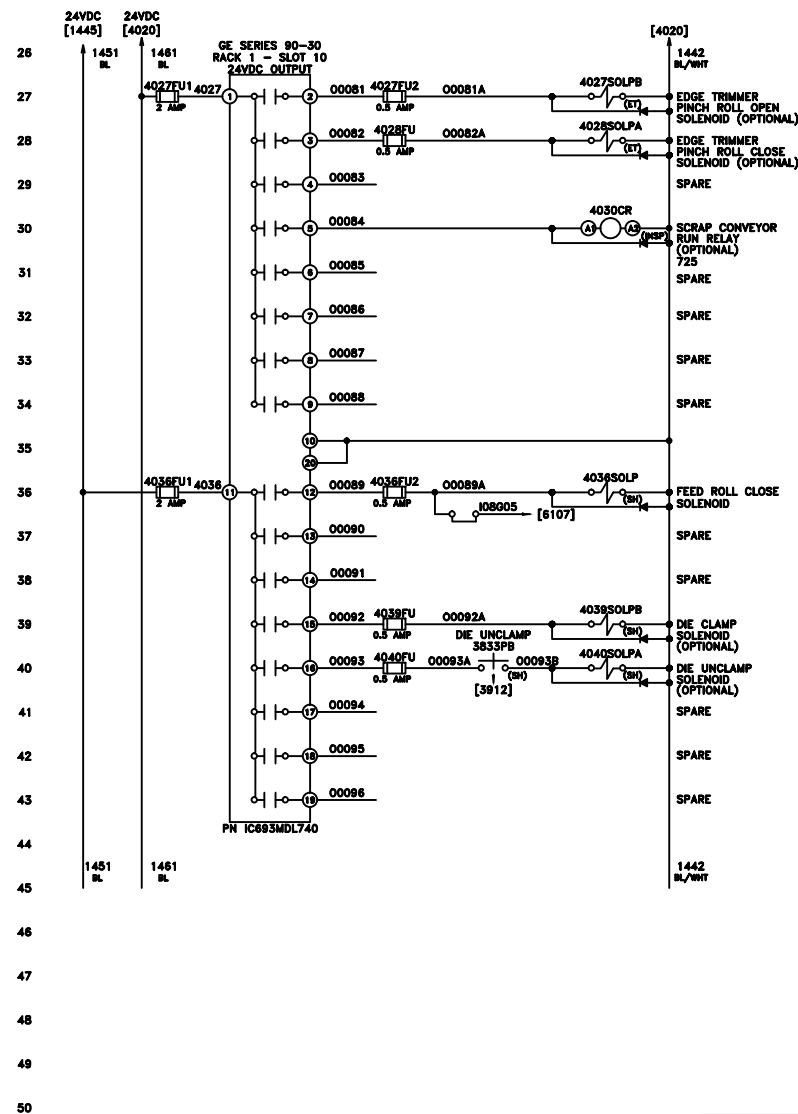
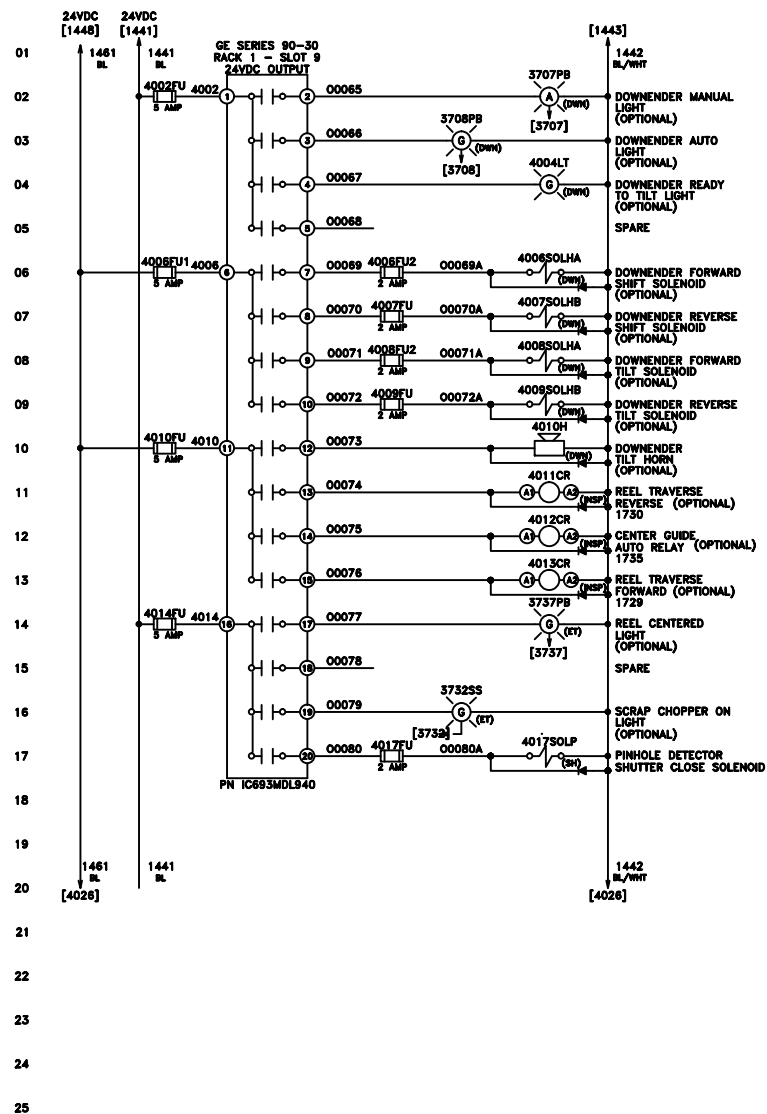


RACK 1 INSPECTION STATION

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 39 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PROYECTO: 5615-23-0028			
DISEÑO: 1152348C			
CONTROLADO: 1152348C			
APROBADO: 1152348C			

DATE	SYM	WAS	BY
		D-305501	

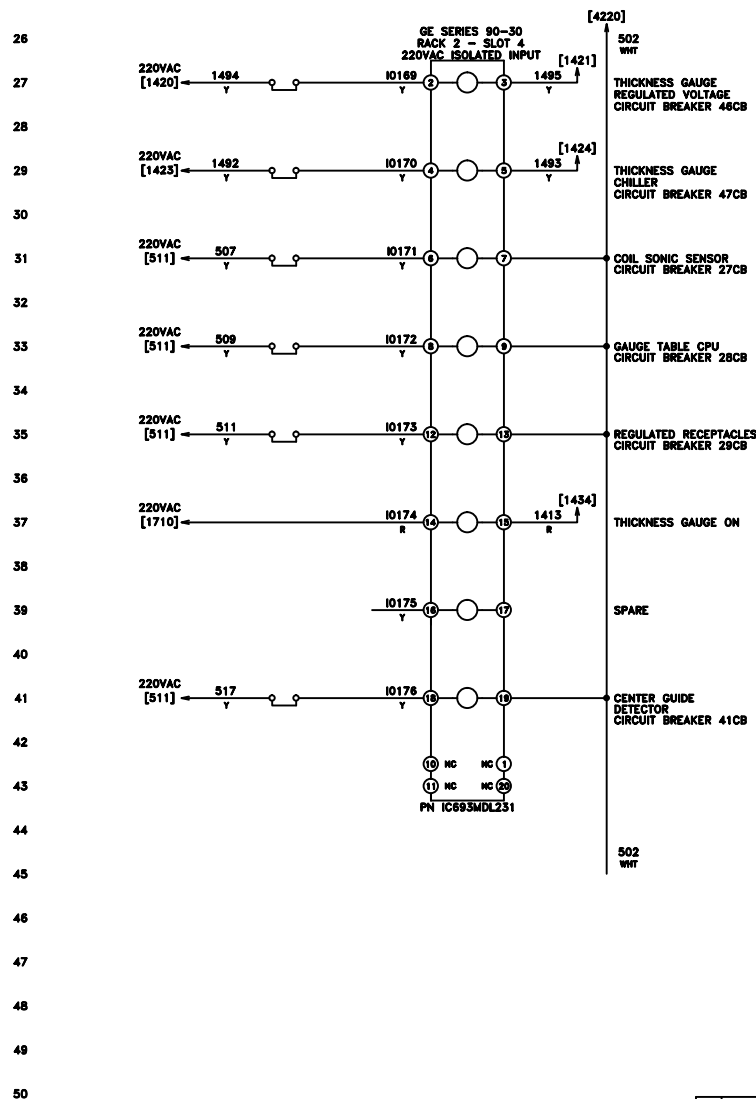
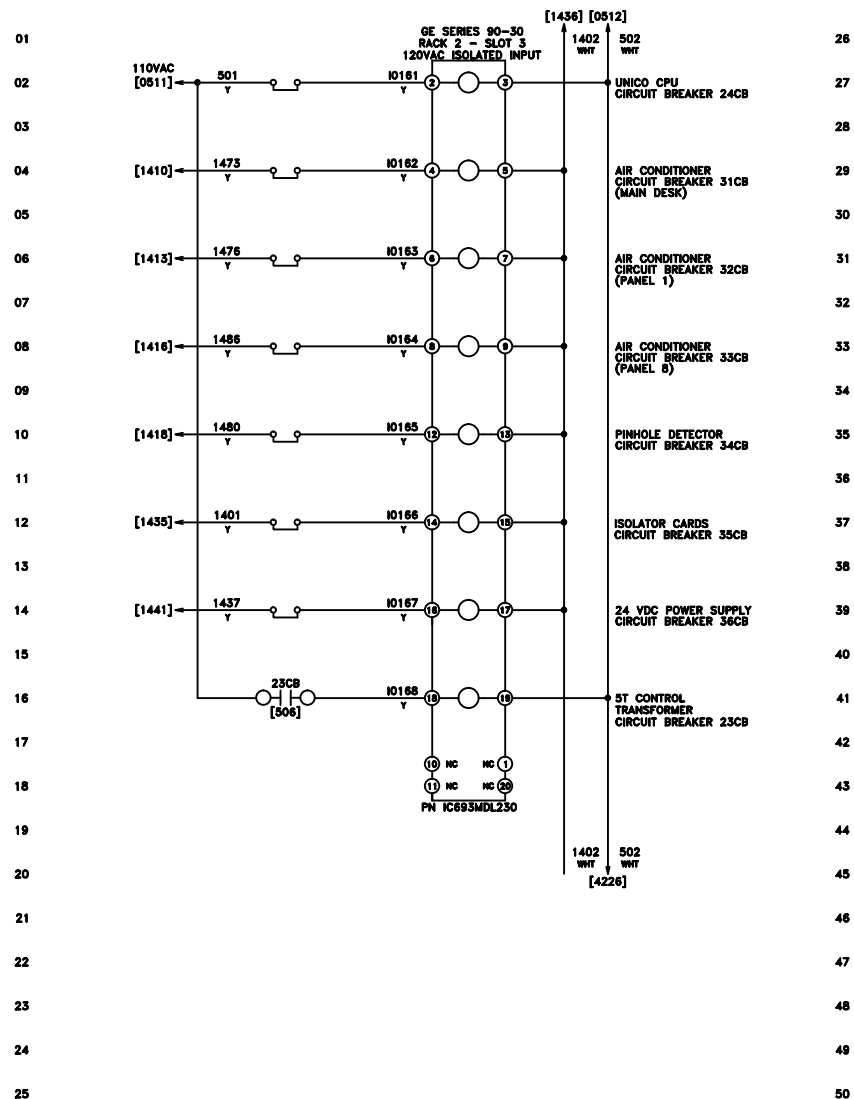


RACK 1 INSPECTION STATION

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APPRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROCROM		HOJA 40 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PROYECTO: 5615-23-0028			
DISEÑO: 1152H040.DWG			
CONTROLADO: 1152348C			

DATE	SYM	WAS	BY
		D-305501	

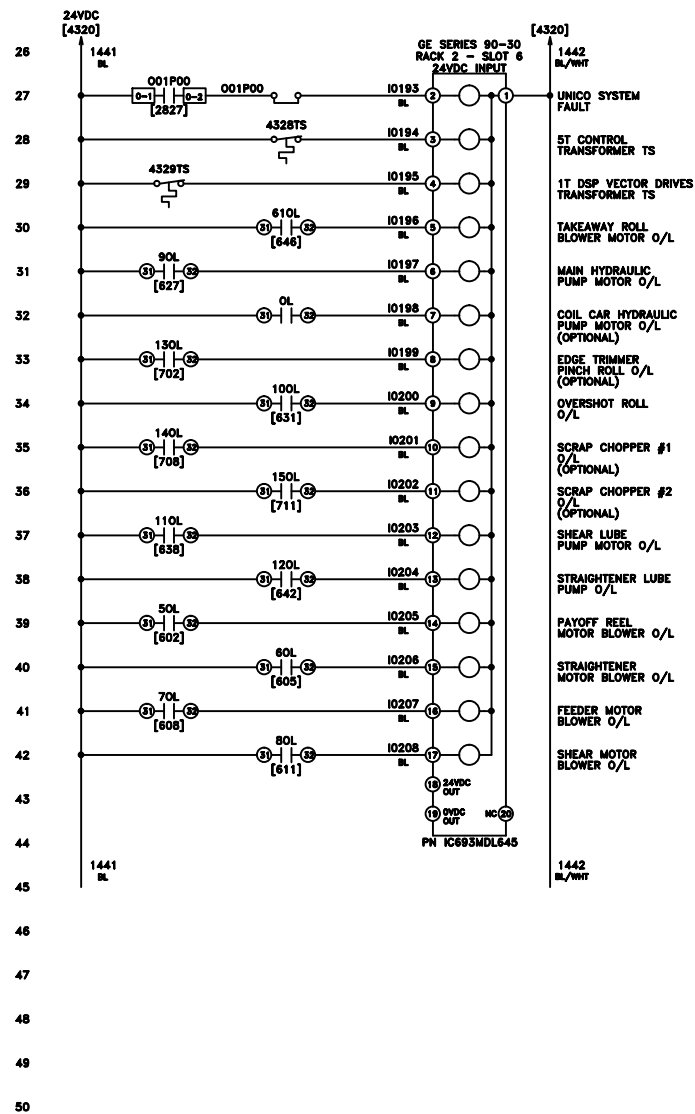
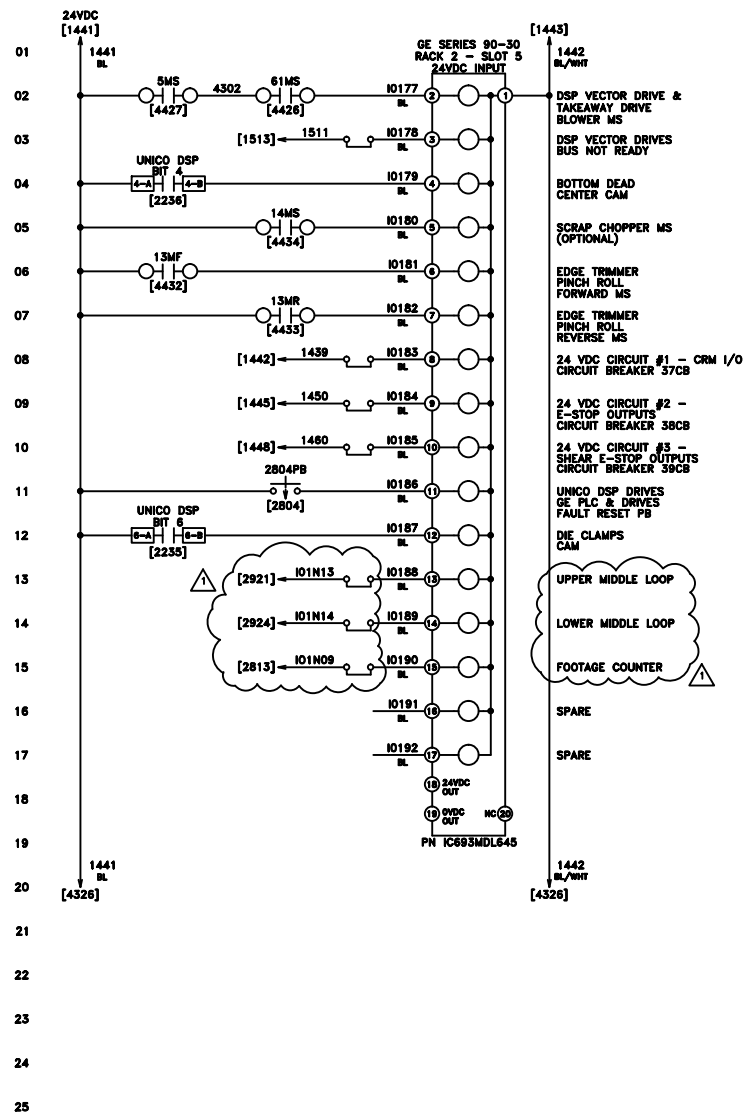


**RACK 2
MAIN CONTROL PANEL
(DOOR 4)**

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 42 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	FECHA:	11/04/82	
APROBADO:	FILED:	11/04/82	

DATE	SYM	WAS	BY
		D-305501	

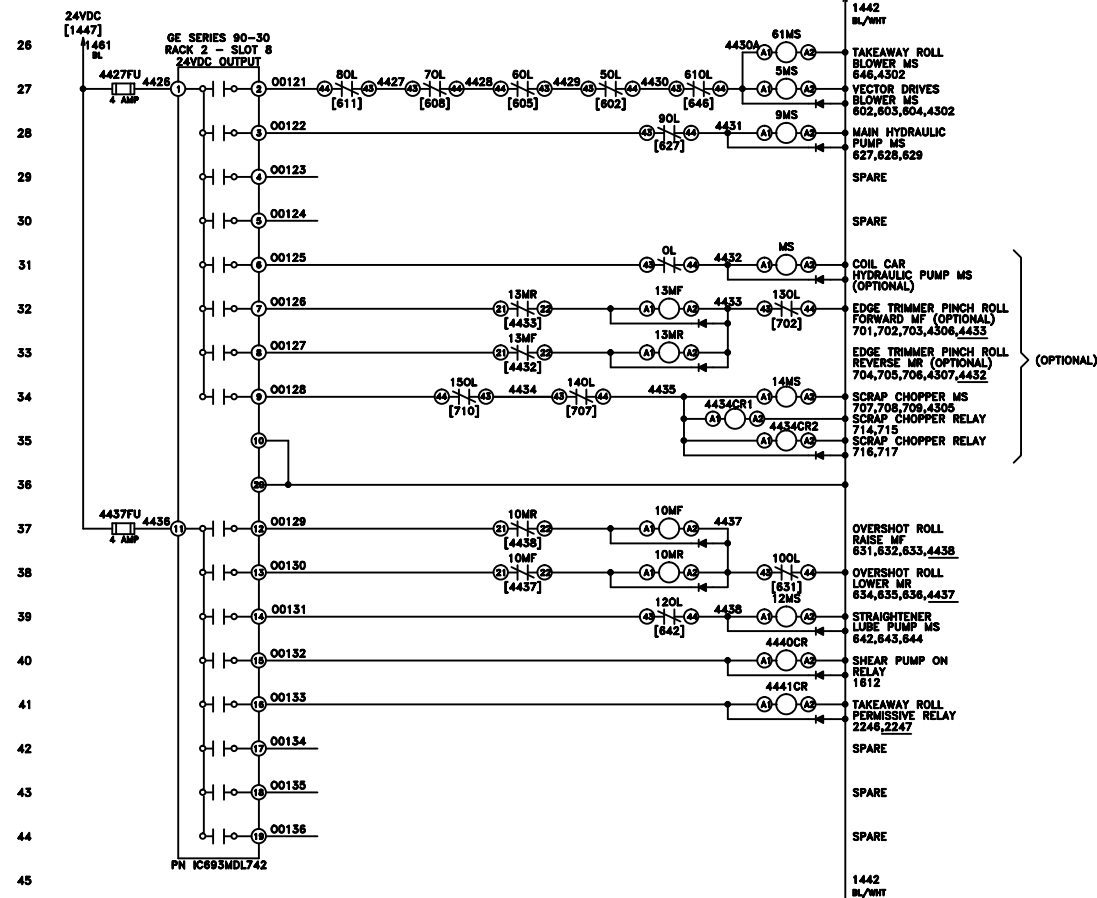
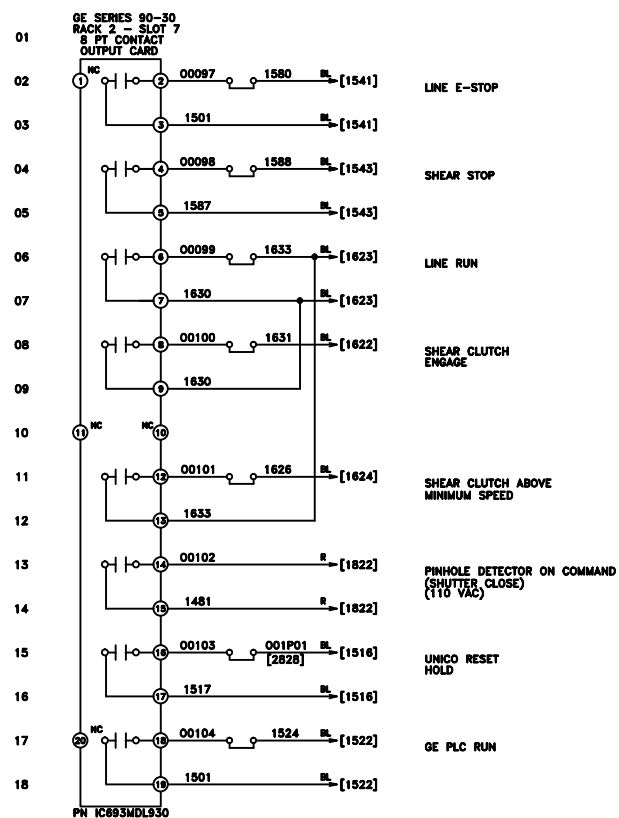


RACK 2 MAIN CONTROL PANEL (DOOR 4)

LITTELL JOB #94228 LINE (R → L)

T SE AGREGAN PUENTES 101N13/0186, 101N14/0189, 101N09/0190 DIVE DIVE DNP 03-08-01	
O EMISION PARA CONSTRUCCION 12-06-01	
REV.	DESCRIPCION
PROVEEDOR:	IT PROVEEDOR:
LITTELL INTERNATIONAL INC.	
ZONA:	SIDERAR
SECTOR:	LINEA LITTELL
OBRA O EQUIPO:	LINEA DE CORTE - BR-4E
PARTE:	SCROLL SHEETING LINE
PROYECTO:	DIAGRAM-ELECTRICAL-SCHEMATIC
DISEÑO:	5615-23-0028
CONTROL:	
APROB:	

DATE	SYM	WAS	BY
		D-305501	



NOTES:

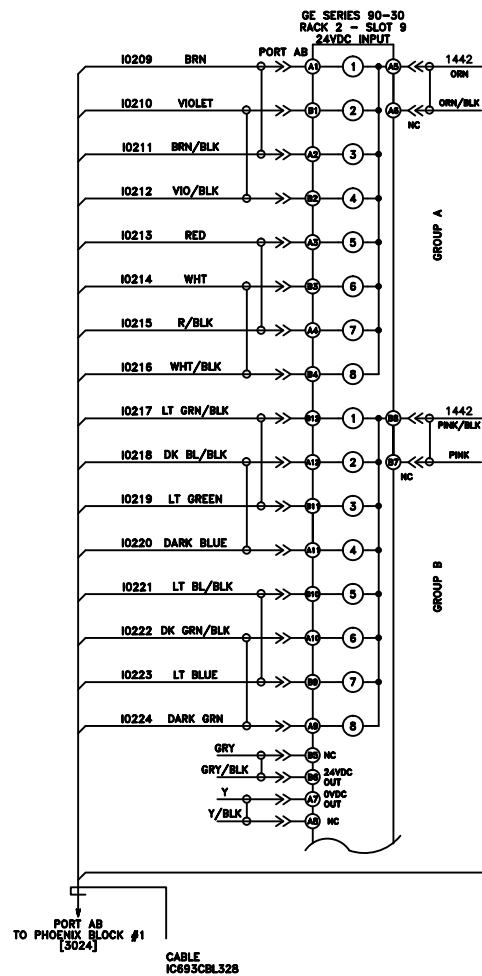
- * - MODULE IS FUSED INTERNALLY
- + - PROVIDE JUMPER ACROSS 8290L WITHOUT FEEDER

RACK 2
MAIN CONTROL PANEL
(DOOR 4)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HOJA 44 DE 67	
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
SCROLL SHEETING LINE		PARTES:	
DIAGRAM - ELECTRICAL - SCHEMATIC		ESCALAS:	
PROYECTO:		5615-23-0028	
DISEÑO:		IMPRESION:	
CONTROL:		HAB	
APROBADO:		CAD: T152H044.DWG FILE: T152H044.DWG	

DATE	SYM	WAS	BY
		D-305501	



PAYOFF DRIVE ON GRANT

PAYOFF DRIVE MANUAL (DSW)

PAYOFF DRIVE AUTO (DSW)

PAYOFF TENSION ON (DSW)

PAYOFF DRIVE JOG FORWARD (DSW)

PAYOFF DRIVE JOG REVERSE (DSW)

STRAIGHTENER DRIVE ON GRANT

STRAIGHTENER DRIVE MANUAL (DSW)

STRAIGHTENER DRIVE AUTO (DSW)

STRAIGHTENER JOG FORWARD (DSW)

STRAIGHTENER JOG REVERSE (DSW)

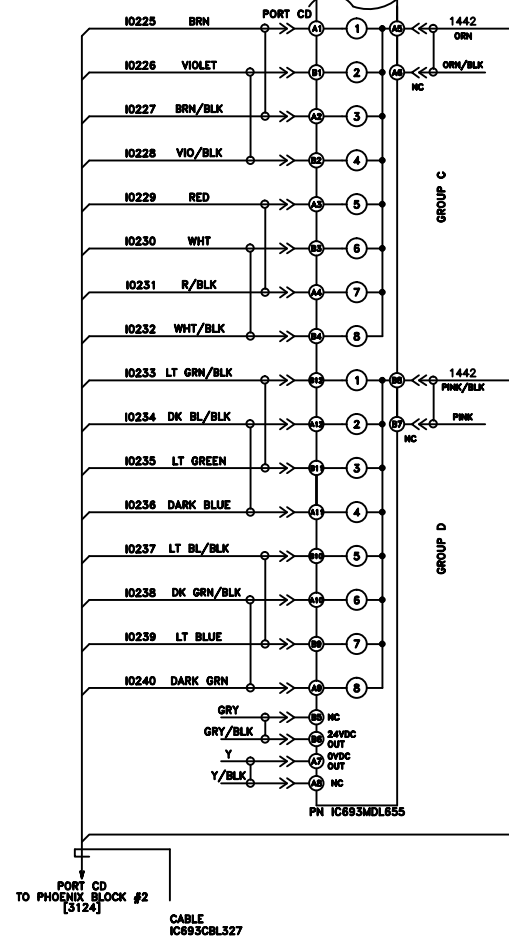
STRAIGHTENER BELOW MIN SPEED (DSW)

STRAIGHTENER FORWARD (DSW)

SPARE

SPARE

SPARE



FEEDER & TAKEAWAY DRIVES ON GRANT

FEEDER & TAKEAWAY DRIVES MANUAL (DSW)

FEEDER & TAKEAWAY DRIVES AUTO (DSW)

FEEDER IN MOTION (DSW)

TAKEAWAY IN MOTION (DSW)

JOG FEED TO POSITION GRANT

SHEAR DRIVE ON GRANT

SHEAR REFERENCED (DSW)

SHEAR DRIVE MIN SPEED (DSW)

SHEAR DRIVE ABOVE MIN SPEED (DSW)

SHEAR ENGAGED (DSW)

BUCKLE DETECT CPU RUN

BUCKLE FAULT

MISFEED BUCKLE-BEFORE SHEAR

REMOVED BUCKLE-AFTER SHEAR

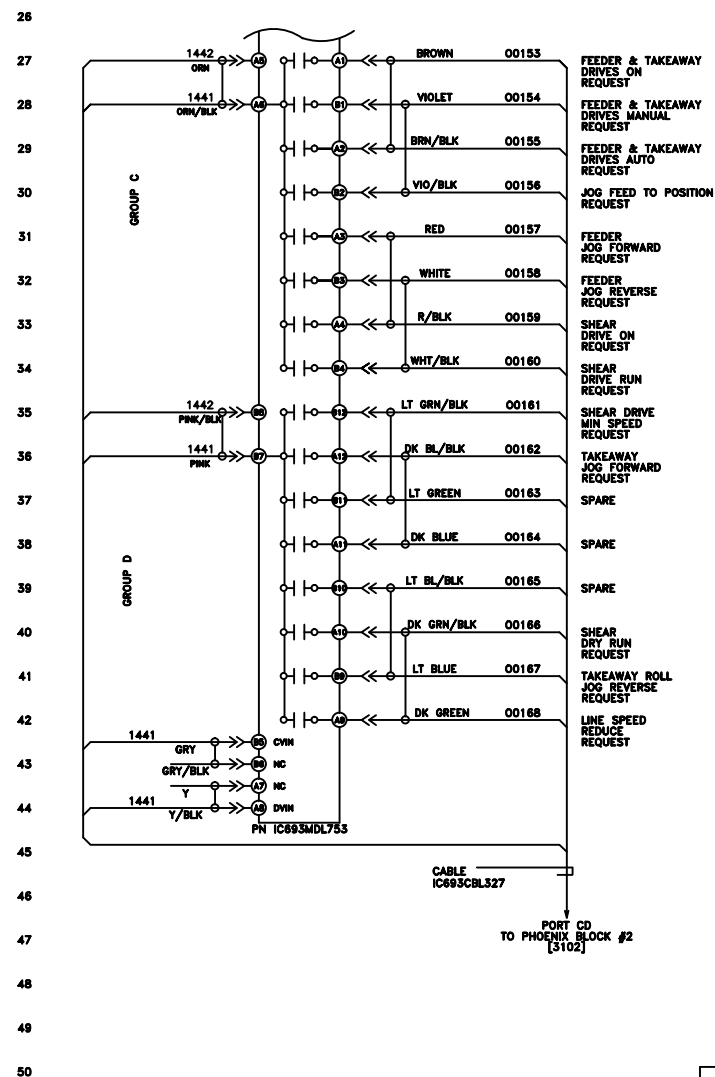
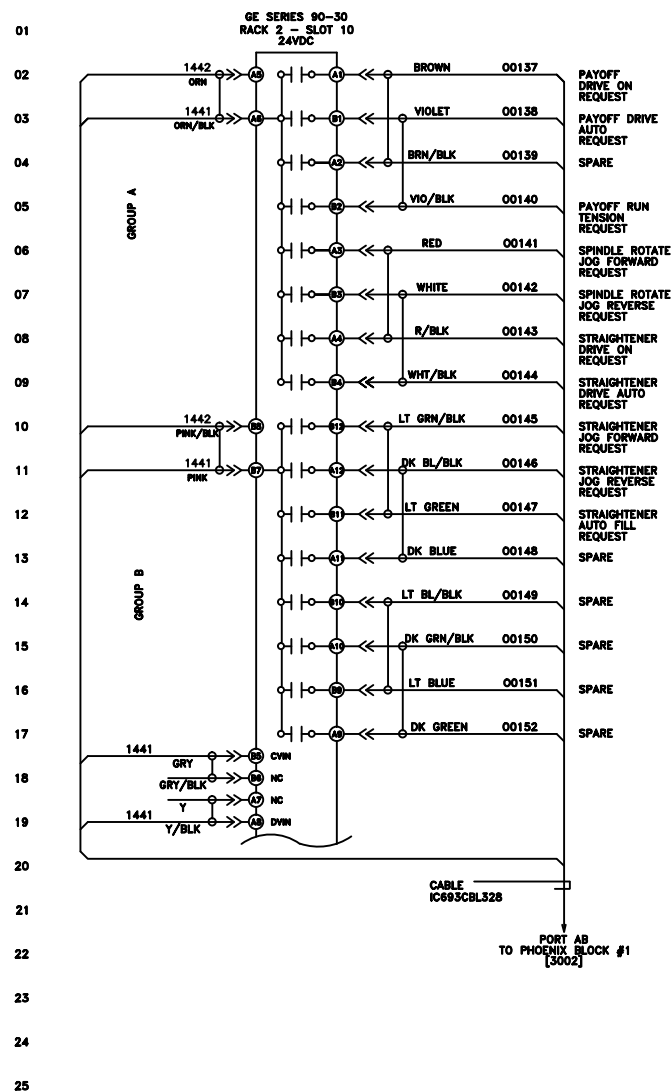
BUCKLE DETECT CAM FAULT

RACK 2
MAIN CONTROL PANEL
(DOOR 4)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 45 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROBADO:			
IMPRESION: HAS		CABLE: T152045.DWG	
		FILE: T152348C	

DATE	SYM	WAS	BY
		D-305501	

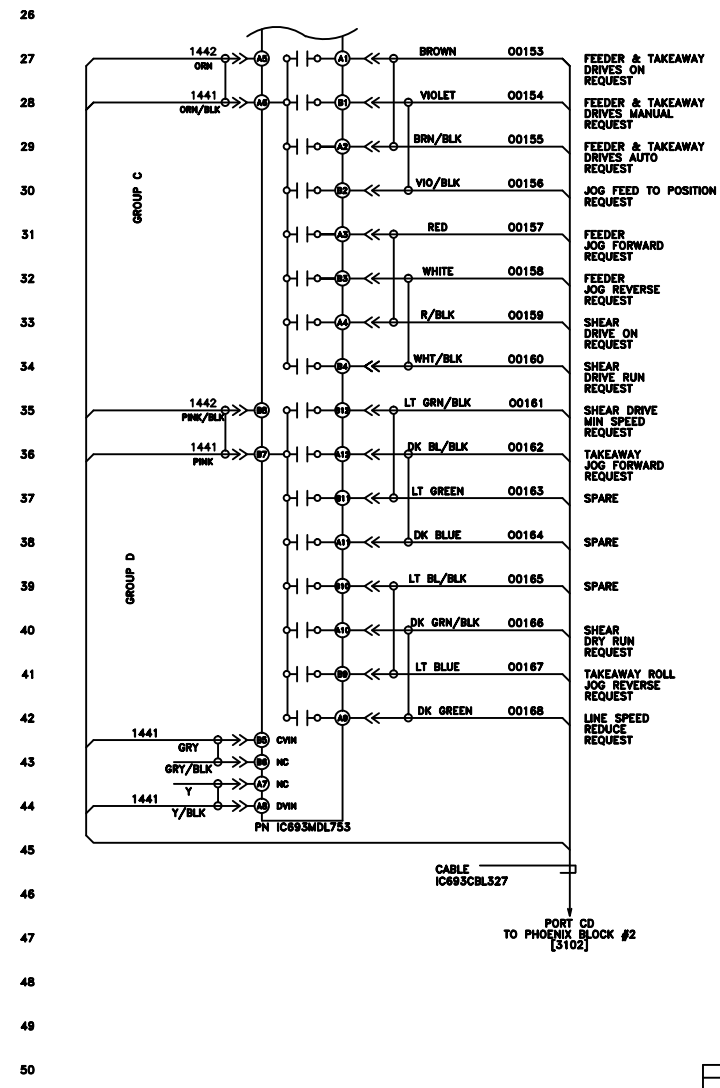
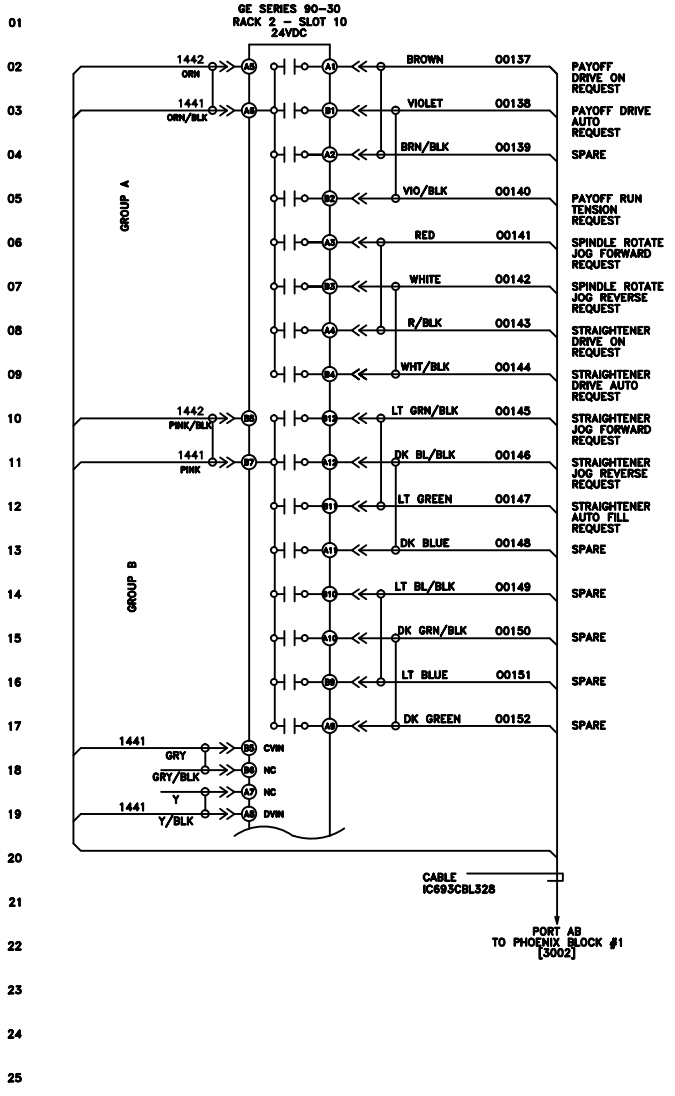


RACK 2
MAIN CONTROL PANEL
(DOOR 4)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERCRON		HOJA 48 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DIBUJO:			
CONTROL:			
APROB:			

DATE	SYM	WAS	BY
		D-305501	

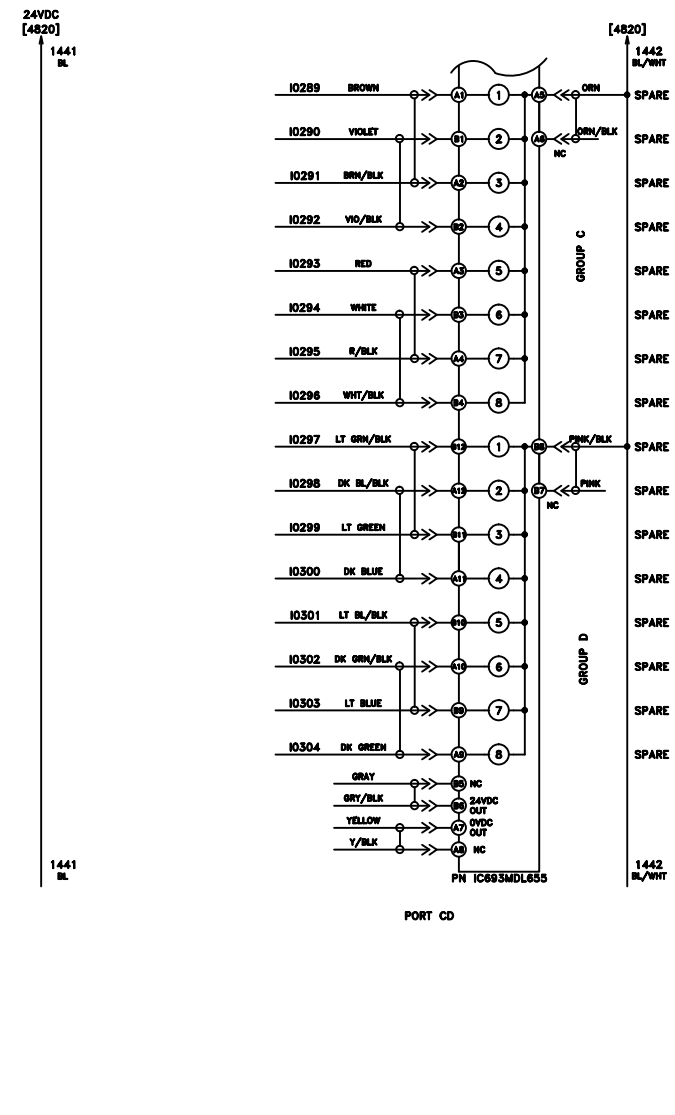
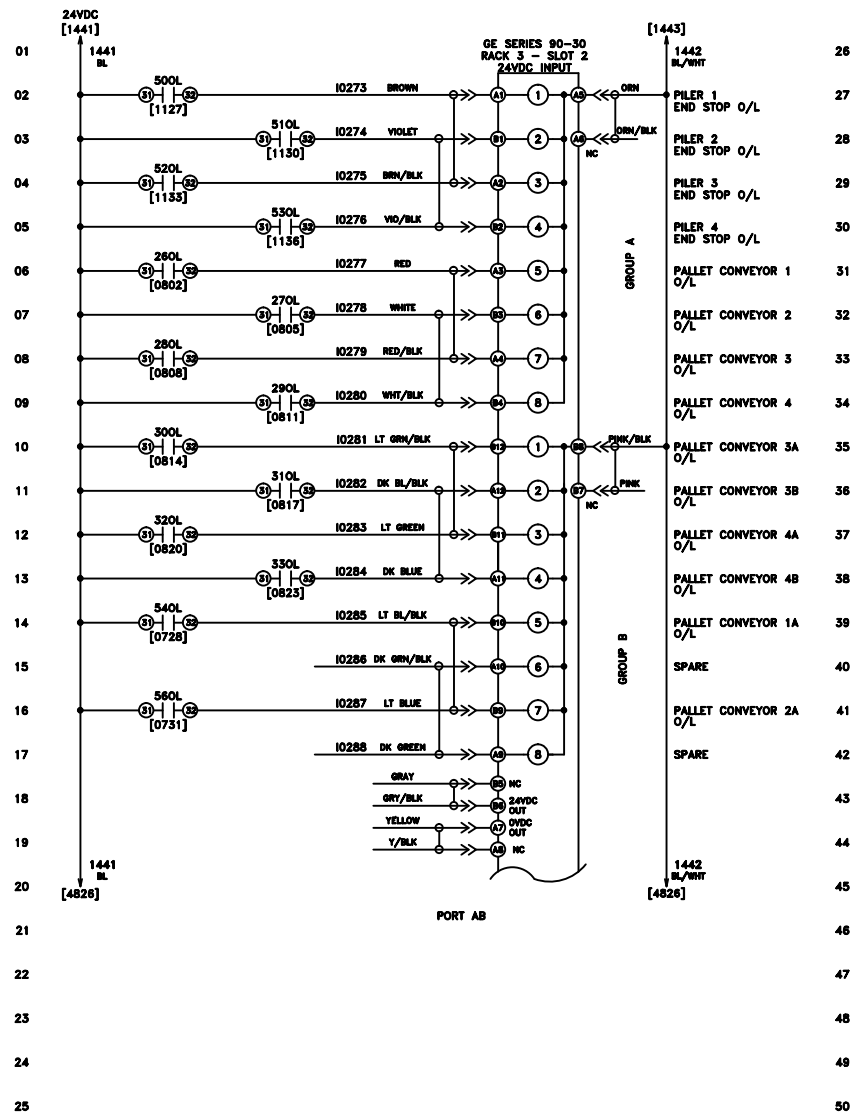


**RACK 2
MAIN CONTROL PANEL
(DOOR 4)**

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691	
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DISEÑO:		ESCALAS	
CONTROL:		5615-23-0028	
APROB:		IMPULSION: H43	
		CAB: T152H047.DWG	
		FILE: T152H48C	

DATE	SYM	WAS	BY
		D-305501	

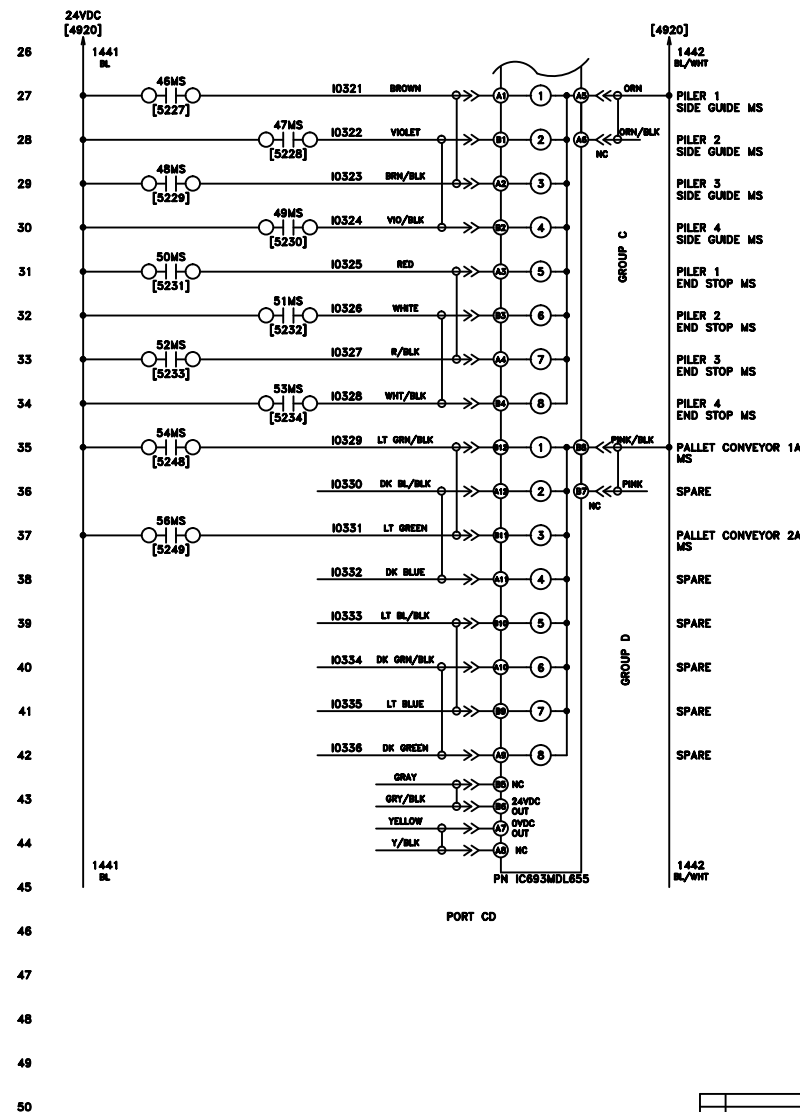
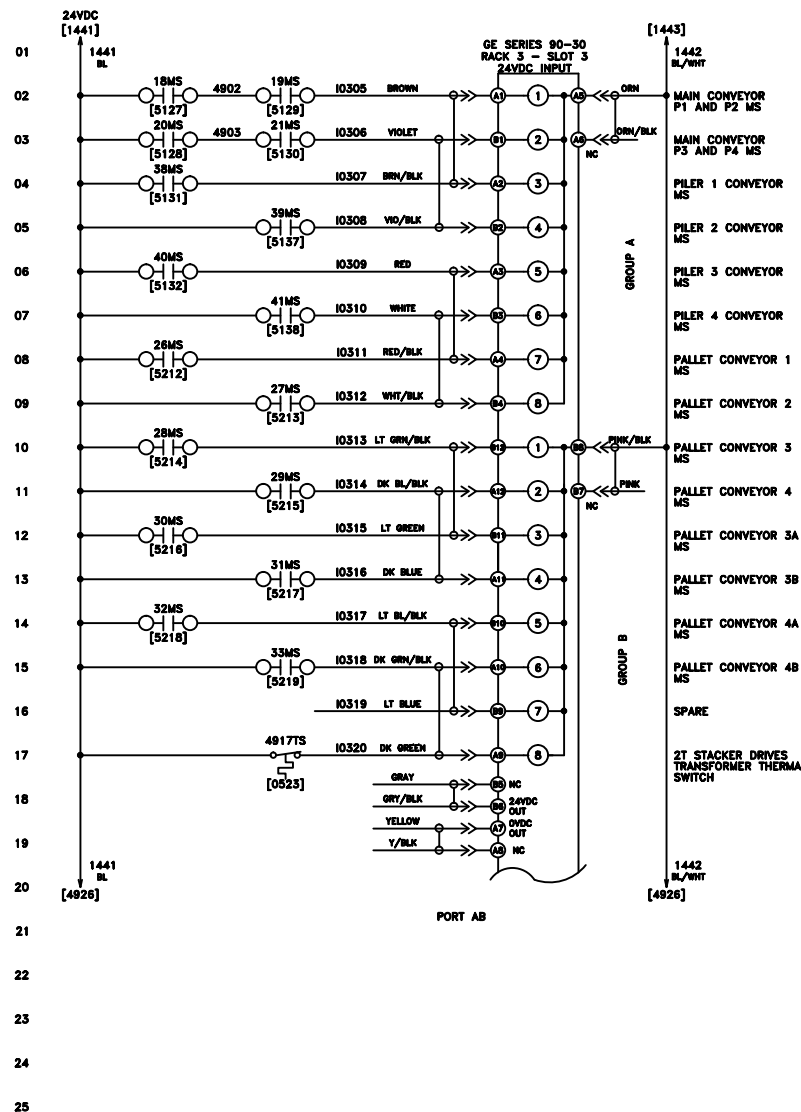


**RACK 3
MAIN CONTROL PANEL
(DOOR 8)**

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-08-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HOJA 48 DE 67	
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DISEÑO:		ESCALAS	
CONTROL:		5615-23-0028	
APROBADO:		IMPULSION: HAS	
		CARG: T152048.DWG	
		FILE: T152348C	

DATE	SYM	WAS	BY
		D-305501	

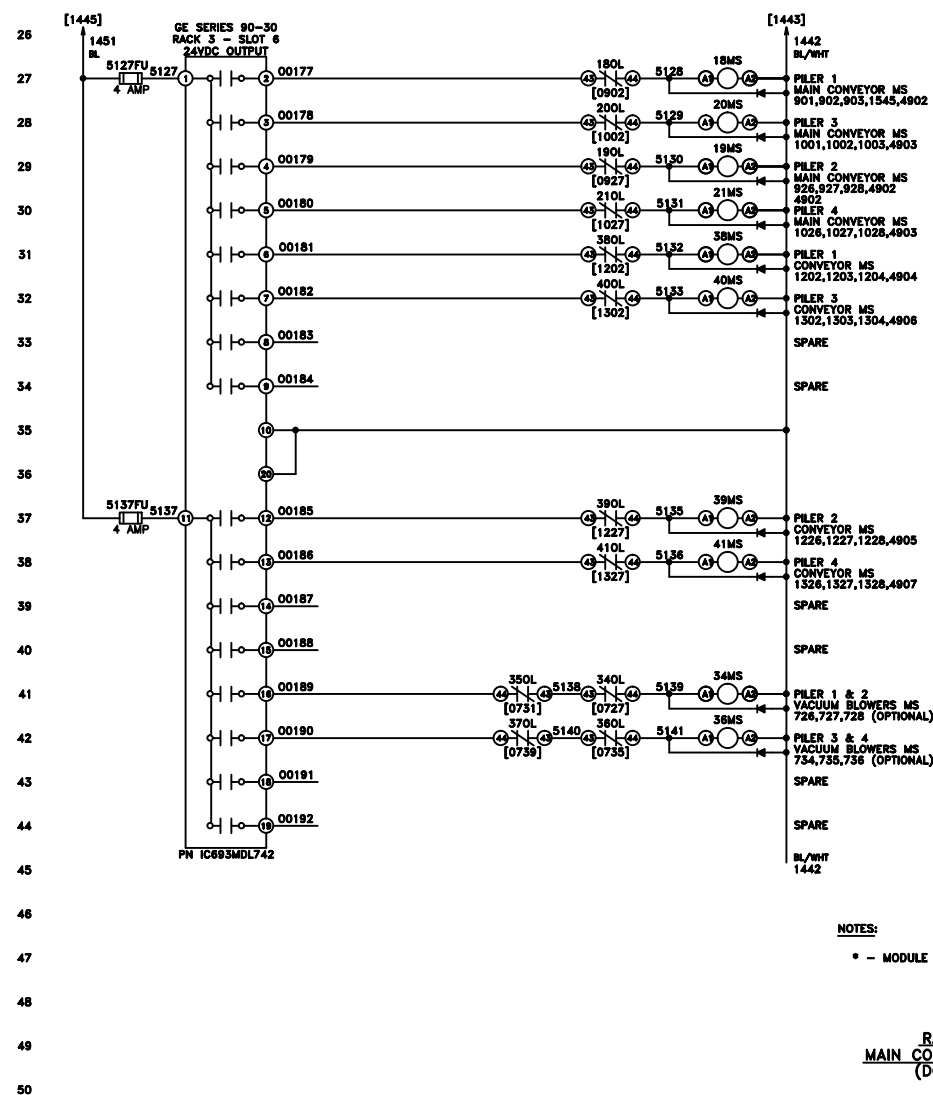
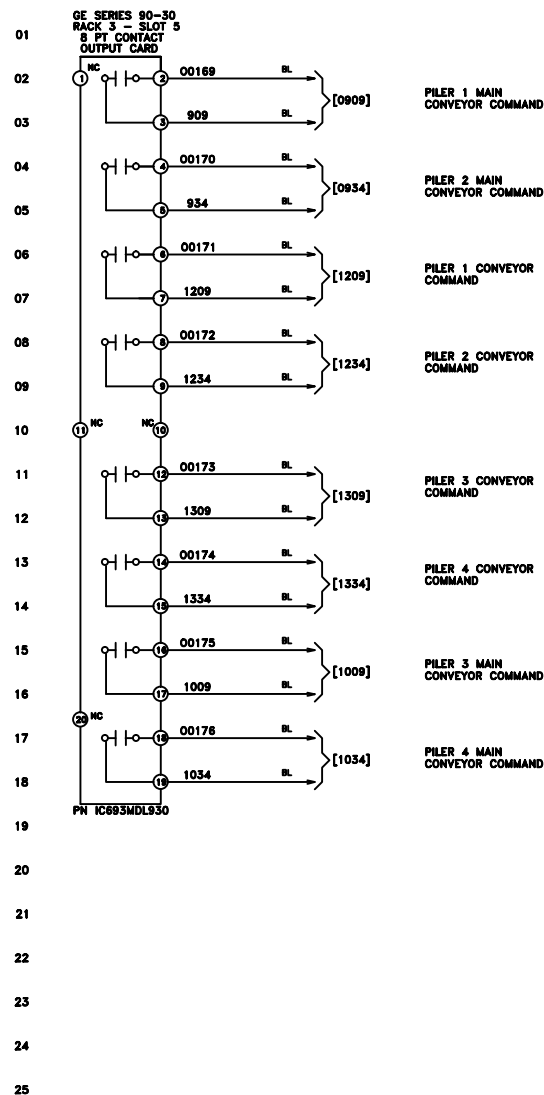


RACK 3 MAIN CONTROL PANEL (DOOR 8)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	IT PROVEEDOR:		
LITTELL INTERNATIONAL INC.		D-308691 HOJA 48 DE 67	
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DISEÑO:		ESCALAS	
CONTROL:		5615-23-0028	
APROBADO:		IMPRESION: HAS	
		CARG: T152H049.DWG	
		FILE: T152348C	

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



NOTES:

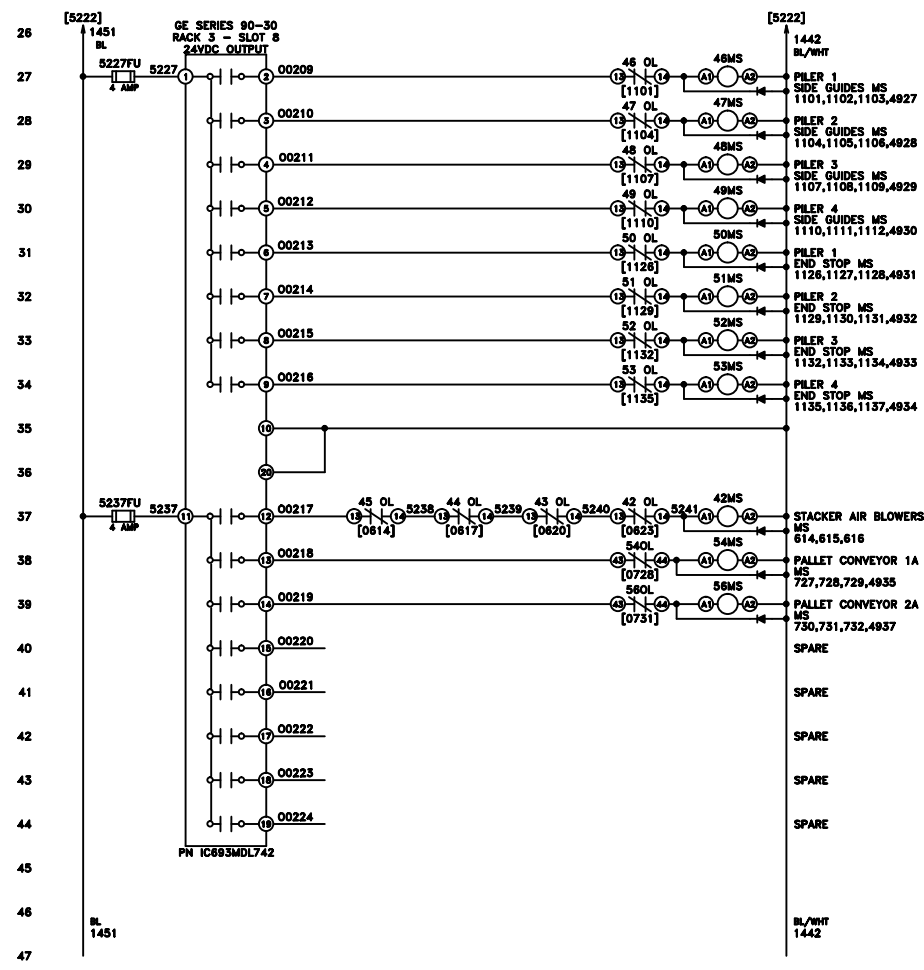
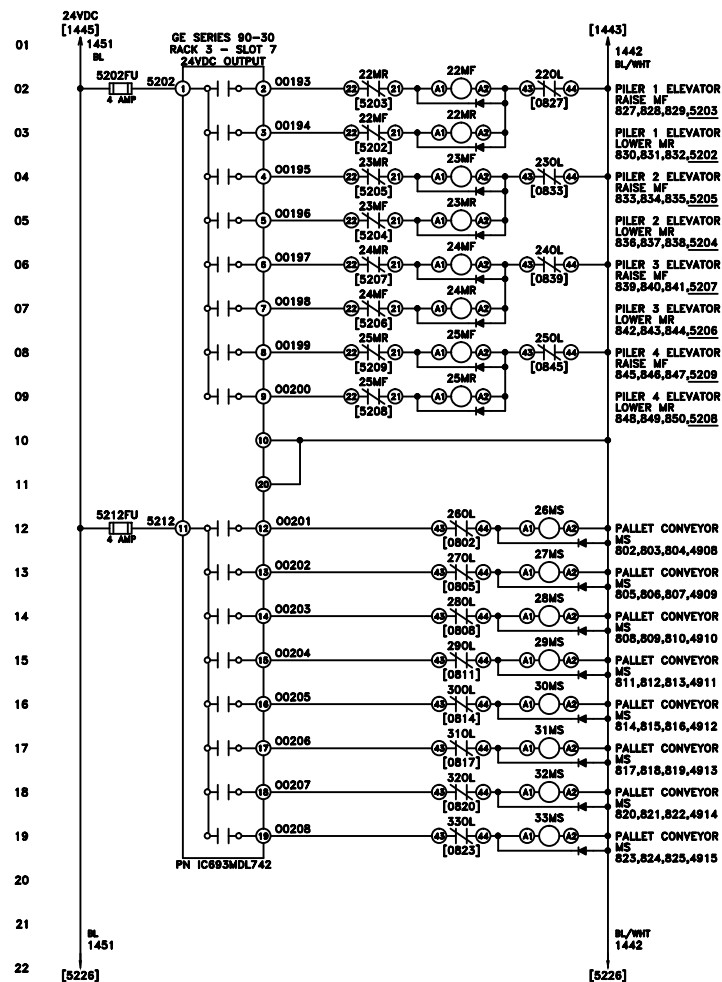
* - MODULE IS FUSED INTERNALLY

RACK 3
MAIN CONTROL PANEL
(DOOR 8)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERCRON		HOJA 51 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAMA-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

DATE	SYM	WAS	BY
		D-305501	

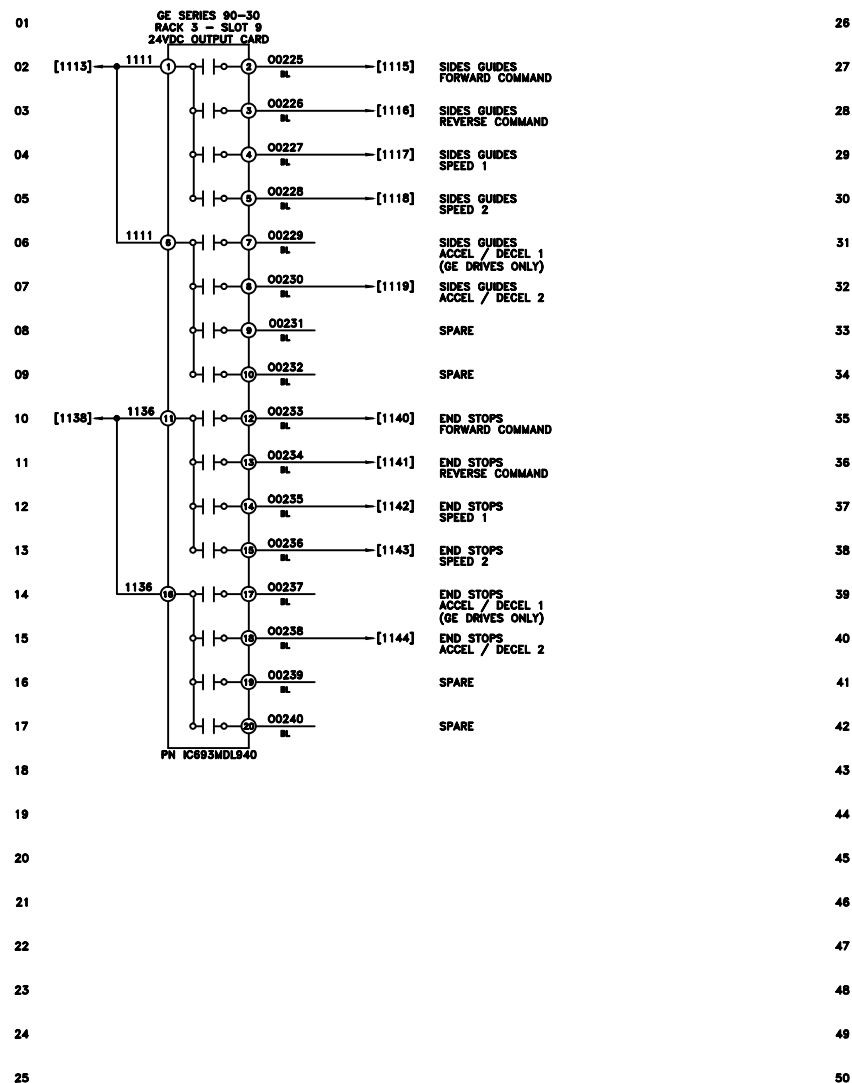


RACK 3 MAIN CONTROL PANEL (DOOR 8)

LITTELL JOB #94228 LINE (R -> L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IF PROVEEDOR:	D-308691 HOJA 52 DE 67
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE - BR-4E	
OBRA O EQUIPO:		SCROLL SHEETING LINE	
PARTES:		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROBADO:			

DATE	SYM	WAS	BY
		D-305501	



**RACK 3
MAIN CONTROL PANEL
(DOOR 8)**

LITTELL JOB #94228 LINE (R → L)

0 EMISION PARA CONSTRUCCION		12-08-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	Nº PROVEEDOR:	D-308691
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DIBUJO	ESCALAS	5615-23-0028	
CONTROL	IMPRESION:	HAB	
APROB	FECHA:	11/03/81	

24VDC [1441] BL

GE SERIES 90-30
RACK 4-SLOT 1
24VDC INPUT CARD

[1443] BL/WHT

01 5402PB1 (P1F) 5402PB2 (P1B) 10369 2

02 5403PB1 (P1F) 5403PB2 (P1B) 10370 3

03 5404LS (P1) 5405SS1 (P1F) 5405SS2 (P1B) 10371 4

04 5407LS (P1) 5408LS (P1) 10372 5

05 5409PB1 (P2F) 5409PB2 (P2B) 10373 6

06 5410PB1 (P2F) 5410PB2 (P2B) 10374 7

07 5411PB1 (P2F) 5411PB2 (P2B) 10375 8

08 5412LS (P2) 5413SS1 (P2F) 5413SS2 (P2B) 10376 9

09 5415LS (P2) 5416LS (P2) 10377 10

10 5417PB1 (P2F) 5417PB2 (P2B) 10378 11

11 24VDC OUT 10379 12

12 0VDC OUT 10380 13

13 24VDC OUT 10381 14

14 0VDC OUT 10382 15

15 24VDC OUT 10383 16

16 0VDC OUT 10384 17

17 24VDC OUT 10385 18

18 0VDC OUT 10386 19

19 24VDC OUT 10387 20

20 0VDC OUT 10388 21

21 24VDC OUT 10389 22

22 0VDC OUT 10390 23

23 24VDC OUT 10391 24

24 0VDC OUT 10392 25

25 24VDC OUT 10393 26

26 0VDC OUT 10394 27

27 24VDC OUT 10395 28

28 0VDC OUT 10396 29

29 24VDC OUT 10397 30

30 0VDC OUT 10398 31

31 24VDC OUT 10399 32

32 0VDC OUT 10400 33

33 24VDC OUT 10401 34

34 0VDC OUT 10402 35

35 24VDC OUT 10403 36

36 0VDC OUT 10404 37

37 24VDC OUT 10405 38

38 0VDC OUT 10406 39

39 24VDC OUT 10407 40

40 0VDC OUT 10408 41

41 24VDC OUT 10409 42

42 0VDC OUT 10410 43

43 24VDC OUT 10411 44

44 0VDC OUT 10412 45

45 24VDC OUT 10413 46

46 0VDC OUT 10414 47

47 24VDC OUT 10415 48

48 0VDC OUT 10416 49

49 24VDC OUT 10417 50

50 0VDC OUT 10418 51

51 24VDC OUT 10419 52

52 0VDC OUT 10420 53

53 24VDC OUT 10421 54

54 0VDC OUT 10422 55

55 24VDC OUT 10423 56

56 0VDC OUT 10424 57

57 24VDC OUT 10425 58

58 0VDC OUT 10426 59

59 24VDC OUT 10427 60

60 0VDC OUT 10428 61

61 24VDC OUT 10429 62

62 0VDC OUT 10430 63

63 24VDC OUT 10431 64

64 0VDC OUT 10432 65

65 24VDC OUT 10433 66

66 0VDC OUT 10434 67

67 24VDC OUT 10435 68

68 0VDC OUT 10436 69

69 24VDC OUT 10437 70

70 0VDC OUT 10438 71

71 24VDC OUT 10439 72

72 0VDC OUT 10440 73

73 24VDC OUT 10441 74

74 0VDC OUT 10442 75

75 24VDC OUT 10443 76

76 0VDC OUT 10444 77

77 24VDC OUT 10445 78

78 0VDC OUT 10446 79

79 24VDC OUT 10447 80

80 0VDC OUT 10448 81

81 24VDC OUT 10449 82

82 0VDC OUT 10450 83

83 24VDC OUT 10451 84

84 0VDC OUT 10452 85

85 24VDC OUT 10453 86

86 0VDC OUT 10454 87

87 24VDC OUT 10455 88

88 0VDC OUT 10456 89

89 24VDC OUT 10457 90

90 0VDC OUT 10458 91

91 24VDC OUT 10459 92

92 0VDC OUT 10460 93

93 24VDC OUT 10461 94

94 0VDC OUT 10462 95

95 24VDC OUT 10463 96

96 0VDC OUT 10464 97

97 24VDC OUT 10465 98

98 0VDC OUT 10466 99

99 24VDC OUT 10467 100

100 0VDC OUT 10468 101

101 24VDC OUT 10469 102

102 0VDC OUT 10470 103

103 24VDC OUT 10471 104

104 0VDC OUT 10472 105

105 24VDC OUT 10473 106

106 0VDC OUT 10474 107

107 24VDC OUT 10475 108

108 0VDC OUT 10476 109

109 24VDC OUT 10477 110

110 0VDC OUT 10478 111

111 24VDC OUT 10479 112

112 0VDC OUT 10480 113

113 24VDC OUT 10481 114

114 0VDC OUT 10482 115

115 24VDC OUT 10483 116

116 0VDC OUT 10484 117

117 24VDC OUT 10485 118

118 0VDC OUT 10486 119

119 24VDC OUT 10487 120

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121 24VDC OUT 10489 122

122 0VDC OUT 10490 123

123 24VDC OUT 10491 124

124 0VDC OUT 10492 125

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126 0VDC OUT 10494 127

127 24VDC OUT 10495 128

128 0VDC OUT 10496 129

129 24VDC OUT 10497 130

130 0VDC OUT 10498 131

131 24VDC OUT 10499 132

132 0VDC OUT 10500 133

133 24VDC OUT 10501 134

134 0VDC OUT 10502 135

135 24VDC OUT 10503 136

136 0VDC OUT 10504 137

137 24VDC OUT 10505 138

138 0VDC OUT 10506 139

139 24VDC OUT 10507 140

140 0VDC OUT 10508 141

141 24VDC OUT 10509 142

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145 24VDC OUT 10513 146

146 0VDC OUT 10514 147

147 24VDC OUT 10515 148

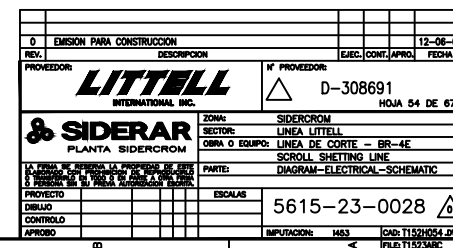
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149 24VDC OUT 10517 150

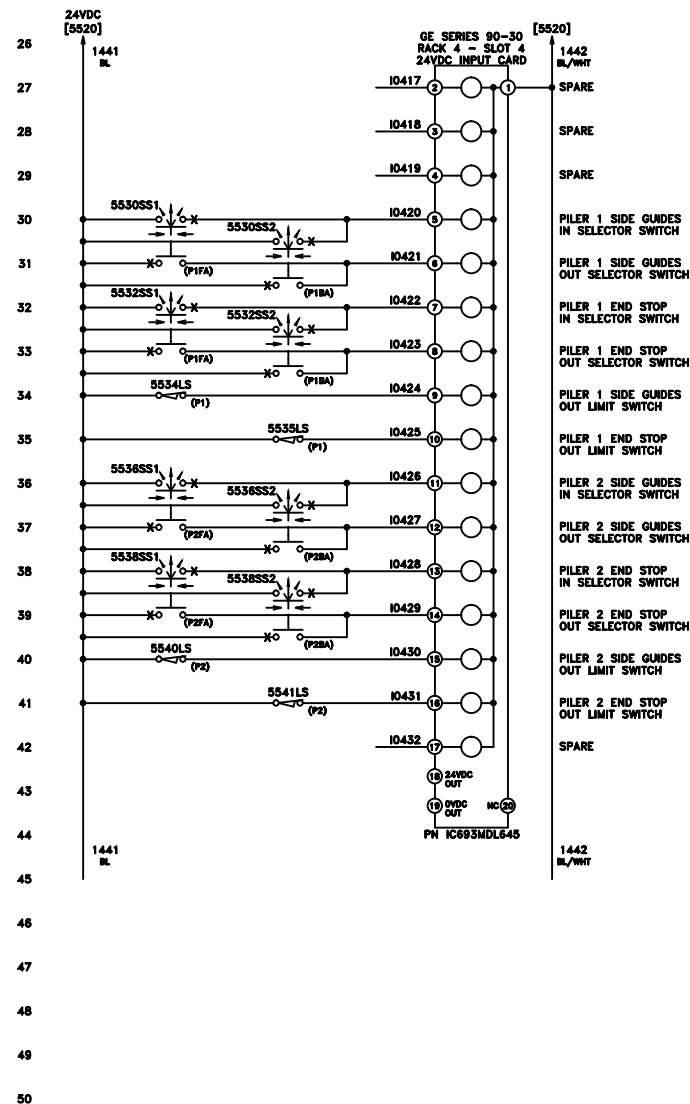
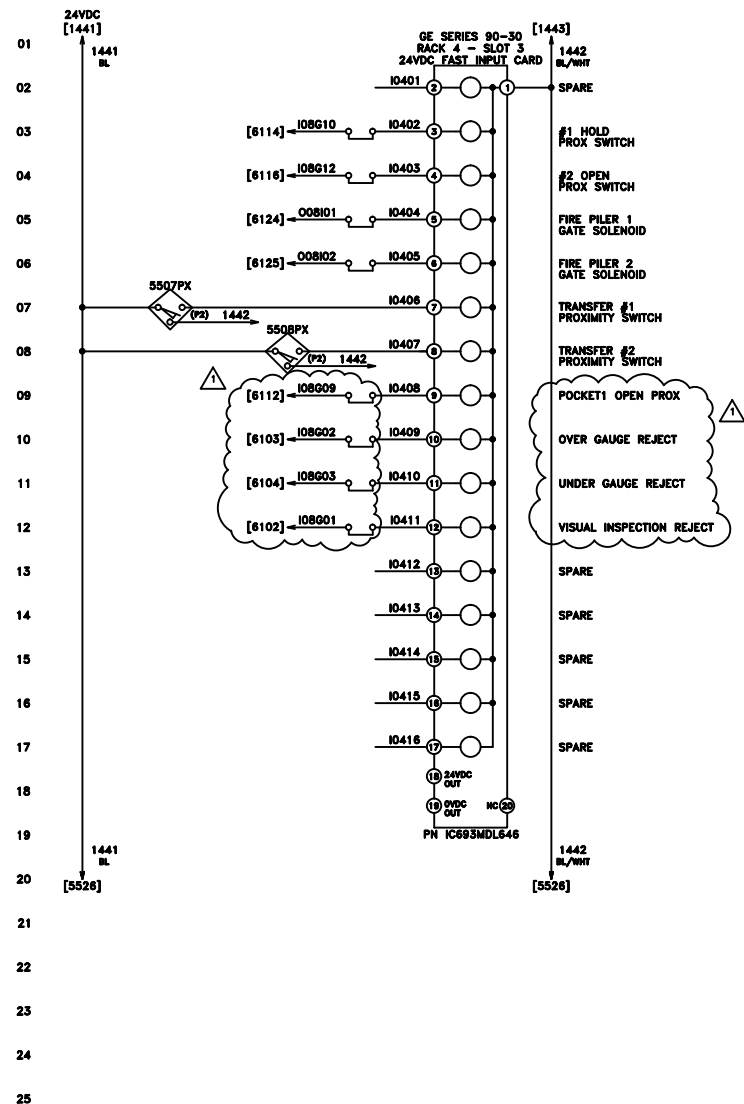
150 0VDC OUT 10518 151

151 24VDC OUT 10519 152

152 0VDC OUT 105



DATE	SYM	WAS	BY
		D-305501	



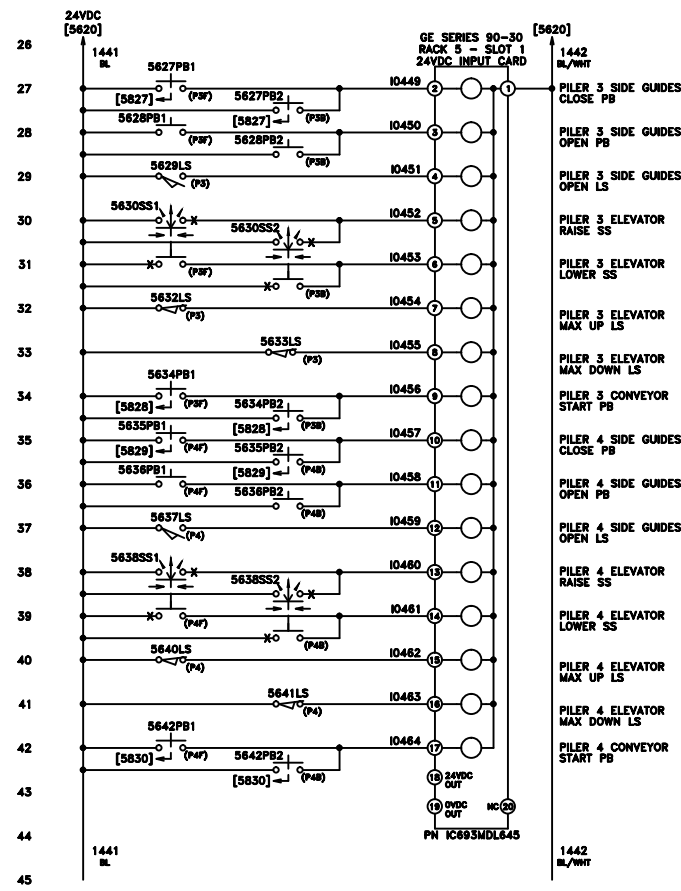
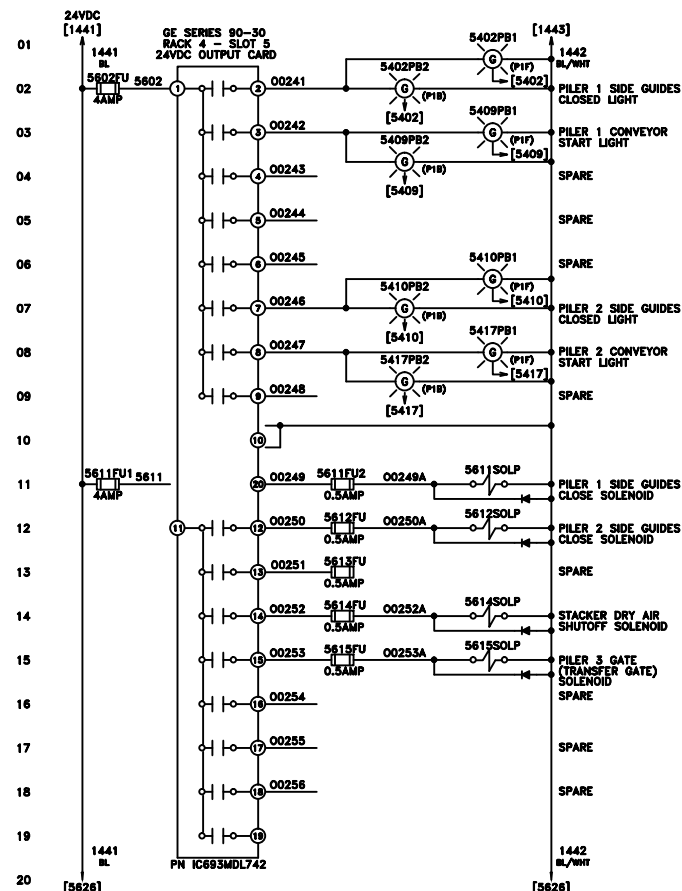
RACK 4
STACKER - PILER 1 & 2

1	PUNTES IOB609/IO406; IOB602/IO406; IOB603/IO410; IOB601/IO411	DNCE	DYCE	DNF	03-08-01
0	EMISION PARA CONSTRUCCION				12-06-01
REV.	DESCRIPCION	EJEC.	CONT.	APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691 HOJA 55 DE 67		
ZONA:		SIDERCROM			
SECTOR:		LINEA LITTELL			
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E			
PARTES:		SCROLL SHEETING LINE			
DIAGRAMA:		ELECTRICAL - SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028			
DISEÑO:					
CONTROL:					
APROB:					

IMPRESION: HAS3 CAD: T155H055.DWG FILE: T155348C

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	

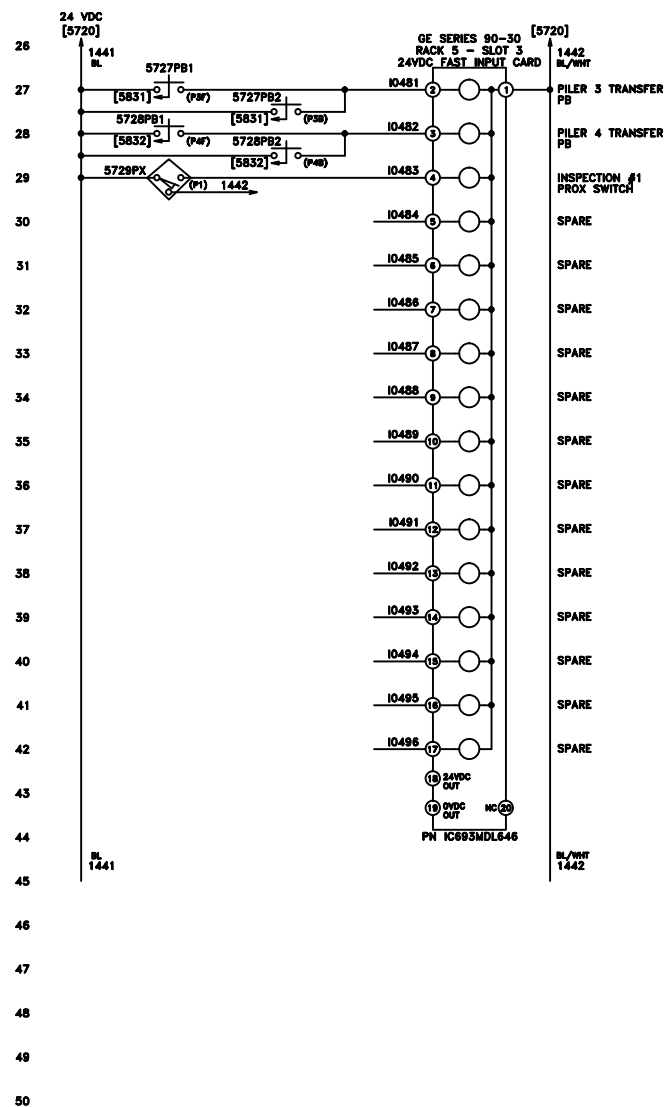
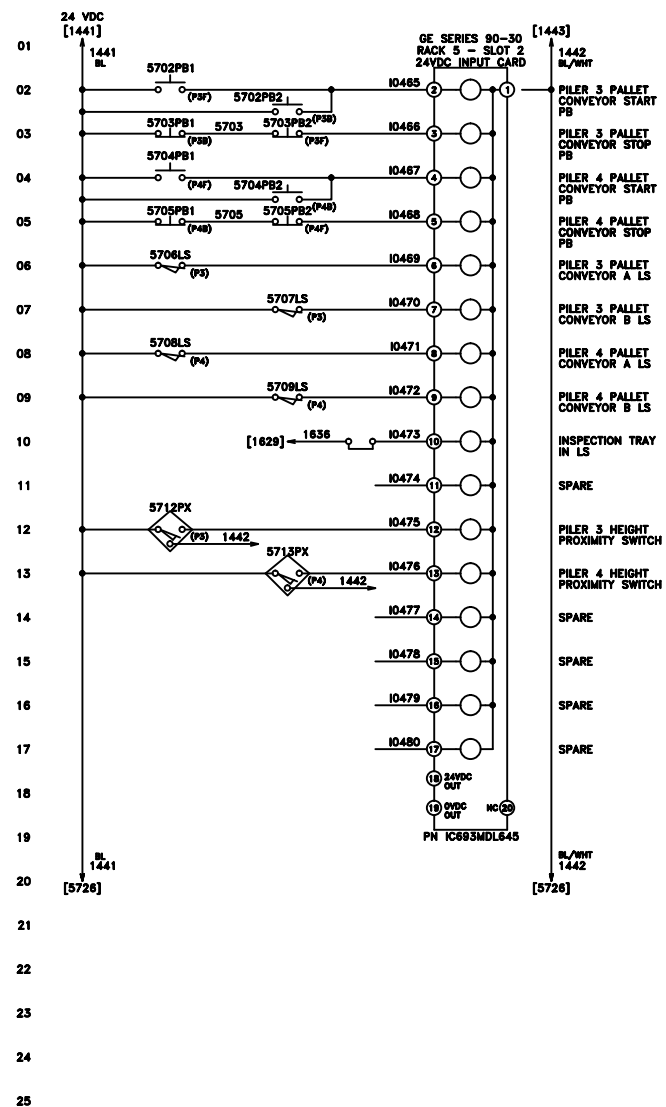


RACK 4 & 5
STACKER - PILER 1,2,3,4

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-91	
REV.	DESCRIPCION	EJEC. CONT. APROV.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 56 DE 67	
SECTOR: LINEA LITTELL		OBRAS O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM - ELECTRICAL - SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROBADO:			

DATE	SYM	WAS	BY
		D-305501	

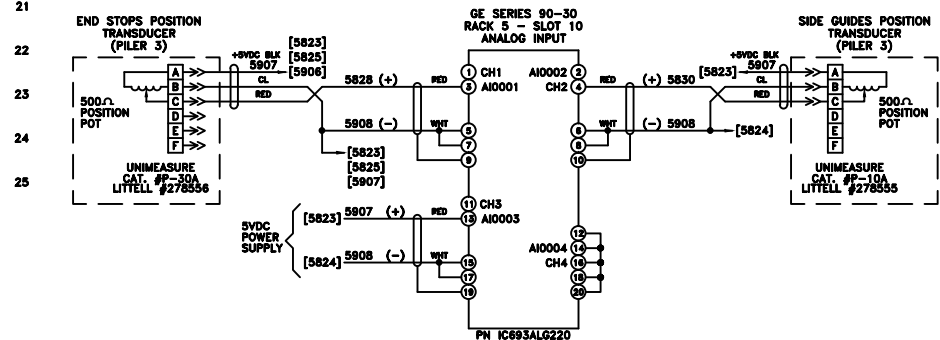
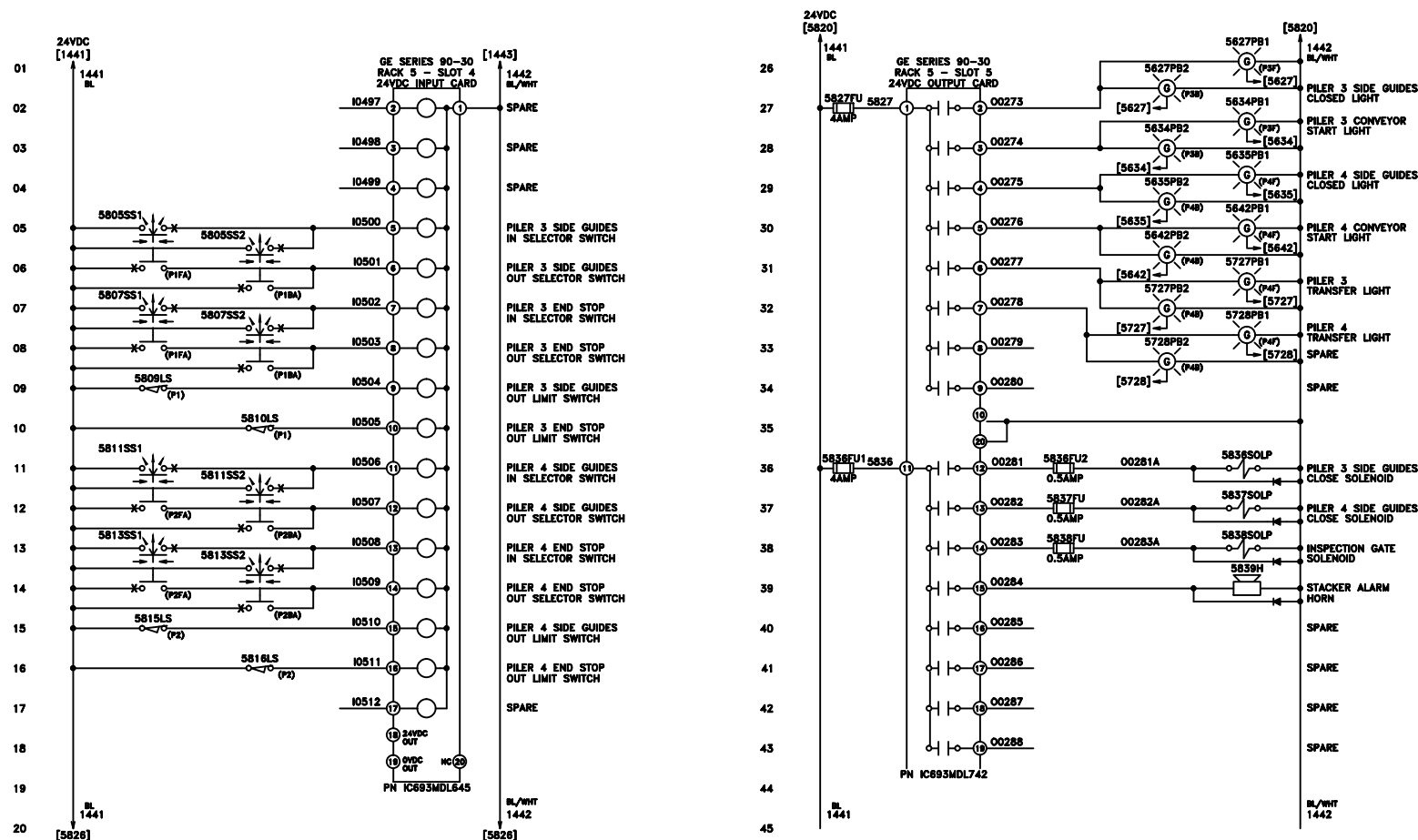


**RACK 5
STACKER - PILER 3 & 4**

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROM		HOJA 57 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROBADO:			
IMPRESION: HAS		CARG: T152H057.DWG	
		FILE: T152348C	

DATE	SYM	WAS	BY
		D-305501	

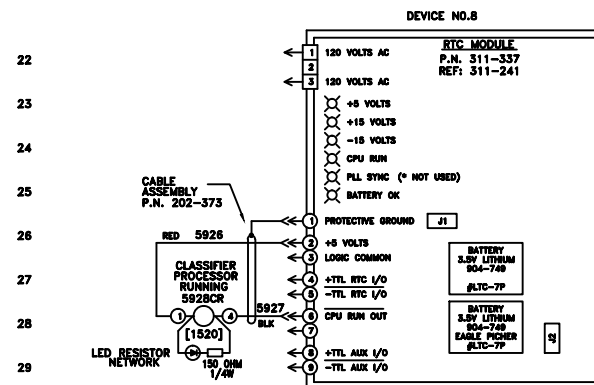
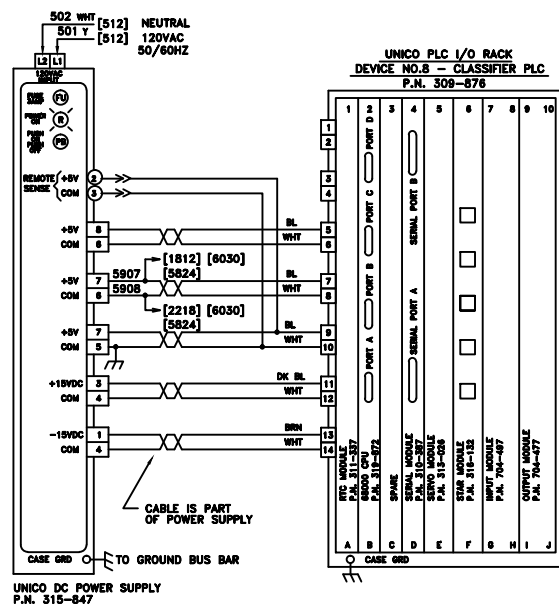


RACK 5 STACKER - PILER 3 & 4

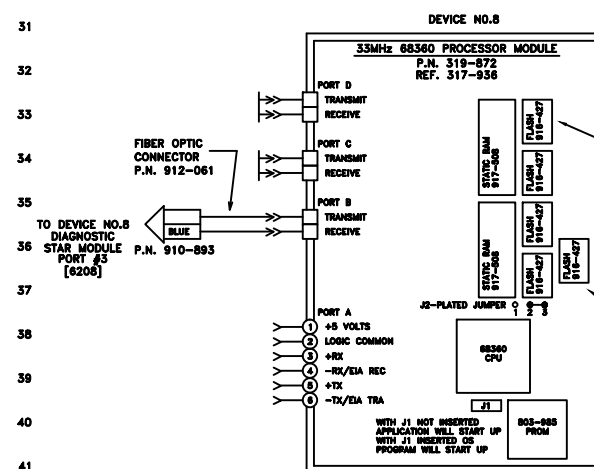
0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDEROCROM		HOJA 58 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: LINEA DE CORTE - BR-4E	
SCROLL SHEETING LINE		PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPRESION:	HAB	
CONTROL:	IMPRESION:	HAB	
APROBADO:	IMPRESION:	HAB	

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	



* NOTE:
THE PLL SYNC LED IS NOT USED WHEN 120VAC IS NOT APPLIED TO TERMINALS 1 AND 5 AT THE TOP OF THE MODULE. THE LED MAY OR MAY NOT BE ON.



APPLICATION PROGRAM
804-898.000
OPERATING SYSTEM PROGRAM
809-012.311

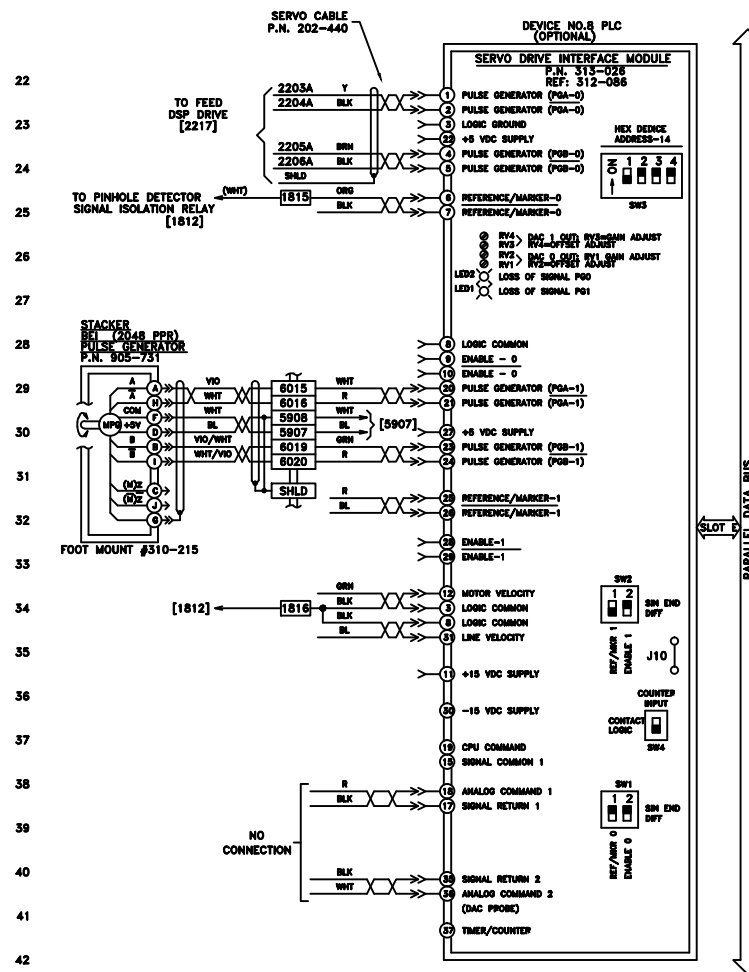
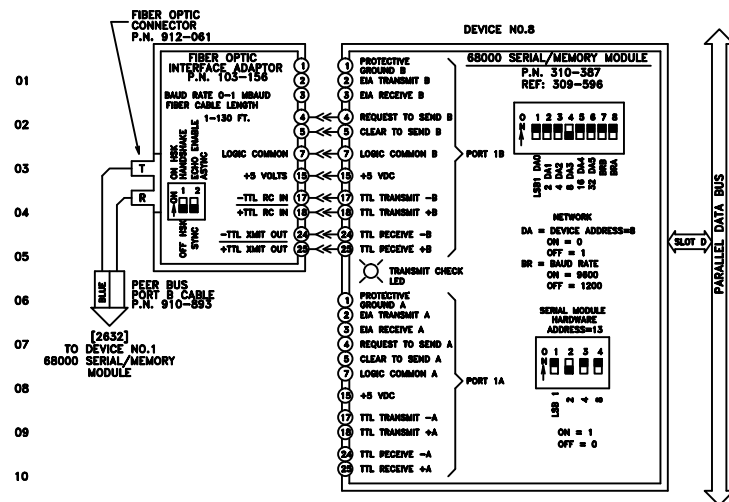
BOOT OS PROGRAM
809-013.311

UNICO PLC I/O RACK
DEVICE NO.8 - CLASSIFIER PLC
SC-2397
(DOOR 5)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REX.	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOLIA 59 DE 67	
SECTOR: LINEA DE CORTE		OBRA O EQUIPO: LINEA DE CORTE - BR-4E	
PARTES: SCROLL SHEETING LINE		DIAGRAM-ELECTRICAL-SCHEMATIC	
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:	IMPULSION:	MA3	IMPULSION: MA3
CONTROL:	IMPULSION:	MA3	IMPULSION: MA3
APROBADO:	IMPULSION:	MA3	IMPULSION: MA3

DATE	SYM	WAS	BY
		D-305501	

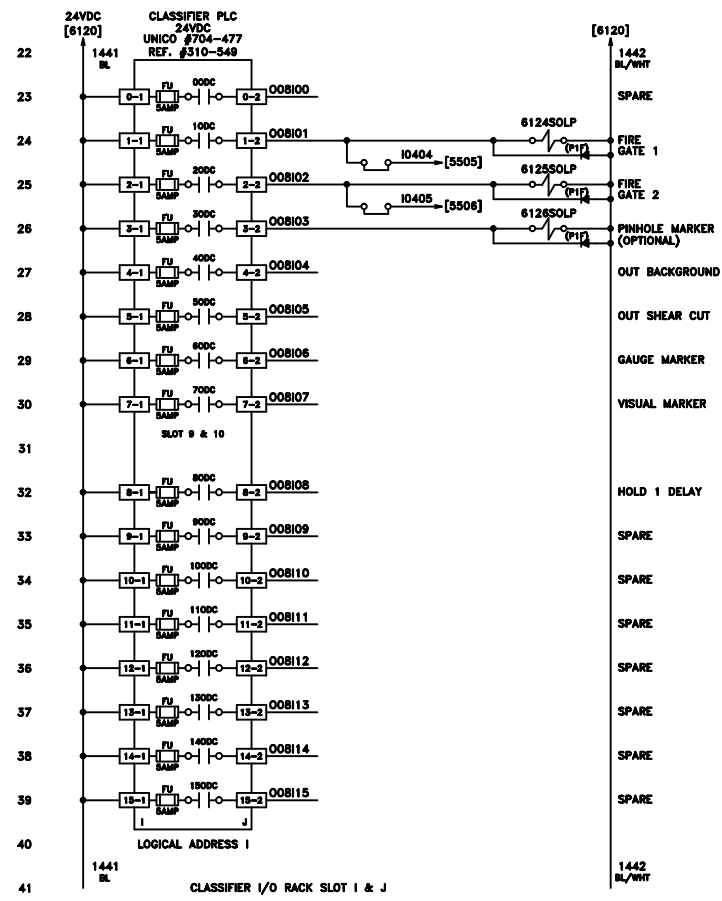
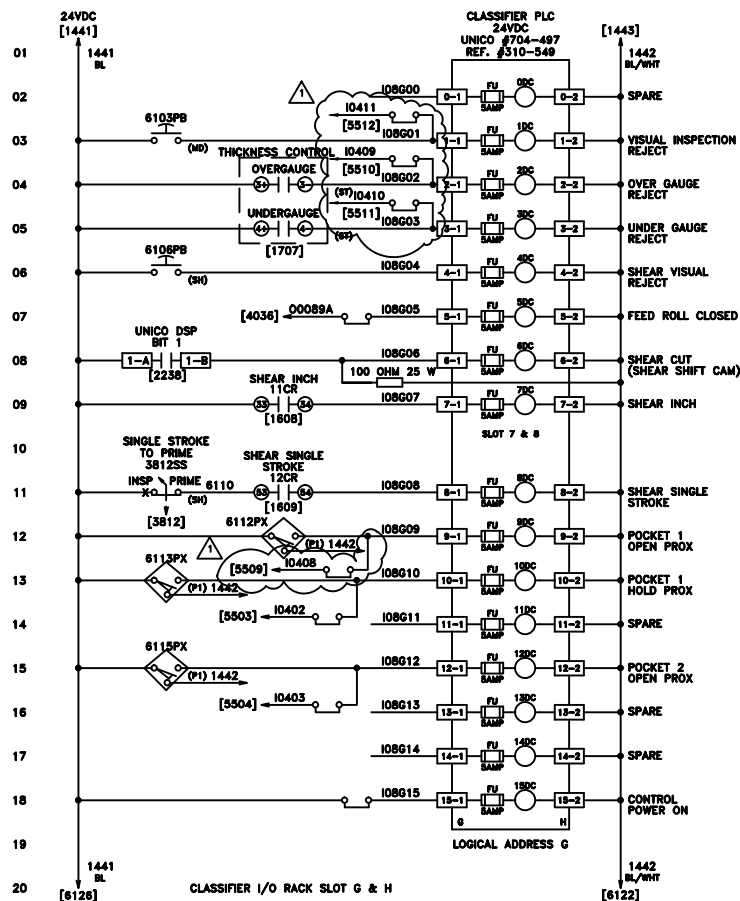


68000 SERIAL/MEMORY MODULE
SERVO DRIVE INTERFACE MODULE
(DEVICE #8)
(DOOR 5)

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APROV.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 60 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PROYECTO: 5615-23-0028			
DISEÑO: 5615-23-0028			
CONTROLADO: 5615-23-0028			
APROBADO: 5615-23-0028			

DATE	SYM	WAS	BY
		D-305501	

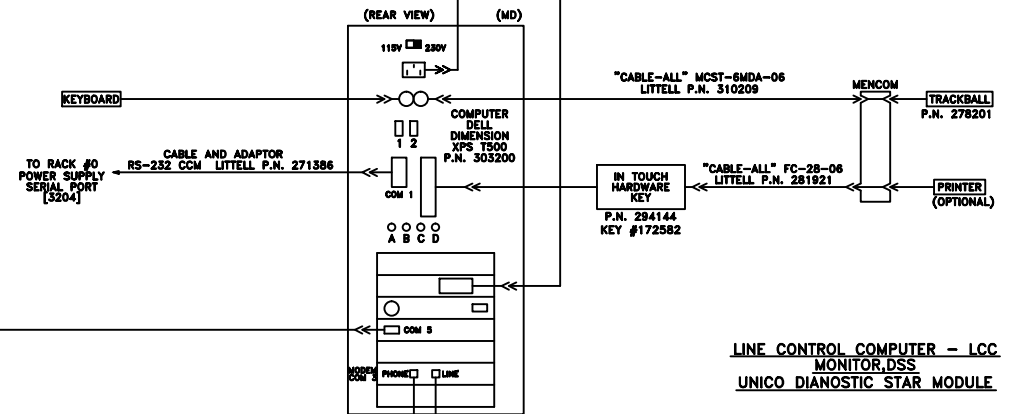
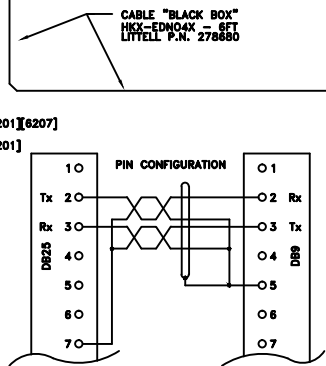
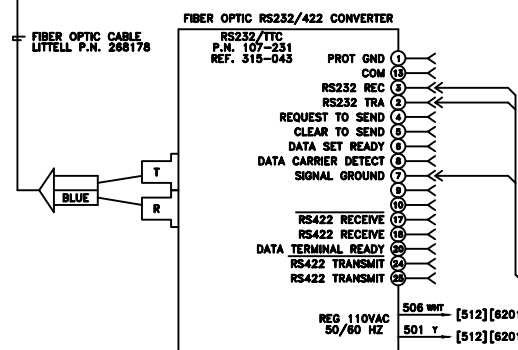
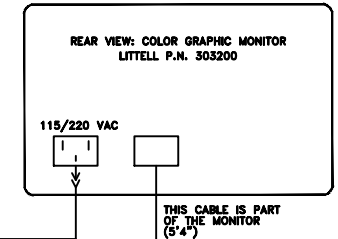
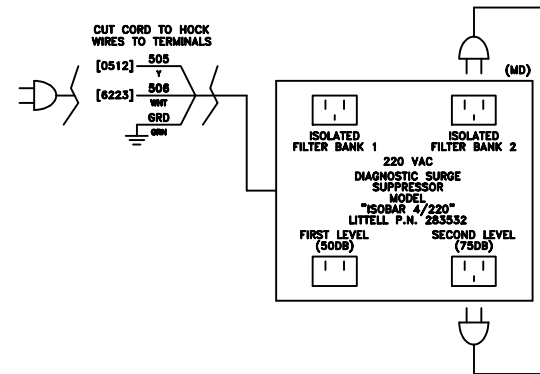
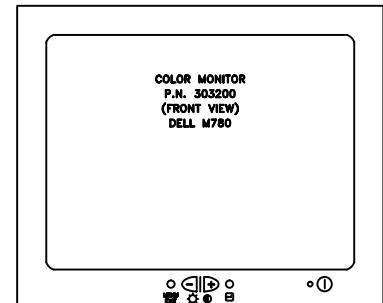
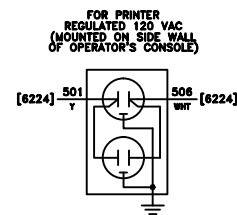
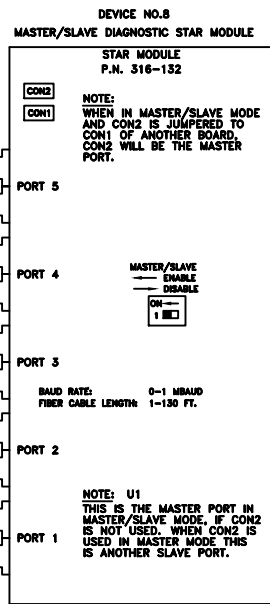


UNICO I/O RACK; DEVICE NO.8-CLASSIFIER PLC
(DOOR 5)

LITTELL JOB #94228 LINE (R → L)

T		PUNTEROS I0411/I08001; I0409/I08002; I0410/I08003; I0408/I08009 DIVE DIVE DNP 03-08-01	
0		EMISION PARA CONSTRUCCION	
REV.	DESCRIPCION	EJEC. CONT. APROV.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691 HOJA 61 DE 67
ZONA:		SIDERCROM	
SECTOR:		LINEA DE CORTE	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		ESCALAS	
DISEÑO:		5615-23-0028	
CONTROL:		IMPRESION: HAS	
APROBADO:		CAD: T1550001.DWG	
		FILE: T155348C	

DATE	SYM	WAS	BY
		D-305501	



LINE CONTROL COMPUTER - LCC
MONITOR, DSS
UNICO DIANOSTIC STAR MODULE

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	ELEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HOJA 82 DE 87	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROB:			
IMPULSION: HAZ		CAB: T152002.DWG	
		FILE: T152348C	

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY
		D-305501	

EXTERNAL WIRING NOTES:

ENCODER CABLE TO BE UNICO P.N. 101-895 (OR EQUIVALENT). REMAINING SIGNAL WIRES ARE TO BE TWISTED PAIR, AND ARE TO BE RUN IN A SEPARATE CONDUIT, AND ROUTED SEPARATE FROM OTHER CONTROL WIRING. IN GENERAL, EACH TRANSDUCER SHOULD HAVE ITS OWN SEPARATE CONDUIT FOR ITS SIGNAL AND BIAS WIRING. SPECIFICALLY ROUTE TRANSDUCER LEADS SEPARATE FROM MOTOR LEADS, FIELD LEADS, AND BLOWER MOTOR WIRES.

RUN TRANSDUCER AND REFERENCE SWITCHES DIRECTLY BACK TO THE OPERATORS CONSOLE USING P.N. 105-819 TRANSDUCER CABLE OR EQUIVALENT. JUNCTION BOXES CAN BE USED AS REQUIRED PROVIDED THE RESULTS ARE EQUIVALENT AS A CONTINUOUS CABLE.

DO NOT ROUTE TRANSDUCER AND REFERENCE SIGNAL WIRES THROUGH TERMINAL JUNCTION BOXES THAT CONTAIN AC CONTROL WIRING.

USE #14 OR #12 TWISTED PAIR FOR TRANSDUCER BIAS WIRES WITH LONG RUNS. THIS REDUCES FUTURE INTERMITTENT SIGNAL PROBLEMS AND LINE VOLTAGE DROP.

ENCLOSURE

NOTE: CONTACT PROTECTION MUST BE PROVIDED ON INDUCTIVE OR HIGH INRUSH CURRENT LOADS.

ARC SUPPRESSORS SHOULD BE USED ON ALL INDUCTIVE LOADS IN CLOSE PROXIMITY TO UNICO CIRCUITS.

CUSTOMER SUPPLIED CONTACTS TO UNICO LOGIC CIRCUITS MUST BE SUITABLE FOR SWITCHING 5 VDC LOGIC SIGNALS, AND MUST BE BOUNCE FREE.

SEPARATE SIGNAL (5-15 VOLT) WIRING FROM CONTROL (120 VOLT) AND (240/480 VOLT) WIRING IN THE POWER CABINET AND ON MACHINE INTERCABLING. IN GENERAL, WIRING WITH DIFFERENT VOLTAGE POTENTIALS SHOULD BE RUN IN SEPARATE WIRE CHANNEL AND TERMINAL STRIPS IN THE ENCLOSURE AND EXCLUSIVE CONDUITS AND JUNCTION BOXES WITH MACHINE INTERCABLING.

WIRING NOTES:

1. PULSE GENERATOR CABLE TO BE UNICO P.N. 101-895 (OR EQUIVALENT). REMAINING SIGNAL WIRES ARE TO BE TWISTED PAIR, ALPHA #2258 (OR EQUIVALENT), AND ARE TO BE RUN IN A SEPARATE CONDUIT AND ROUTED SEPARATE FROM OTHER CONTROL WIRING. IN GENERAL, EACH TRANSDUCER SHOULD HAVE ITS OWN SEPARATE CONDUIT FOR ITS SIGNAL AND BIAS WIRING.

2. REMAINING WIRING, INCLUDING BIAS WIRING, CAN BE #14 AWG.

3. N/A

4. ARC SUPPRESSORS SHOULD BE USED ON ALL INDUCTIVE LOADS IN CLOSE PROXIMITY TO UNICO CIRCUITS.

5. CUSTOMER SUPPLIED CONTACTS MUST BE SUITABLE FOR SWITCHING 5 VDC LOGIC SIGNALS, AND MUST BE BOUNCE FREE.

6. UNICO SYSTEMS SUPPLIED IN ENCLOSURES OR ON PANELS MUST HAVE ENCLOSURES, PANELS, TRANSFORMERS, AND LOOSE ELECTRICAL COMPONENTS CONNECTED TO EARTH ACCORDING TO NEC AND ALL LOCAL CODES.

7. UNICO LOOSE PARTS SYSTEMS MUST HAVE ALL ELECTRICAL COMPONENTS, ESPECIALLY TRANSFORMERS, CONNECTED TO EARTH ACCORDING TO NEC AND ALL LOCAL CODES.

ENCLOSURE WIRING STANDARDS

1. FOLLOW JOINT INDUSTRIAL COUNCIL ELECTRICAL STANDARDS (J.I.C.) AND CONSULT THE NATIONAL ELECTRICAL CODE (N.E.C.)
2. PANEL SHOULD BE LAID OUT TO KEEP POWER AND CONTROL CIRCUITRY ISOLATED.
3. LAY OUT THE WIRE TERMINAL BLOCKS SO THAT INTERNAL AND EXTERNAL WIRING DOES NOT CROSS OVER TERMINALS.
4. SHOULD HAVE A MINIMUM OF 1 1/2" CLEARANCE ON BOTH SIDES OF TERMINALS.
5. KEEP PULSE GENERATOR CONNECTIONS CLOSE TOGETHER BY USING CONSECUTIVE TERMINALS.
6. WATCH THAT PULSE GENERATOR AND ENCODER WIRING DOES NOT SPLIT BETWEEN TIERS OF TERMINALS.
7. USE TWISTED PAIR WIRES FOR SIGNAL WIRES AS DESIGNATED ON THE ELECTRICAL DRAWINGS.
8. SIGNAL AND GROUND WIRES SHOULD ORIGINATE AT ADJACENT TERMINALS.
9. LAY OUT +5VDC AND GROUND WIRES CONSIDERING THE LOADING REQUIREMENTS OF VARIOUS DEVICES.
10. ON PANEL DRAWINGS, DESIGNATE AND LABEL WIRE CHANNEL FOR HIGH VOLTAGE, LOW VOLTAGE, AND CUSTOMER WIRING.
11. TRY TO LEAVE EMPTY WIRE CHANNEL AVAILABLE FOR CUSTOMER'S WIRES. (TRY NOT TO EXCEED 50% USAGE)
12. ON THE ENCLOSURE DRAWINGS SHOW THE POSITION OF EQUIPMENT SO THE CUSTOMER CAN FIGURE WHERE TO ENTER THE ENCLOSURE WITH HIS WIRING.
13. DO NOT PANEL MOUNT ANY EQUIPMENT LOWER THAN 18" FROM THE FLOOR.
14. KEEP AC LINES SEPARATE FROM DC LINES, CROSS AT RIGHT ANGLES ONLY, AND KEEP THE RED 110VAC AND BLACK HIGH VOLTAGE WIRES AWAY FROM OTHER WIRES.
15. MAXIMUM OF TWO WIRES PER SIDE OF TERMINAL, EXCLUDING BUS BARS.
16. TWIST AC LINES WHEN POSSIBLE.
17. WHEN RUNNING HIGH POWER ON SUB-PANEL, WIRE SHOULD LIE ON SUB-PANEL EXCEPT WHEN CROSSING OVER WIRE CHANNEL. THEN 1/2" HIGHER.
18. KEEP UNICO WIRING TO ONE SIDE OF TERMINALS. (TOP AND INSIDE)
19. NO SPLICING OF WIRE EXCEPT FOR INTERLOCKS, ETC.
20. TRANSFORMER 110VAC WIRING TO TOP OF FUSE BLOCKS, LOAD TO BOTTOM.
21. SAME WIRE NUMBER ON BOTH SIDES OF TERMINALS. ALL WIRES TO HAVE WIRE NUMBERS.

LITTELL JOB #94228 LINE (R → L)

0		EMISION PARA CONSTRUCCION		12-08-81	
REV.	DESCRIPCION	ELEC. CONT. APROB.		FECHA	
PROVEEDOR:	LITTELL INTERNATIONAL INC.		N° PROVEEDOR: D-308691 HOJA 63 DE 67		
ZONA:		SIDERCROM			
SECTOR:		LINEA LITTELL			
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E			
PARTES:		SCROLL SHEETING LINE DIAGRAM-ELECTRICAL-SCHEMATIC			
LA FIRMA DEL INGENIERO LA PROYECCION DE SER EL PROYECTO, EL PROYECTO DE SER EL PROYECTO Y EL PROYECTO DE SER EL PROYECTO DE SER EL PROYECTO		ESCALAS			
PROYECTO			5615-23-0028		
DISEÑO					
CONTROL					
APROBADO					
IMPRESION:		HAB		CAD: T1524063.DWG FILE: T152348C	

DATE	SYM	WAS	BY
		D-308501	

MODEL

OUTPUT SPECIFICATIONS

LINE VOLTAGE
3-80 Vdc
MAX. OFF-STATE LEAKAGE
1.5 mA @ 80Vdc
MINIMUM PEAK BLOCKING VOLTAGE
MAXIMUM TURN-ON TIME
MAXIMUM TURN-OFF TIME
LOAD CURRENT RANGE (rms)
POWER DISSIPATION
MAXIMUM SURGE CURRENT
MAXIMUM ZERO VOLTAGE OFFSET

STATIC dV/dt
ON STATE VOLTAGE DROP
CLAMPING VOLTAGE
THERMAL RESISTANCE
TRANSIENT POWER DISSIPATION
(P_t) FOR PULSING ($t=0.1ms$)

INPUT SPECIFICATIONS

NOMINAL LOGIC VOLTAGE (V_{in})
LOGIC VOLTAGE RANGE: STD
OS
MAX. LOGIC SUPPLY CURRENT
AT NOMINAL V_{in} STD
NOMINAL INPUT RESISTANCE (R_{in}): STD
OS
MIN. DROP OUT VOLTAGE
MAXIMUM REVERSE LOGIC VOLTAGE

GENERAL SPECIFICATIONS

STORAGE TEMPERATURE RANGE
OPERATING TEMPERATURE RANGE
ISOLATION RESISTANCE (INPUT TO OUTPUT;
INPUT TO OUTPUT TO GND)
DIELECTRIC STRENGTH INPUT TO OUTPUT
INPUT TO OUTPUT CAPACITANCE
VIBRATION

MECHANICAL SHOCK

DC OUTPUT MODULE 70-00GS AND 700-00GS

60 Vdc max.
3-80 Vdc
1.5 mA @ 80Vdc

0.20 μ Sec
0.50 μ Sec
0.03-5.5 Amperes
1.5 Watt/Amp
5 Amperes max for 1 Sec

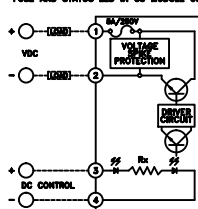
1.2 V max.
80 Vdc max.
20°C/Watt typical
400 Watts @ 1mSec non-recurring

5Vdc
2.5-10Vdc
-5Vdc
15 mA
15 mA
750 Ohms
150 Ohms
1Vdc
-5Vdc

-40° TO +125° C
-40° TO +100° C

10¹⁰ Ohms
4000Vdc (rms) min.
6pF typical
20 G's PEAK OR 0.5" DOUBLE AMPLITUDE
10-2000 Hz PER MIL-STD-202, METHOD 204,
CONDITION B
1800 G's 0.5ms HALF-SINE PER MIL-STD-202,
METHOD 215, CONDITION F

DC OUTPUT FUSE AND STATUS LED IN GS MODULE ONLY



AC OUTPUT MODULE 70-0AGS AND 700-0AGS

120 Vac nominal
24-140 Vac
2 mA @ 80Hz
400 V
8.3 mSec @ 60 Hz
8.3 mSec @ 60 Hz
0.03-5.5 Amperes
1.0 Watt/Amp
80 Amperes @ 80Hz, 1 cycle (peak)
5 V peak
3000 V/ μ Sec
1.5 V max.

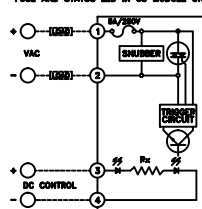
20°C/Watt typical
35 A² per Sec. min.

5Vdc
2.5-10Vdc
-5Vdc
15 mA
15 mA
750 Ohms
150 Ohms
1Vdc
-5Vdc

-40° TO +125° C
-40° TO +100° C

10¹⁰ Ohms
4000Vdc (rms) min.
6pF typical
20 G's PEAK OR 0.5" DOUBLE AMPLITUDE
10-2000 Hz PER MIL-STD-202, METHOD 204,
CONDITION B
1800 G's 0.5ms HALF-SINE PER MIL-STD-202,
METHOD 215, CONDITION F

AC OUTPUT FUSE AND STATUS LED IN GS MODULE ONLY



MODEL

INPUT SPECIFICATIONS

INPUT VOLTAGE
INPUT VOLTAGE RANGE
INPUT CURRENT AT MAX INPUT VOLTAGE
MAXIMUM TURN-ON TIME
MAXIMUM TURN-OFF TIME
NOMINAL INPUT RESISTANCE (R_{in})
MAX. PICK UP VOLTAGE
MIN. DROP OUT VOLTAGE

OUTPUT SPECIFICATIONS
NOMINAL LOGIC VOLTAGE (V_{in})
LOGIC VOLTAGE RANGE: STD
OS
MAX. LOGIC SUPPLY CURRENT
AT NOMINAL V_{in}
OUTPUT CURRENT RANGE
BREAKDOWN VOLTAGE
OFF STATE LEAKAGE CURRENT
ON STATE VOLTAGE DROP

GENERAL SPECIFICATIONS

STORAGE TEMPERATURE RANGE
OPERATING TEMPERATURE RANGE
ISOLATION RESISTANCE (INPUT TO OUTPUT;
INPUT TO OUTPUT TO GND)
DIELECTRIC STRENGTH INPUT TO OUTPUT
INPUT TO OUTPUT CAPACITANCE
VIBRATION

MECHANICAL SHOCK

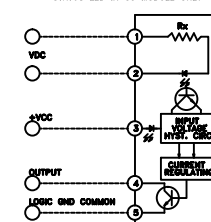
DC INPUT MODULE 70-0DS AND 700-0DS

32 Vdc max.
3-32 Vdc
18 mA
0.20 mSec
0.40 mSec
1.85Ohm
3Vdc
1Vdc
5Vdc
2-5Vdc
1.5-5Vdc
10 mA
1-50mA
50Vdc min.
1uA max.
0.45Vdc @ 50mA max.

-40° TO +125° C
-40° TO +100° C

10¹⁰ Ohms
4000Vdc (rms) min.
6pF typical
20 G's PEAK OR 0.5" DOUBLE AMPLITUDE
10-2000 Hz PER MIL-STD-202, METHOD 204,
CONDITION B
1800 G's 0.5ms HALF-SINE PER MIL-STD-202,
METHOD 215, CONDITION F

DC INPUT STATUS LED IN GS MODULE ONLY



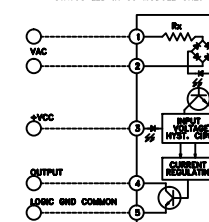
AC INPUT MODULE 70-0AS AND 700-0AS

120 Vac nominal
90-140 Vac/Vdc
8 mA rms
20 mSec
20 mSec
2250Ohm
90Vdc
28Vdc
5Vdc
2-5Vdc
1.5-5Vdc
10 mA
1-50mA
50Vdc min.
1uA max.
0.45Vdc @ 50mA max.

-40° TO +125° C
-40° TO +100° C

10¹⁰ Ohms
4000Vdc (rms) min.
6pF typical
20 G's PEAK OR 0.5" DOUBLE AMPLITUDE
10-2000 Hz PER MIL-STD-202, METHOD 204,
CONDITION B
1800 G's 0.5ms HALF-SINE PER MIL-STD-202,
METHOD 215, CONDITION F

AC INPUT STATUS LED IN GS MODULE ONLY



LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IF PROVEEDOR: D-308691
ZONA:		SIDERCROM	
SECTOR:		LINEA LITTELL	
OBRA O EQUIPO:		LINEA DE CORTE - BR-4E	
PARTES:		SCROLL SHEETING LINE	
PROYECTO:		DIAGRAM-ELECTRICAL-SCHEMATIC	
DIBUJO	ESCALAS	5615-23-0028	
CONTROL	IMPULSION:	HAB	
APROB	FILE:	T152348C	

DATE	SYM	WAS	BY
		D-305501	

* ABB DRIVE SETTINGS

PARAMETER	MAIN CONV.	PILER CONV. DRIVES 1-4	SIDE GUIDES & END STOPS	COMMENT
9901	1 = ENGLISH (US)	1 = ENGLISH (US)	1 = ENGLISH (US)	LANGUAGE
9902	1 = ABB STANDARD	1 = ABB STANDARD	1 = ABB STANDARD	APPLICATION MACRO
9905	460	460	460	MOTOR VOLTAGE
9906	12.5	4	0.7	MOTOR CURRENT
9907	60	60	60	MOTOR FREQUENCY
9908	1785	1760	1725	MOTOR SPEED
9909	10	3	1/3	MOTOR POWER
9910	82	80	58	MOTOR COS PWJ (PF)
1001	0 = NOT SEL	0 = NOT SEL	9 = DI1F,2R	EXT1 COMMANDS
1003	1 = FORWARD	1 = FORWARD	3 = REQUEST	DIRECTION
1105	60	60	60	EXT REF1 MAX (HZ)
1201	0 = NOT SEL	0 = NOT SEL	7 = DI3,4	CONST SPEED SEL
1202	0 = NOT SEL	0 = NOT SEL	30	CONST SPEED 1 (HZ)
1203	0 = NOT SEL	0 = NOT SEL	60	CONST SPEED 2 (HZ)
1204	0 = NOT SEL	0 = NOT SEL	0 = NOT SEL	CONST SPEED 3 (HZ)
1402	7 = SUPRV1 OVER	7 = SUPRV1 OVER	7 = SUPRV1 OVER	RELAY OUTPUT 2 (SUPERVISE PARAM. 3201)
1602	0 = LOCKED	0 = LOCKED	0 = LOCKED	PARAMETER LOCK
1604	5 = DI5	5 = DI5	0 = KEYPAD	FAULT RESET SELECT
1605	1 = LOCKED	1 = LOCKED	1 = LOCKED	LOCAL LOCK (EXTERNAL CONTROL ONLY)
2008	60	60	60	MAXIMUM FREQUENCY (HZ)
2102	1 = COAST	1 = COAST	1 = COAST	STOP FUNCTION
2201	0 = NOT SEL	0 = NOT SEL	5 = DI5	ACC/DEC 1/2 SEL
2202	3	3	1	ACCELER TIME 1 (SEC)
2203	60	60	1	DECELER TIME 1 (SEC)
2204	1	1	1	ACCELER TIME 2 (SEC)
2205	1	1	1	DECELER TIME 2 (SEC)
3003	3 = DI3	3 = DI3	0 = NOT SEL	EXTERNAL FAULT
TO SET UP RELAY 2 FUNCTION				
3201	103	103	103	SUPERVISE 1 PARAMETER
3202	29	20	10	SUPERVISE 1 LIMIT LOW (OF PARAM. 3201)
3203	32	23	12	SUPERVISE 1 LIMIT HI (OF PARAM. 3201)

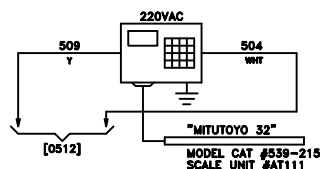
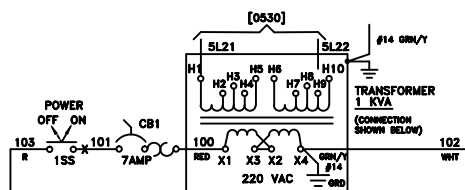
* NOTE: PARAMETERS WHICH ARE SHOWN HAVE VALUES DIFFERENT FROM FACTORY SETTINGS. FOR FULL LISTING AND DESCRIPTION OF PARAMETERS AND DRIVE OPERATIONS, SEE ABB USER'S MANUAL IN LITTELL SERVICE MANUAL.

STACKER DRIVE SETTINGS

LITTELL JOB #94228 LINE (R -> L)

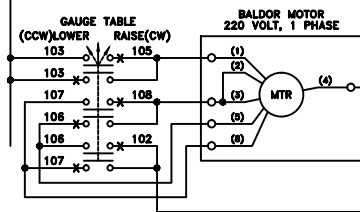
0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691 HOJA 65 DE 67
SIDERAR INTERNATIONAL INC.		ZONA:	SIDERCROM
		SECTOR:	LINEA LITTELL
		OBRA O EQUIPO:	LINEA DE CORTE - BR-4E
			SCROLL SHEETING LINE
		PARTE:	DIAGRAM-ELECTRICAL-SCHEMATIC
LA FIRMA DE APROBACION LA PRODUCCION DE ESTE DIBUJO DEBE SER AUTOGRAFICA Y DEBE INCLUIR EL NOMBRE DEL INGENIERO RESPONSABLE DEL DISEÑO.		ESCALAS	
PROYECTO			5615-23-0028
DIBUJO			
CONTROLADO			
APROBADO			
		IMPRESION:	MAS
		ARCHIVO:	T1524065.DWG
		FILE:	T152348C

DATE	SYM	WAS	BY
		D-305501	



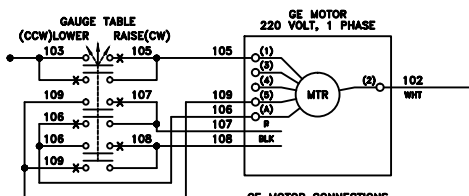
DIGITAL
READOUT
UNIT
(SEE NOTES)

LINEAR
ENCODER



BALDOR MOTOR CONNECTIONS
110/220 VOLT, 1 PHASE

	110VAC		220VAC	
	CW	CCW	CW	CCW
L1	1-3-8	1-3-5	1	1
L2	2-4-5	2-4-8	4-5	4-8
			2-3-8	2-3-5



	110VAC	220VAC
BL	5	4
WHT	2	5
R	5	5
BLK	A	A

NOTES:

COMPUTER MODEL UDR-200-1L
DIP SWITCH SETTINGS

RESOLUTION 0.0005"/0.005MM

POSITION 1 2 3 4 5 6 7 8
SETTINGS on off on off on off on off

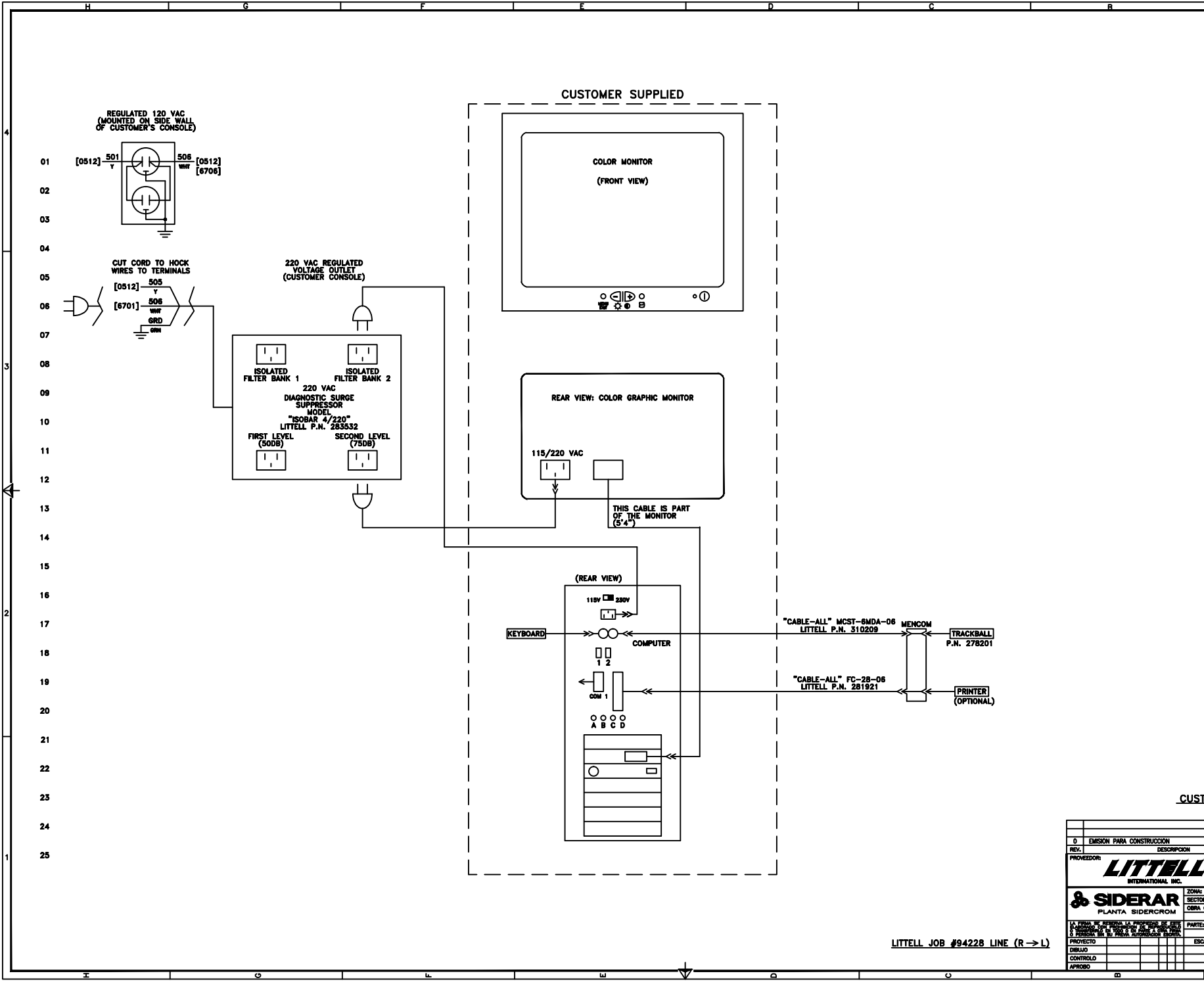
TRANSFORMER CONNECTIONS (50/60 HZ)

PRIMARY VOLT	H1	H2	H3	H4	H5	H6	H7	H8	H9	H10
200	1L1	1L2				1L1	1L2			
220	1L1				1L2	1L1				1L2
380	1L1		JUMPER				1L2			
400	1L1		JUMPER				1L2			
415	1L1		JUMPER				1L2			
440	1L1		JUMPER				1L2			
480	1L1	JUMPER		1L2						1L2

LITTELL JOB #94228 LINE (R → L)

0 EMISSION PARA CONSTRUCCION		12-06-81	
REV.	DESCRIPCION	EJEC. CONT. APRO.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	IT PROVEEDOR:	D-308691
ZONA: SIDEROCROM		HOJA 66 DE 67	
SECTOR: LINEA LITTELL			
OBRA O EQUIPO: LINEA DE CORTE - BR-4E			
PARTES: SCROLL SHEETING LINE			
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC			
PROYECTO:	ESCALAS:	5615-23-0028	
DISEÑO:			
CONTROL:			
APROBADO:			

DATE	SYM	WAS	BY
		D-305501	

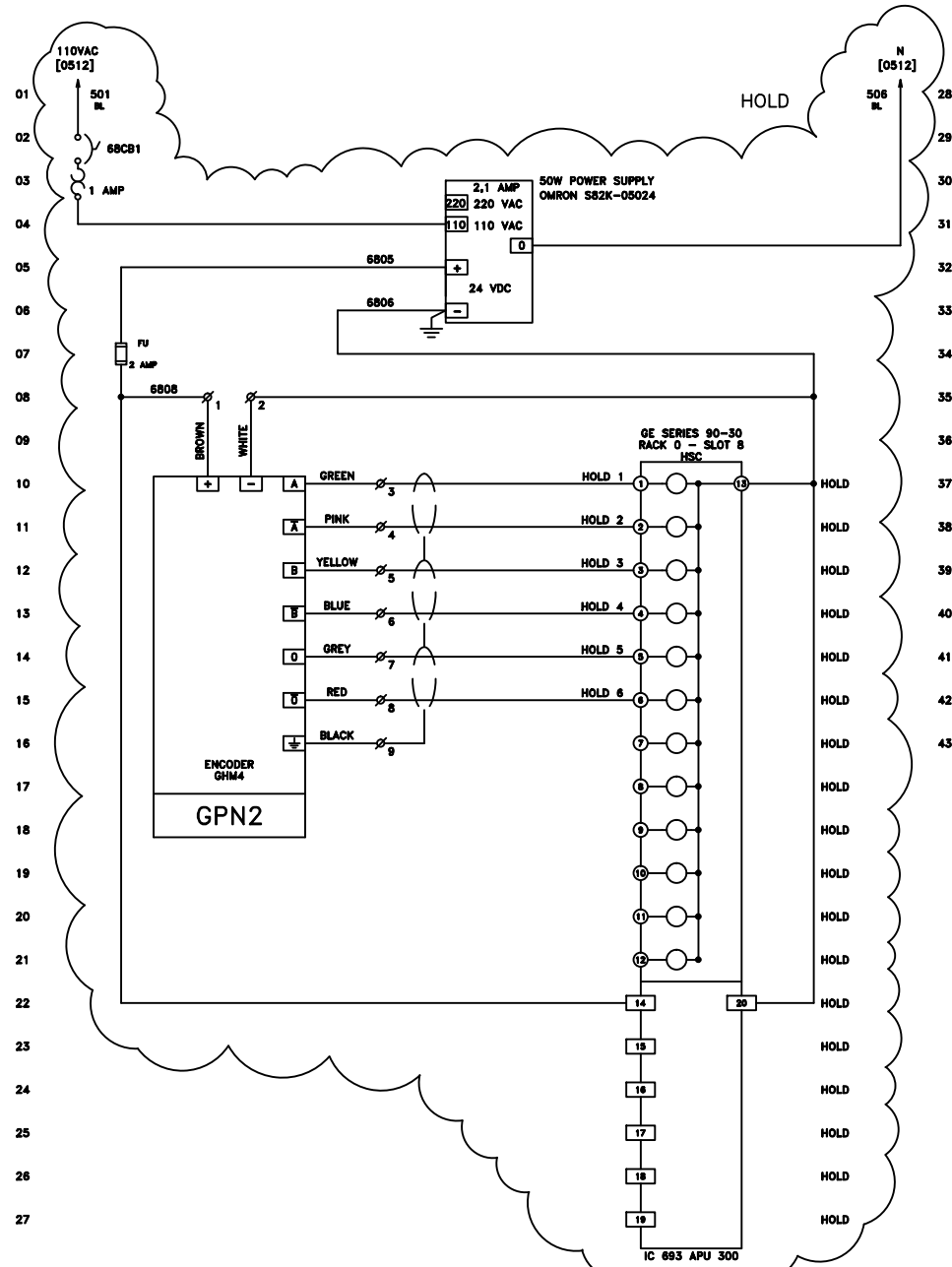


CUSTOMER CONSOLE

0 EMISSION PARA CONSTRUCCION		12-08-81	
REV.	DESCRIPCION	EJEC. CONT. APROB.	FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.		IT PROVEEDOR: D-308691
ZONA: SIDERAR		HORA 67 CONT 68	
SECTOR: LINEA DE CORTE		HORA 67 CONT 68	
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		HORA 67 CONT 68	
PARTES: SCROLL SHEETING LINE		HORA 67 CONT 68	
PARTES: DIAGRAM-ELECTRICAL-SCHEMATIC		HORA 67 CONT 68	
PROYECTO	ESCALAS	5615-23-0028	
DISEÑO			
CONTROL			
APROB			
IMPRESION: H43		CAB: T1524067.DWG	
		FILE: T1524067	

LITTELL JOB #94228 LINE (R → L)

DATE	SYM	WAS	BY



**RACK 0
OPERATOR'S CONSOLE**

1 SE AGREGA SLOT 8		DATE: 03-08-81
REV.	DESCRIPCION	EJEC. CONT. APROV. FECHA
PROVEEDOR:	LITTELL INTERNATIONAL INC.	Nº PROVEEDOR: D-308691 HOJA 68 CONT -
ZONA: SIDEROCROM		SECTOR: LINEA LITTELL
OBRA O EQUIPO: LINEA DE CORTE - BR-4E		PARTE: SCROLL SHEETING LINE
PROYECTO: DYCESA 01 8 01		ESCALAS: 5615-23-0028
DISEÑO: DYCESA 01 8 01		APPROBADO: DYCESA 03 8 01
CONTROLADO: DRP 08 8 01		IMPRESION: HAZ
APPROBADO: DYCESA 03 8 01		ICAD: T1554008.DWG FILE: T155348C